
Library Management System

*Library Management System
System Documentation*

Subject: Advanced Software Engineering

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1. Summary

1.1 Overview

This document outlines the development of a software application named **Library Management System (LMS)**, a desktop-based application developed using JavaFX and a Java backend, aimed at modernizing and optimizing the operations of a physical library. The system provides an integrated environment to manage:

- *Book inventory and catalog management*
- *Member registration and profile management*
- *Borrowing and returning transactions*
- *Book reservation handling*
- *Fine calculation and overdue tracking*
- *User access control*
- *Reports and library statistics*

This centralized platform offers tailored access to three types of users: Administrator, Librarian, and Member (Patron), each with clearly defined roles and permissions. Administrators can manage users, books, and reports, librarians can handle book circulation and member activities, while members can search books, reserve available titles, and view their borrowing history.

This software application aims to improve efficiency, reduce manual paperwork, and provide real-time access to library information for institutions that wish to digitize and simplify their library operations.

1.2 Project purpose

The goal of the **Library Management System (LMS)** is to digitize and simplify the daily operations of a library. Traditional library environments rely heavily on manual record keeping, paper-based tracking systems, and time-consuming administrative processes. The system addresses these issues by providing:

- *Centralized access to book and member information*
- *Automated borrowing, returning, and due-date tracking*
- *Digital reservation management for unavailable books*
- *Automatic fine calculation for overdue returns*
- *Detailed reporting and statistics for library performance analysis*
- *Role-based access to ensure secure and appropriate interaction with system data*

Additionally, the system aims to increase operational efficiency, minimize human errors, reduce paperwork, and provide an intuitive and user-friendly interface for administrators, librarians, and members.

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1.3 Stakeholders

- **Core Stakeholders**

The core stakeholders of the Library Management System (LMS) include the administrator users, librarians, and members (patrons). Administrators are the primary operators of the system with full access to the management dashboard, where they can manage user accounts, maintain book inventory, generate reports, monitor borrowing and returning activities, and configure system settings. Their continuous interaction with all system modules makes their experience and feedback essential for ensuring system efficiency, security, and reliability. Librarians are also fundamental users who rely on the platform to manage day-to-day library operations such as issuing and returning books, handling reservations, tracking overdue items, and assisting members in accessing library resources. The software should enable librarians to perform their duties efficiently while reducing manual paperwork and improving transaction accuracy. Members or patrons represent the main end-user group who require an easy-to-use and transparent interface through which they can search the book catalog, borrow or reserve books, view due dates, monitor fines, and access their borrowing history. Ensuring a positive member experience is essential for improving user satisfaction, encouraging active library usage, and supporting the digital transformation of library services.

- Intermediate Stakeholders

The intermediate stakeholders are those who support and maintain the functionality of the Library Management System without necessarily using all operational modules directly. This includes the IT Department and library operational staff. The IT Department is responsible for ensuring the technical stability and security of the platform by handling database maintenance, system updates, backups, bug fixes, and cybersecurity measures. They are also responsible for maintaining system availability and ensuring that the application performs efficiently within the institution's infrastructure. Library operational staff or assistants, if present, may support librarians in organizing books, updating records, and managing circulation activities. Their involvement helps streamline daily library workflows and improve operational efficiency. Their feedback regarding system usability and performance is important for identifying process improvements and ensuring smooth system operation.

- Outer Stakeholders

The outer stakeholders include the university or institution management, educational authorities, and third-party technology providers. The university or institution that owns the library relies on the system to support educational objectives, improve resource management, and provide efficient access to academic materials for students and staff. Institutional management may use reports and statistical data generated by the system to evaluate library performance, monitor resource utilization, and support strategic decision-making. Educational authorities or accreditation bodies may also be considered stakeholders, as they depend on accurate records, reporting, and compliance with institutional standards. Additionally, third-party software developers, consultants, or technology vendors involved in the development, maintenance, or future expansion of the platform play an important role in ensuring long-term system reliability and scalability. Their ability to provide technical support, implement new features, and resolve system issues contributes significantly to the continued success and modernization of the Library Management System.

2. Legal Framework and Regulatory Compliance for the Library Management System

2.1 Overview

The Library Management System (LMS) is a specialized desktop-based digital platform engineered to support the administrative, operational, and management functions of a modern library environment. The system integrates modules for book inventory management, member registration, borrowing and returning transactions, reservation handling, fine calculation, reporting, and secure user access control. Due to the importance of protecting user information, maintaining accurate records, and ensuring secure access to educational resources, the LMS has been developed in accordance with applicable legal, ethical, and technical standards within Albania and internationally.

The Library Management System is designed to be secure, reliable, user-friendly, and compliant with modern information management practices. Its legal and technical framework encompasses copyright protection, data privacy and security regulations, software licensing standards, and institutional information governance policies. The platform aims to ensure transparency, accountability, and operational efficiency while supporting the digital transformation of library services in educational institutions.

2.2 Intellectual Property and Licensing

All intellectual property related to the Library Management System, including its software architecture, source code, database structure, user interface design, documentation, and deployment configuration, is considered the property of the developing institution or project owners. Any external contributors or developers involved in the development process are required to operate under agreements that protect intellectual property rights and ensure the proper transfer or licensing of ownership.

The Library Management System is protected under the Berne Convention for the Protection of Literary and Artistic Works, the WIPO Copyright Treaty, and Law No. 35/2016 of the Republic of Albania on Copyright and Related Rights. These legal frameworks protect the software against unauthorized copying, distribution, modification, or commercial misuse.

The system may utilize third-party software libraries and frameworks such as JavaFX, MySQL, JDBC, and Apache Commons, each operating under their respective software licenses including MIT, Apache 2.0, or GPL licenses. All third-party components are reviewed to ensure compatibility with the project requirements and compliance with open-source licensing obligations. Documentation regarding software dependencies, versions, and licensing terms is maintained as part of the project records.

The Library Management System is intended for educational and institutional use under a non-transferable and limited-use software agreement. Unauthorized redistribution, reverse engineering, sublicensing, or commercial exploitation of the system is prohibited. Access to the platform is restricted to authorized users including administrators, librarians, and registered members, all of whom are required to follow institutional policies regarding acceptable system use.

2.3 Data Protection, Security, and Privacy Compliance

The Library Management System is designed in accordance with the principles of the General Data Protection Regulation (EU 2016/679 – GDPR) and the Albanian Law No. 9887/2008 on the Protection of Personal Data. The system also follows modern information security and privacy management practices to ensure the confidentiality, integrity, and availability of user and institutional data.

Members, librarians, administrators, and other authorized users whose personal data is processed by the system are entitled to privacy rights including access to their information, correction of inaccurate records, restriction of unauthorized processing, and protection against misuse of personal data. The system collects and processes information only for legitimate library management purposes, such as membership registration, borrowing records, reservation tracking, and reporting activities.

To protect sensitive information and ensure system security, the platform implements security measures including password authentication, Role-Based Access Control (RBAC), encrypted database connections, and secure data storage practices. Access permissions are restricted according to user roles to ensure that users can only interact with data relevant to their responsibilities. System activities such as borrowing transactions, returns, reservations, and account updates are logged to maintain accountability and support auditing processes.

The system also includes backup and recovery mechanisms to protect against accidental data loss or technical failures. The IT Department or system administrators are responsible for maintaining database integrity, applying security updates, monitoring system performance, and ensuring continuous availability of the platform.

Where applicable, the Library Management System supports institutional compliance requirements related to educational data management and information security standards. Any future integration with cloud services or external platforms must comply with applicable data protection regulations and institutional authorization procedures.

Cross-border data transfers, if necessary, are subject to GDPR Chapter V obligations and will only occur under valid legal mechanisms such as Standard Contractual Clauses (SCCs), explicit user consent, or appropriate regulatory authorizations. A&S Law Firm is registered as a Data Controller with the Albanian Information and Data Protection Commissioner (IDP).

2.4 Confidentiality, Legal Ethics, and Professional Obligations

The Library Management System (LMS) implements appropriate safeguards to uphold data confidentiality, secure access control, and responsible management of library information. Since the platform stores member records, borrowing history, reservations, and administrative data, maintaining privacy and protecting institutional information are essential system requirements.

Sensitive user information and administrative records are accessible only to authorized personnel based on their assigned roles and permissions. Administrators, librarians, and members are granted different levels of access through Role-Based Access Control (RBAC), ensuring that users can only view or modify information relevant to their responsibilities. User authentication mechanisms and secure login procedures help prevent unauthorized access to the platform.

All important system activities, including book issuance, returns, reservations, fine updates, and account modifications, are recorded with timestamps and user identification details to maintain accountability and support auditing processes. These logs help monitor system usage, identify unauthorized activities, and improve operational transparency within the library environment.

The system also supports secure data storage and backup procedures to reduce the risk of data loss or corruption. Regular database backups and maintenance operations are performed by the IT Department to ensure system reliability and recovery in case of technical failures. In addition, institutional policies regarding the retention and management of member records are respected to ensure compliance with privacy regulations and educational data management standards.

Where applicable, the Library Management System follows professional and ethical responsibilities related to the handling of educational and personal information. All users are expected to use the system responsibly, maintain the confidentiality of login credentials, and comply with institutional rules governing the proper use of library resources and digital services.

2.5 Disclaimers, Warranty, and User Responsibilities

The Library Management System (LMS) is provided “as is” without any implied warranties of merchantability, uninterrupted availability, or fitness for a particular purpose unless otherwise specified in written agreements by the developing institution or authorized administrators. The system is intended strictly for educational and institutional library management purposes and should not be repurposed for unauthorized commercial or unrelated deployments without proper authorization.

The institution, developers, and technical contributors involved in the development of the Library Management System disclaim liability for damages, data loss, operational interruptions, or security issues resulting from improper usage, unauthorized modifications, third-party software conflicts, or negligent user behavior. Any misuse of the platform, including unauthorized access attempts, manipulation of records, or violation of institutional policies, is the responsibility of the users involved and may result in disciplinary or legal consequences.

Users of the system, including administrators, librarians, and members, are required to follow institutional policies regarding secure system usage, password confidentiality, and responsible handling of library data. Administrators and IT personnel are expected to maintain regular database backups, monitor access permissions, and ensure that the system environment remains secure and operational. Proper user training and awareness regarding system functionality and security practices are also important responsibilities to ensure safe and efficient use of the platform.

2.6 Compliance Maintenance and Incident Response

The Library Management System is supported through an ongoing maintenance and compliance framework intended to ensure continued reliability, security, and alignment with institutional and legal requirements. The platform undergoes periodic updates, system testing, database maintenance, and security reviews to maintain proper functionality and protect against evolving cybersecurity risks.

System updates, policy changes, and modifications affecting user access, data processing, or operational procedures are managed through a structured change management process. Any significant updates to system functionality or security controls are documented and communicated to authorized users and administrators to ensure transparency and operational continuity.

The IT Department is responsible for monitoring system performance, maintaining data integrity, applying security patches, and responding to technical issues or vulnerabilities. Security practices such as access monitoring, backup verification, and user activity logging are regularly reviewed to ensure compliance with institutional standards and privacy regulations.

In the event of a technical failure, security breach, or data-related incident, the Library Management System follows a predefined Incident Response Procedure. This procedure includes incident identification, risk assessment, containment measures, recovery actions, and investigation of the root cause. Where required by applicable privacy or institutional regulations, affected parties and relevant authorities may be notified within an appropriate timeframe. Incident reports and recovery records are maintained to support accountability, future prevention measures, and continuous improvement of system security and reliability.

3. Product Description

3.1 Context

The Library Management System (LMS) is designed to address the operational and administrative challenges faced by educational institutions and libraries in managing their daily activities. Traditionally, many libraries rely on manual record keeping, paper-based tracking systems, or disconnected digital tools to manage books, borrowing records, member information, reservations, and overdue fines. These outdated methods often result in inefficiencies, misplaced records, delayed services, and difficulties in maintaining accurate inventory and transaction data. The Library Management System aims to modernize these traditional processes by providing a comprehensive desktop-based platform that centralizes all essential library functions into a single integrated system. By doing so, it reduces administrative workload, improves accuracy, and enhances the overall efficiency of library operations.

This centralized digital environment ensures that administrators, librarians, and members have secure and timely access to accurate information. It minimizes manual errors in book tracking and member management, streamlines borrowing and returning procedures, and improves communication between library staff and users. Members can easily search for books, reserve unavailable materials, monitor due dates, and view borrowing history, while librarians and administrators can efficiently manage inventory, transactions, fines, and reports. Additionally, the system provides management with valuable statistical insights and reporting capabilities to support better decision-making, resource allocation, and overall library performance monitoring.

3.2 User Features

The Library Management System (LMS) is designed specifically for three distinct user roles—Administrator, Librarian, and Member (Patron)—with each role provided a customized interface and a set of functionalities tailored to their responsibilities within the library environment.

The Administrator has full control over the system and is responsible for managing overall library operations. Administrators can create and manage user accounts, maintain the book inventory, configure system settings, monitor borrowing and returning activities, and generate analytical and operational reports. They can oversee all platform activities to ensure that permissions are correctly assigned, records remain accurate, and the system operates efficiently according to institutional requirements.

Librarians are provided with tools that support efficient management of daily library operations. They can issue and return books, manage reservations, update book availability, monitor overdue materials, and calculate fines for late returns. Librarians also assist members in locating and accessing library resources while maintaining accurate transaction records. Their interface is designed to simplify routine tasks, reduce manual paperwork, and improve operational productivity within the library.

Members or patrons benefit from a secure and user-friendly interface through which they can search the library catalog, borrow available books, reserve unavailable materials, and monitor their borrowing history and due dates. Members can also view outstanding fines and manage their personal account information. This digital access improves convenience and transparency for library users while reducing the need for manual inquiries or administrative assistance.

3.3 System Assumptions

The Library Management System (LMS) assumes that users will primarily access the platform through desktop or laptop computers within the library or institutional environment. The system is developed as a desktop-based JavaFX application and therefore requires devices capable of running Java applications efficiently. It is also assumed that users will have basic computer literacy and familiarity with graphical user interfaces, allowing administrators, librarians, and members to navigate the system without extensive technical training.

From an operational perspective, the system assumes that the institution has the necessary technical

infrastructure to support the application, including database management, system maintenance, and regular data backups. The institution is also expected to provide IT support for software installation, updates, troubleshooting, and security monitoring. Additionally, it is assumed that all book records, member information, and transaction data entered into the system are accurate and maintained properly by authorized users.

The system also assumes that library staff will follow institutional procedures for issuing and returning books, handling reservations, and managing overdue fines. Since the current version of the platform operates within a local or institutional environment, internet connectivity may only be required for updates, backups, or future integrations rather than for all core functionalities.

3.4 Constraints

While the Library Management System provides comprehensive functionality for managing library operations, several constraints exist within the current system design. The platform supports three primary user roles—Administrator, Librarian, and Member—and does not currently support more complex organizational hierarchies or advanced permission customization beyond these predefined roles.

The system is designed as a desktop-based application and currently does not provide dedicated support for mobile devices, tablets, or web-based access. Although the modular architecture allows for future expansion, mobile compatibility and remote cloud access are outside the scope of the initial release.

Book reservations, borrowing, and fine calculations are managed within the platform; however, certain advanced automation features are limited. For example, notifications for overdue books or reservation availability may require manual communication unless future notification modules are implemented. Additionally, barcode scanning, RFID integration, and automated inventory tracking are not included in the current version of the system.

The platform also depends on accurate manual data entry by administrators and librarians. Incorrect or incomplete information entered into the system may affect reporting accuracy, inventory tracking, and transaction management. Furthermore, database maintenance, backup procedures, and system security rely on institutional IT support, meaning that inadequate technical maintenance could impact overall system performance and reliability.

These limitations are acknowledged as part of the current development scope, with future enhancements planned to expand functionality, automation capabilities, and platform accessibility.

3.5 Dependencies

From a technical perspective, the Library Management System (LMS) is built using a technology stack that includes Java and JavaFX for the graphical user interface, JDBC for database connectivity, and a relational database management system such as MySQL or PostgreSQL for persistent data storage. These technologies provide the foundation required for managing book records, member information, borrowing transactions, reservations, and reporting functionalities within the platform.

The system depends on a properly configured database environment to ensure reliable data storage, retrieval, and transaction processing. Stable hardware resources and institutional computer systems are required to support smooth application performance. In addition, the platform relies on the Java Runtime Environment (JRE) and compatible operating systems capable of running JavaFX applications effectively.

For secure and reliable operation, the deployment environment must support appropriate security configurations, including user authentication mechanisms, encrypted database connections where applicable, and routine backup procedures. Proper access control measures are required to ensure that administrators, librarians, and members can only access features and data relevant to their assigned roles.

The system also depends on institutional IT support for software installation, maintenance, troubleshooting, database administration, and security monitoring. Regular updates, system maintenance, and backup management are essential to preserving system stability, protecting against data loss, and ensuring the integrity and confidentiality of library records.

Where notification or communication features are implemented in future versions, the system may also depend on external email or messaging services for sending alerts related to overdue books, reservations, or account updates. Overall, reliable IT infrastructure and maintenance policies are critical for sustaining the long-term performance, availability, and security of the Library Management System.

4. Requirements

The success and usability of the Law Firm Management System (LFMS) depend heavily on a comprehensive understanding and fulfillment of both functional and non-functional requirements. These requirements have been meticulously defined to ensure the system not only meets the operational demands of legal professionals but also adheres to high standards of security, performance, and user experience. This chapter elaborates on these critical requirements, presenting them in an integrated and descriptive manner to convey their significance within the system's architecture.

4.1 Functional Requirements

Functional requirements define the core behaviors and operations that the Library Management System (LMS) must perform to effectively support the daily activities and management processes of a library environment. They encompass all user interactions, book and member management operations, borrowing and returning procedures, reservation handling, fine calculations, reporting functions, and system outputs necessary to achieve the intended operational and educational objectives of the institution.

4.1.1 Requirement Table

ID	Requirement Description	Priority	Category
FR-01	Admin shall be able to add, edit, and remove book records	High	Book Mgmt
FR-02	Admin shall be able to manage user accounts (Librarians, Members)	High	User Mgmt
FR-03	Admin shall generate reports (books, members, transactions, fines)	High	Reporting
FR-04	Admin shall configure system settings (fine rates, borrowing limits)	Medium	Configuration
FR-05	Librarian shall issue books to members and record the transaction	High	Borrowing
FR-06	Librarian shall record book returns and trigger fine calculation	High	Returning
FR-07	Librarian shall manage and confirm reservation requests	Medium	Reservations
FR-08	Librarian shall search books by title, author, category, or ISBN	High	Search

FR-09	Member shall be able to register a new account	High	Authentication
FR-10	Member shall log in and view their borrowing history	High	Authentication
FR-11	Member shall search the book catalog and check availability	High	Search
FR-12	Member shall reserve a book that is currently unavailable	Medium	Reservations
FR-13	Member shall view their active fines and fine history	Medium	Fines
FR-14	System shall automatically calculate fines based on overdue days	High	Fines
FR-15	System shall notify members when a reserved book becomes available	Low	Notifications
FR-16	System shall enforce borrowing limits per member	Medium	Borrowing
FR-17	System shall track multiple physical copies of the same book title	High	Book Mgmt
FR-18	System shall maintain an audit log of all transactions	Medium	Audit

5.1 Non-Functional Requirements

Beyond delivering the essential functionalities, LMS must satisfy several non-functional requirements that ensure the system's overall quality, security, and sustainability.

ID	Requirement Description	Priority	Category
NFR-01	The system shall respond to any user action within 2 seconds	High	Performance
NFR-02	All passwords shall be stored using bcrypt hashing (encrypted at rest)	High	Security
NFR-03	The system shall be available during library operating hours with 99.5% uptime	High	Availability
NFR-04	The interface shall be intuitive and consistent for non-technical users	Medium	Usability
NFR-05	The system shall support at least 1,000 concurrent book records without degradation	Medium	Scalability
NFR-06	Database backups shall be performed automatically on a daily schedule	High	Reliability
NFR-07	The codebase shall be fully documented (Javadoc) for future maintenance	Medium	Maintainability
NFR-08	The system shall run on Windows 10+ with Java 17+ and JavaFX 21+	High	Compatibility

5.1.1 Security

Security is a paramount concern due to the sensitive nature of data handled by the Library Management System (LMS), including member information, borrowing records, reservations, and administrative data. The system implements strict access control mechanisms to ensure that only authorized users can access specific functionalities based on their assigned roles (Administrator, Librarian, and Member). Data security is further reinforced through encrypted password storage and secure authentication processes that prevent unauthorized access to the system.

In addition, the system maintains audit logging capabilities that record all critical user actions such as book issuance, returns, reservations, and account modifications. These logs ensure accountability, traceability, and support system monitoring in case of suspicious or unauthorized activities. By enforcing these security measures, the LMS ensures the confidentiality, integrity, and protection of institutional library data.

5.1.2 Performance

Performance requirements ensure that the Library Management System operates efficiently and provides a responsive user experience under normal and increased workloads. The system is designed to handle a growing number of users, books, and transactions without significant delays or performance degradation. Key operations such as searching the catalog, issuing books, returning items, and generating reports are optimized to ensure fast processing times.

Database queries are structured for efficiency, and the system architecture is designed to minimize latency during user interactions. As a result, users experience smooth navigation across dashboards and modules, even in environments with standard desktop hardware. This ensures that library operations can continue without disruption or delays in service delivery.

5.1.3 Usability

Usability is a critical requirement for ensuring user adoption and satisfaction across all stakeholder groups. The Library Management System is designed with a clean, intuitive, and user-friendly interface that simplifies interaction for administrators, librarians, and members. Each user role is provided with a tailored dashboard that displays only the relevant features and information needed for their tasks.

The system minimizes the learning curve by using clear navigation structures, consistent layouts, and straightforward workflows. Additionally, feedback messages and contextual prompts guide users during operations such as borrowing books, managing reservations, or updating records. This reduces the likelihood of user errors and improves overall efficiency while interacting with the system.

5.1.4 Maintainability

Maintainability and scalability are key design considerations in the Library Management System to ensure long-term usability and adaptability. The system is built using a modular architecture that separates core functionalities such as user management, book management, transactions, and reporting. This structure allows developers to easily update or extend specific components without affecting the entire system.

The application follows best coding practices and includes proper documentation to support future maintenance and development efforts. This ensures that IT staff can efficiently debug, upgrade, and enhance the system when needed. The design also allows for future expansion, such as mobile application support, cloud integration, or additional library services, without requiring major system redesigns.

5.1.5 Reliability

Reliability is a key requirement of the Library Management System (LMS) to ensure continuous and dependable access to library services and data. The system is designed to operate consistently without frequent failures, providing stable performance for administrators, librarians, and members during daily operations such as book management, borrowing, returning, and reservations.

To ensure data integrity and prevent loss of information, the system implements fault tolerance strategies including regular database backups, structured error handling mechanisms, and recovery procedures in case of system failures. These measures help restore system functionality quickly and minimize downtime in the event of unexpected issues such as software errors, hardware failures, or data corruption.

In addition, the system is designed to handle exceptions gracefully, ensuring that errors do not disrupt the entire application workflow. By incorporating these reliability mechanisms, the LMS maintains user trust, ensures uninterrupted access to library resources, and supports the smooth operation of institutional library services.

6. Software Design

6.1 BPMN

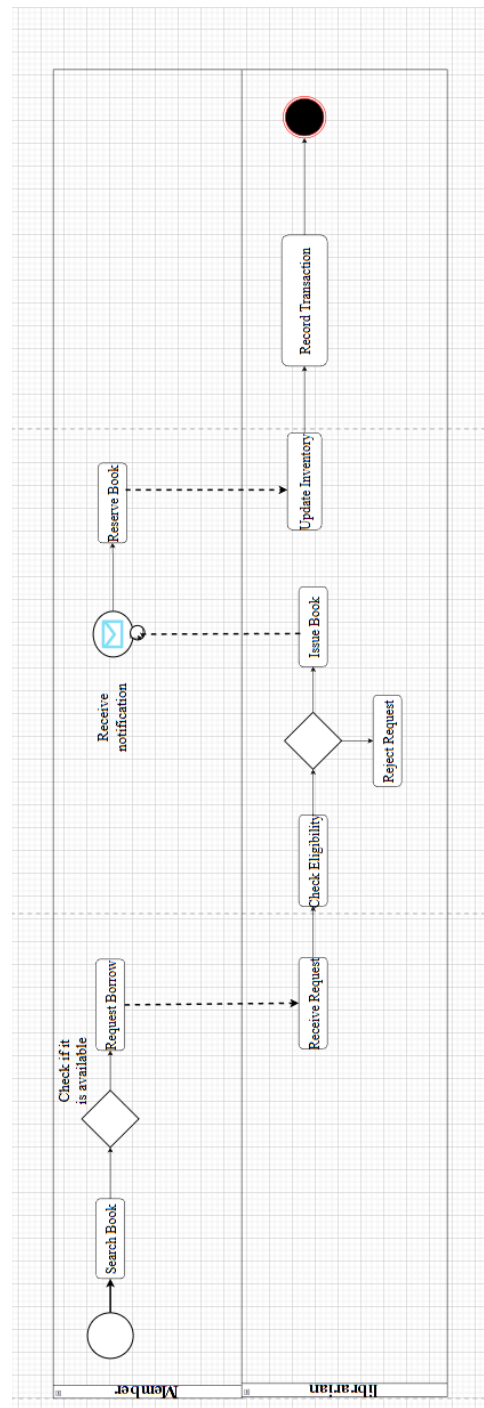


Fig 1: Borrow

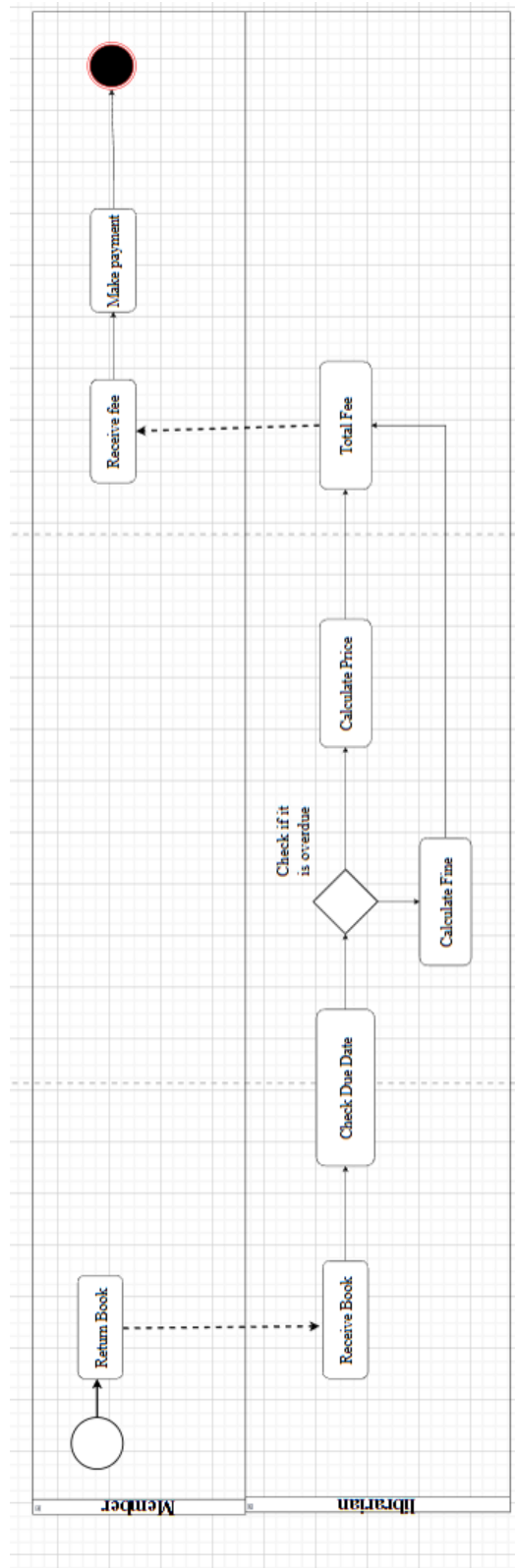


Fig 2: Return Process

6.2 User Scenarios

Scenario Title: Register New

- A librarian receives a request to register a new member.
- The librarian enters personal details into the system.
- The system creates a member account with a unique ID.
- Membership details and borrowing limits are assigned.

Scenario Title: Update Member Account

- A librarian searches and opens a member profile.
- The librarian updates required information.
- The system saves and validates the changes.
- The system updates the member record.

Scenario Title: Add New Book

- A librarian enters new book details into the system.
- The system validates and checks for duplicates.
- The system creates a book record and updates inventory.

Scenario Title: Remove or Decommission Book

- A librarian selects a book to remove.
- The librarian chooses a removal reason.
- The system updates the book status and inventory.

Scenario Title: Search / Browse Catalog

- A user searches for a book using keywords or filters.
- The system displays matching results.

- The user views book details or selects an action.

Scenario Title: Place Hold / Reserve Book

- A member selects a book to reserve.
- The system checks eligibility and places the hold.
- The system notifies the member when available.

Scenario Title: Check Out Book (Borrow)

- A member presents books for borrowing.
- The system verifies and processes the checkout.
- The system assigns due dates and updates records.

Scenario Title: Return Item and Process Fine

- A member returns a borrowed book.
- The system checks for overdue or damage.
- The system updates records and applies fines if needed.

Scenario Title: Pay Fine or Membership Fee

- A member requests to pay fines or fees.
- The system processes the payment.
- The system updates the account and issues a receipt.

Scenario Title: Generate Reports & Financial Oversight

- A user selects a report type.
- The system generates and displays the report.
- The report can be exported or saved.

6.3 List of Use Cases

UC ID	Use Case Name	Primary Actor	Priority
UC-01	Register Member Account	Member	High
UC-02	Update Member Account	Librarian	Medium
UC-03	Add New Book	Librarian	High
UC-04	Remove or Decommission Book	Librarian, System Administrator	High
UC-05	Search / Browse Catalog	Members, Librarian	High
UC-06	Place Hold / Reserve Book	Member, Librarian	High
UC-07	Check Out Book (Borrow)	Librarian, Member (self-checkout)	Medium
UC-08	Return Item and Process Fine	Librarian, Member	High
UC-09	Pay Fine or Membership Fee	Librarian, Member	High
UC-10	Generate Reports & Financial Oversight	Admin	Medium
UC-11	View Borrowing History	Member, Librarian	Medium
UC-12	Send Availability Notification	System	Low
UC-13	Update Profile	Member, Librarian	Low
UC-14	Configure System Settings	Admin	Medium

6.4 Extended Use Cases

UC1: *Register New Member*

Use case name:	Register New Member	
Scenario:	New patron registers for library membership	
Triggering event:	A person requests to join the library.	
Brief description:	A librarian collects required member details, verifies identity, creates a member account in the system, assigns membership type and initial borrowing limits.	
Actors:	Librarian	
Stakeholders:	Librarian, System Administrator, Member, Accounting (if membership fee)	
Preconditions:	Person provides required ID/details	
Postconditions:	Member account created with unique ID, assigned membership type, borrowing limits, and (if applicable) membership fee recorded.	
Flow of activities:	Actor	System
	1. Member requests registration. 2. Librarian collects name, contact, ID, address, age, affiliation (student/staff/public), and any required documentation. 3. Librarian selects membership type and any paid options. 4. Librarian submits registration.	4.1. System validates input, checks for duplicates, assigns member ID, sets initial borrowing/quota and due rules, and records membership start date. 4.2. If membership fee applies, system creates a payment transaction or invoice. 4.3. System issues membership card/credentials and prints/stores welcome information.
Exception conditions:	If required information missing or invalid, system prompts for correction. Registration is paused until resolved. If duplicate member found, system alerts librarian. Librarian may merge or cancel. If payment fails, membership may be provisional or blocked until payment resolved.	

UC2: Update Member Account

Use case name:	Update Member Account	
Scenario:	Modify member details or status	
Triggering event:	Member requests an update or staff needs to change account (address change, lost card, suspensions).	
Brief description:	Librarian updates member personal details, contact, membership type, or status (suspend/reactivate/terminate).	
Actors:	Librarian	
Stakeholders:	Librarian, Member, System Administrator	
Preconditions:	Member exists in system. Staff authenticated with permission.	
Postconditions:	Member record updated. Any status changes enforced.	
Flow of activities:	Actor	System
	1. Librarian searches and retrieves member record. 2. Librarian edits fields (address, phone, email, membership type, expiry, notes) or changes status. 3. Librarian saves changes.	3.1. System validates edits, records audit trail, updates borrowing limits and notifications as needed. 3.2. If status changed to suspended/blocked, system prevents new checkouts and flags outstanding obligations.
Exception conditions:	If member not found, librarian may re-search or escalate. If change violates system rules (ex: downgrade below outstanding loans), system warns and prevents change until resolved.	

UC3: Add New Book

Use case name:	Add New Book	
Scenario:	Adding a new book to library collection	
Triggering event:	Library acquires a new book (purchase, donation).	
Brief description:	Librarian records bibliographic metadata, assigns call number/ISBN, sets copy count, location, and item status.	
Actors:	Librarian	
Stakeholders:	Librarian, System Administrator, Members	
Preconditions:	Request confirmed. Access to cataloging tools and classification rules present.	
Postconditions:	Bibliographic record created. One or more inventory items (copies) created with barcodes, location, status = available.	
Flow of activities:	Actor 1. Librarian starts new catalog entry and enters metadata (title, author, publisher, edition, ISBN, genre, language, subjects). 2. Librarian assigns classification/call number, shelf location, and number of copies. 3. Librarian attaches barcodes/labels to physical copies and enters barcode numbers. 4. Librarian sets initial status (available, on-hold, in-processing).	System 4.1. System validates ISBN/metadata (optional external lookup). 4.2. System creates bibliographic record, generates item records, and updates inventory counts.
Exception conditions:	If ISBN or metadata conflicts (duplicate), system prompts to merge or link to existing record. If barcode conflicts, staff must correct barcode before saving. If external lookup fails, librarian enters metadata manually.	

UC4: Remove or Decommission Book

Use case name:	Remove or Decommission Book	
Scenario:	Remove damaged copy from collection	
Triggering event:	Librarian identifies book copy to withdraw (lost, damaged, outdated).	
Brief description:	Librarian marks item as withdrawn/lost/damaged and optionally records reason, updates inventory and accounting (write-off).	
Actors:	Librarian, System Administrator	
Stakeholders:	Librarian, System Administrator, Accounting	
Preconditions:	Item exists in system. Librarian authorized.	
Postconditions:	Item status changed to withdrawn/damaged/lost. Inventory counts updated, if applicable, disposal or transfer recorded.	
Flow of activities:	Actor	System
	1. Librarian searches and selects item record. 2. Librarian selects removal reason and action (discard, sell, transfer, repair). 3. Librarian confirms action.	3.1. System updates item status, adjusts inventory counts, logs audit trail. 3.2. If required creates accounting entry or disposal record.
Exception conditions:	If item currently on loan or reserved, system warns and prevents removal until resolved. If removal requires admin approval per policy, system routes request to System Administrator.	

UC5: Search / Browse Catalog

Use case name:	Search / Browse Catalog	
Scenario:	Member or staff searches for books	
Triggering event:	A search query by title, author, subject, or ISBN.	
Brief description:	User searches catalog. System returns matching bibliographic records with availability, location, and copy details.	
Actors:	Member, Librarian, System Administrator (as viewer)	
Stakeholders:	Members, Librarian	
Preconditions:	Catalog data exists. Search system operational.	
Postconditions:	Search results displayed. User may initiate actions (reserve, view details, request).	
Flow of activities:	Actor	System
	1. User enters search parameters (keyword, title, author, ISBN, genre, filter by availability). 2. User selects a record to view details (summary, copies, due dates, holds).	1.1 System performs search, ranks results, and displays list with availability, location, and call number 2.1 System displays detailed bibliographic and copy-level info and available actions (reserve, request, place hold).
Exception conditions:	If no results, system suggests alternative queries (fuzzy match, spelling corrections). If external catalog integration fails, system displays only local results and notifies user.	

UC6: Place Hold / Reserve Book

Use case name:	Place Hold / Reserve Book	
Scenario:	Member reserves a checked-out or available book	
Triggering event:	Member requests to hold or reserve a book copy.	
Brief description:	Member places a hold on a bibliographic record or specific copy; system queues hold, notifies when available, and enforces pickup window.	
Actors:	Member, Librarian	
Stakeholders:	Member, Librarian	
Preconditions:	Member account in good standing (no blocking fines or suspensions, unless policy allows). Item exists.	
Postconditions:	Hold record created with position in queue. Notifications scheduled, when available, item set aside with expiry.	
Flow of activities:	Actor	System
	1. Member (or librarian on behalf) initiates place-hold on title or specific copy. 2. Member picks up item within pickup window. Librarian checks out to member and removes hold.	1.1 System verifies member eligibility, determines hold queue position, estimates availability, and confirms hold placement. 1.2 System records pickup location and expected hold status. 1.3 When item is returned or becomes available, system assigns item to next hold, changes item status to On Hold, reserves item, and sends notification to member (email/SMS).
Exception conditions:	If member is blocked (excessive fines or suspensions), system denies hold and instructs to resolve. If item cannot be reserved (reference-only), system informs user. If member does not pick up within window, system cancels hold and notifies next in queue.	

UC7: Check Out Book (Borrow)

Use case name:	Check Out Book (Borrow)	
Scenario:	Member borrows one or more items	
Triggering event:	Member requests to borrow items at circulation desk or via self-checkout.	
Brief description:	Librarian or self-checkout scans member ID and item barcodes. System verifies eligibility, calculates due dates, updates item status, and prints/records loan receipt.	
Actors:	Librarian, Member (self-checkout)	
Stakeholders:	Members, Librarian, System Administrator	
Preconditions:	Member active and eligible (no blocks beyond allowed). Items available and not reserved for others (unless checkout by reservation).	
Postconditions:	Loan transactions created with due dates. Item status = Checked Out, member account updated.	
Flow of activities:	Actor	System
	1. Member presents items and membership ID (or uses self-checkout). 2. Librarian scans member ID and item barcodes.	2.1. System verifies member standing, item availability, and reservation status. 2.2. For each item, system calculates due date based on membership type and item rules, updates item status to Checked Out, and creates loan record. 2.3. System updates member's borrowed list and available quota and prints/emailed receipt with due dates and fine policy.
Exception conditions:	If member has reached borrowing limit, system denies further checkouts and suggests holds or extensions. If item reserved for another member, system prevents checkout unless policy allows. If item damaged or barcode unreadable, librarian intervenes to resolve.	

UC8: Return Item and Process Fine

Use case name:	Return Item and Process Fine	
Scenario:	Member returns borrowed item; system processes return and fines if overdue or damaged	
Triggering event:	Member returns item to desk or dropbox.	
Brief description:	Librarian or automated system records return, updates inventory, checks for damage, calculates fines for overdue/damage, and collects payment or adds fine to account.	
Actors:	Librarian, Member	
Stakeholders:	Member, Librarian, Accounting, System Administrator	
Preconditions:	Loan record exists for item, return method available.	
Postconditions:	Item status updated (Available, In-Process, Repair), loan closed, fines created or waived as appropriate, notifications/receipts issued.	
Flow of activities:	Actor	System
	1. Member returns item to desk.	1.1. System locates loan record, calculates overdue days and fines per policy, and updates item status. 1.2. If item damaged, librarian documents damage and marks item status (Repair/Damaged) and the system creates a damage charge. 1.3. System posts fines/charges to member account and, when payment occurs, records payment transaction. 1.4. System updates inventory and, if item becomes available, notifies waiting holds.
Exception conditions:	If item barcode unreadable, librarian searches by title/other ID to process return manually. If item reported lost earlier, system reconciles (remove lost charge or adjust). If member disputes fine, librarian can place charge on hold/pending and escalate per policy.	

UC9: Pay Fine or Membership Fee

Use case name:	Pay Fine or Membership Fee	
Scenario:	Member pays outstanding fines or membership fees	
Triggering event:	Member requests to pay or receives invoice/notification.	
Brief description:	Librarian processes payment (cash, card, online), system records payment, updates member balance, and clears blocks if resolved.	
Actors:	Librarian, Member	
Stakeholders:	Member, Librarian, Accounting, System Administrator	
Preconditions:	Fine or fee exists on member account, payment method available.	
Postconditions:	Payment transaction recorded, member balance updated, account blocks removed if cleared.	
Flow of activities:	Actor	System
	1. Member presents invoice or account, requests to pay. 2. Librarian selects outstanding items to pay or member chooses online payment. 3. Librarian/Member selects payment method and provides payment details.	3.1. System processes payment (authorize card or record cash), updates accounting, applies payment to charges, and issues receipt. 3.2. If balance cleared, system clears holds/borrowing blocks as per policy.
Exception conditions:	If card authorization fails, system prompts for alternate payment or voids transaction. If partial payment accepted, system allocates payment according to policy (oldest-first) and leaves balance. If payment needs manager approval for waiver/refund, hold and escalate.	

UC10: Generate Reports & Financial Oversight

Use case name:	Generate Reports & Financial Oversight	
Scenario:	Administrator or librarian requests operational or financial reports	
Triggering event:	Need for periodic reporting (daily circulation, overdue list, revenue, payroll).	
Brief description:	System Administrator or Librarian requests preconfigured or custom reports; system aggregates data and produces downloadable/printable reports.	
Actors:	System Administrator, Librarian	
Stakeholders:	System Administrator, Accounting department, Library management, Librarians	
Preconditions:	Data collected (transactions, loans, payments); user has permission to access reports.	
Postconditions:	Report generated in requested format; saved or exported if requested.	
Flow of activities:	Actor	System
	1. User selects report type (circulation summary, overdue items, revenue by period, staff payroll, inventory status) and filter parameters (date range, branch). 2. User reviews and may schedule recurring reports or set notifications.	1.1 System validates permissions, queries database, aggregates metrics, formats results, and displays or exports report (PDF, CSV).
Exception conditions:	If user lacks permission, system denies access. If data source unavailable, system returns partial report and logs error. If report generation times out for large queries, system suggests narrower filters or background scheduling.	

7. Behavioral Diagrams

7.1 Use Case Diagram

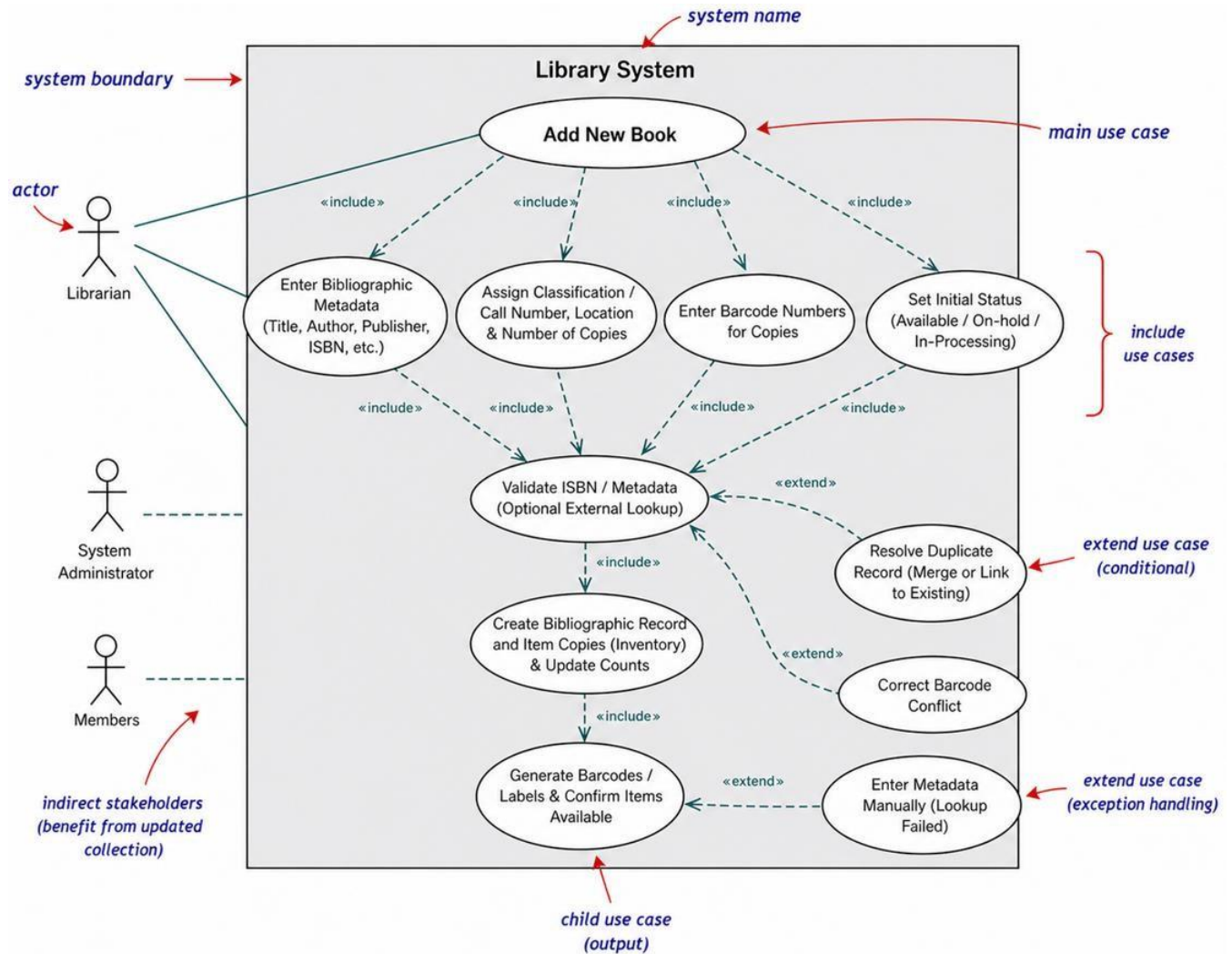


Figure 1: Add New Book

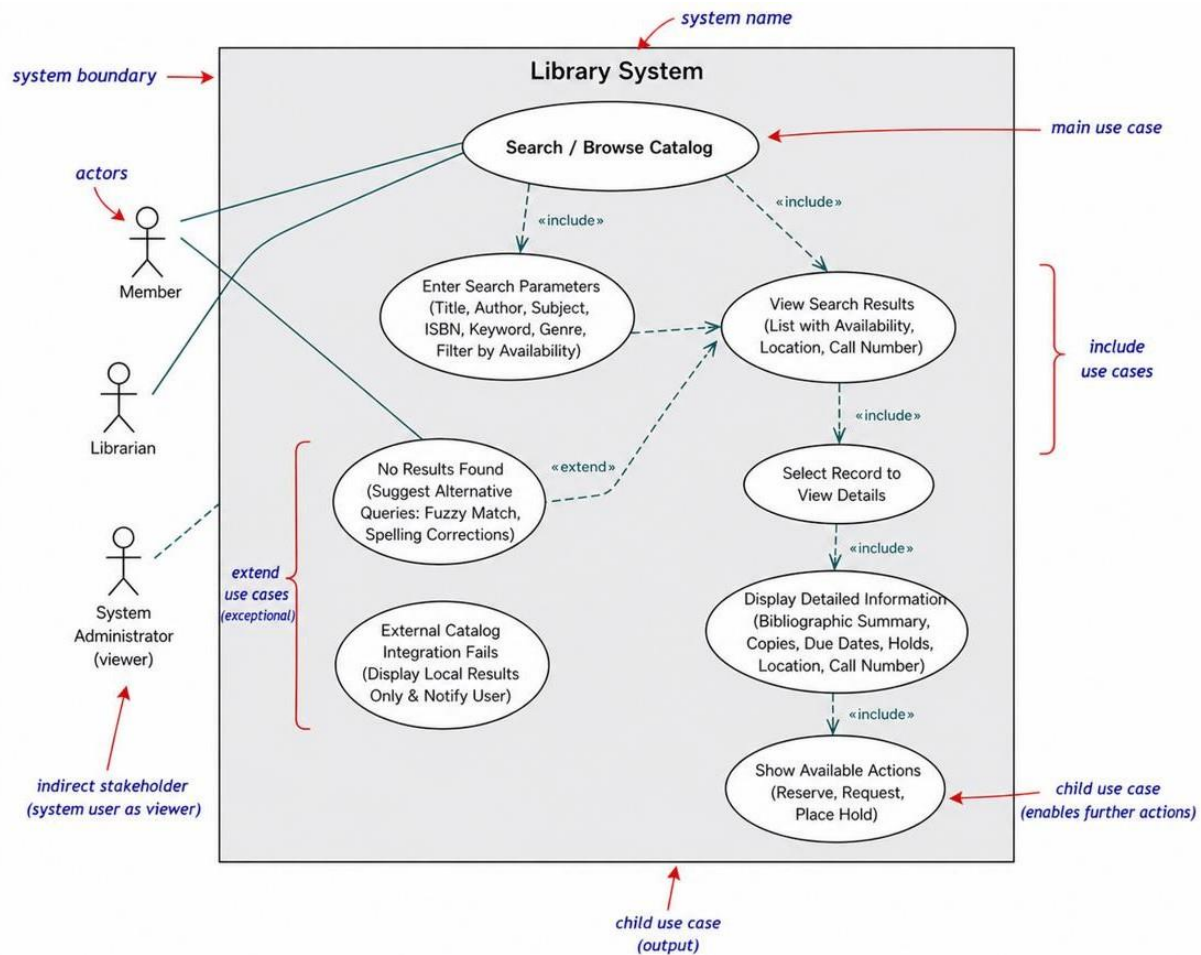


Figure 2: Browse Catalog

Figure 3: Check out Book

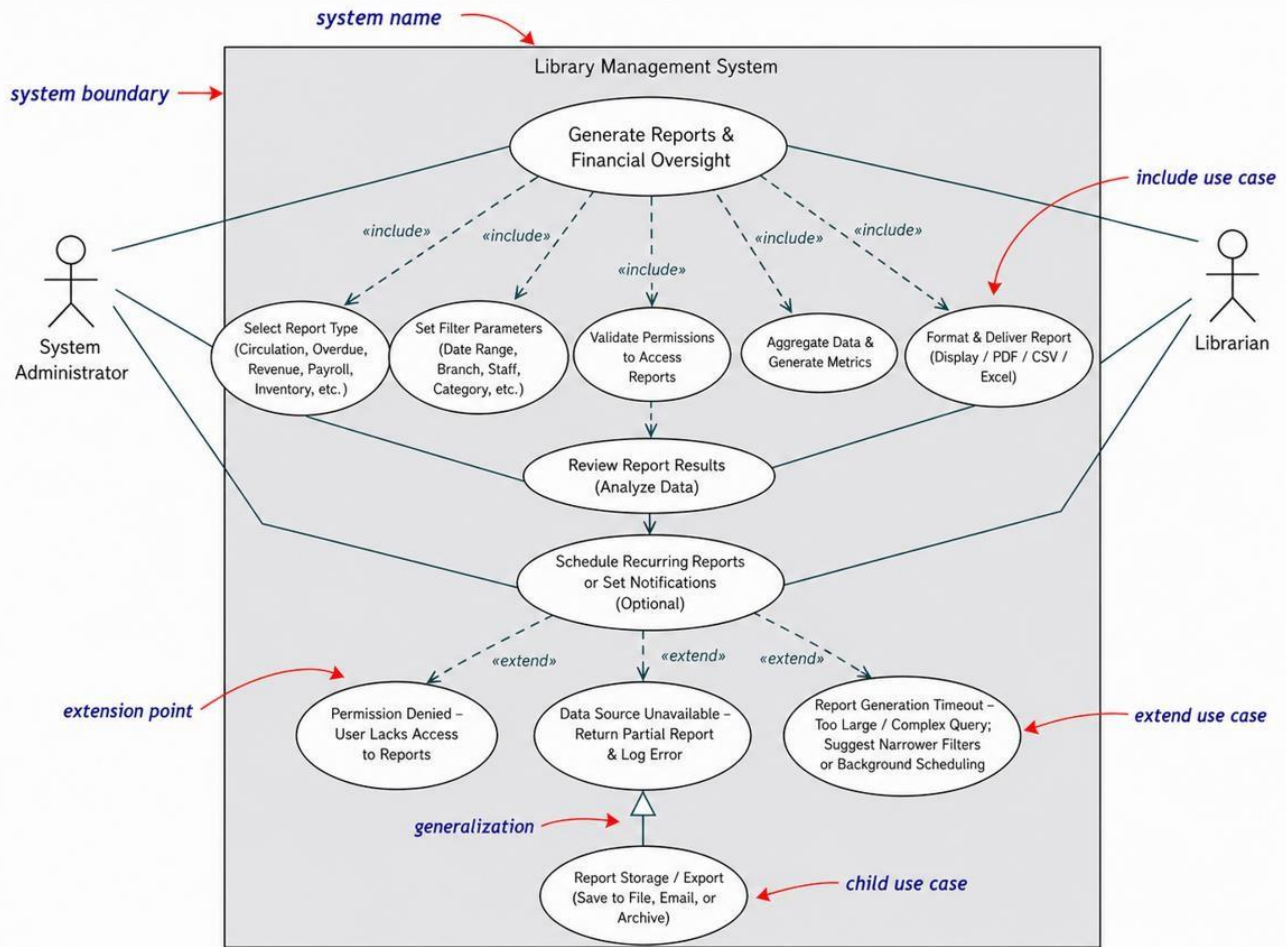


Figure 4: Generate Report

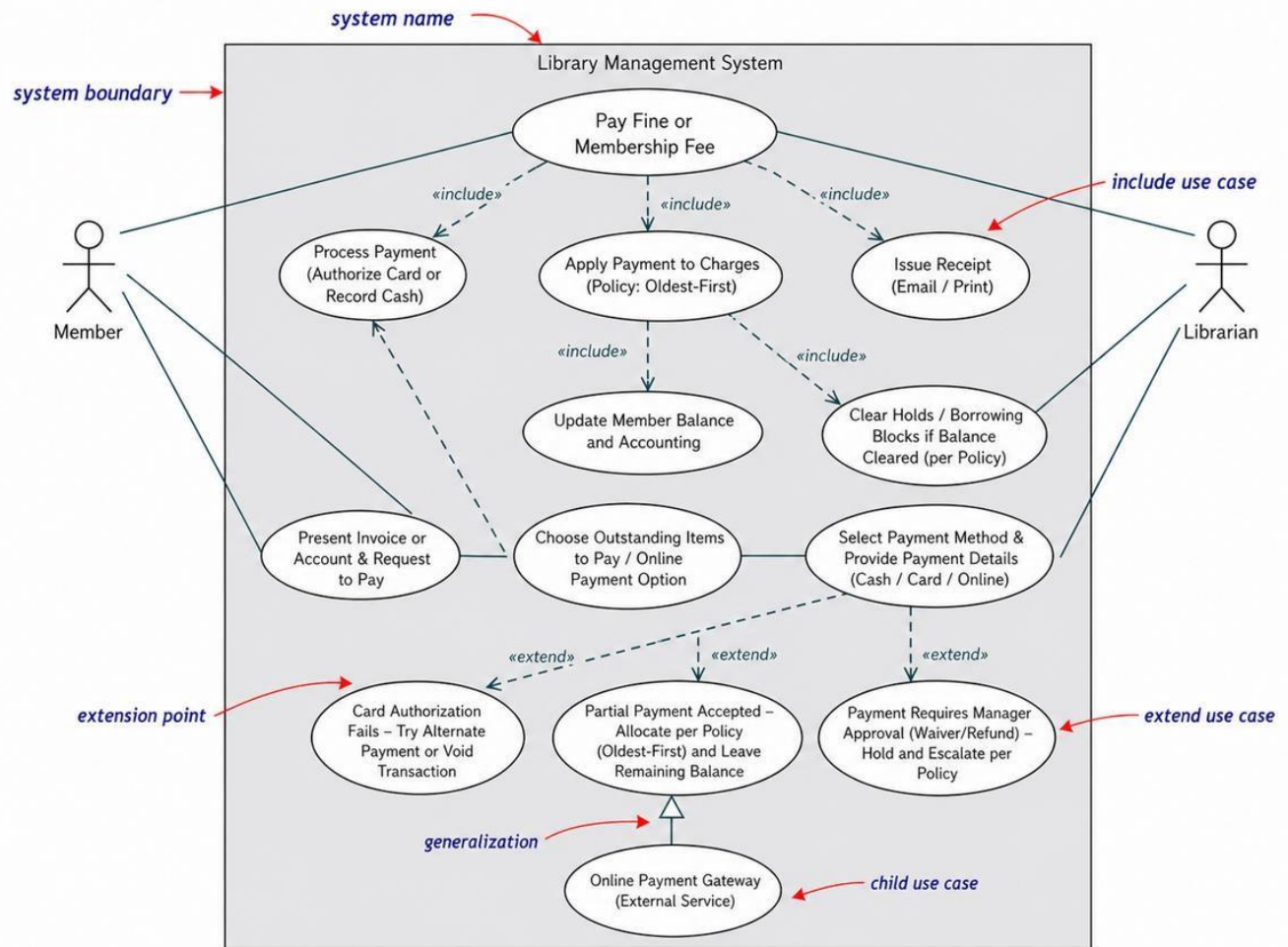


Figure 5: Pay Fine

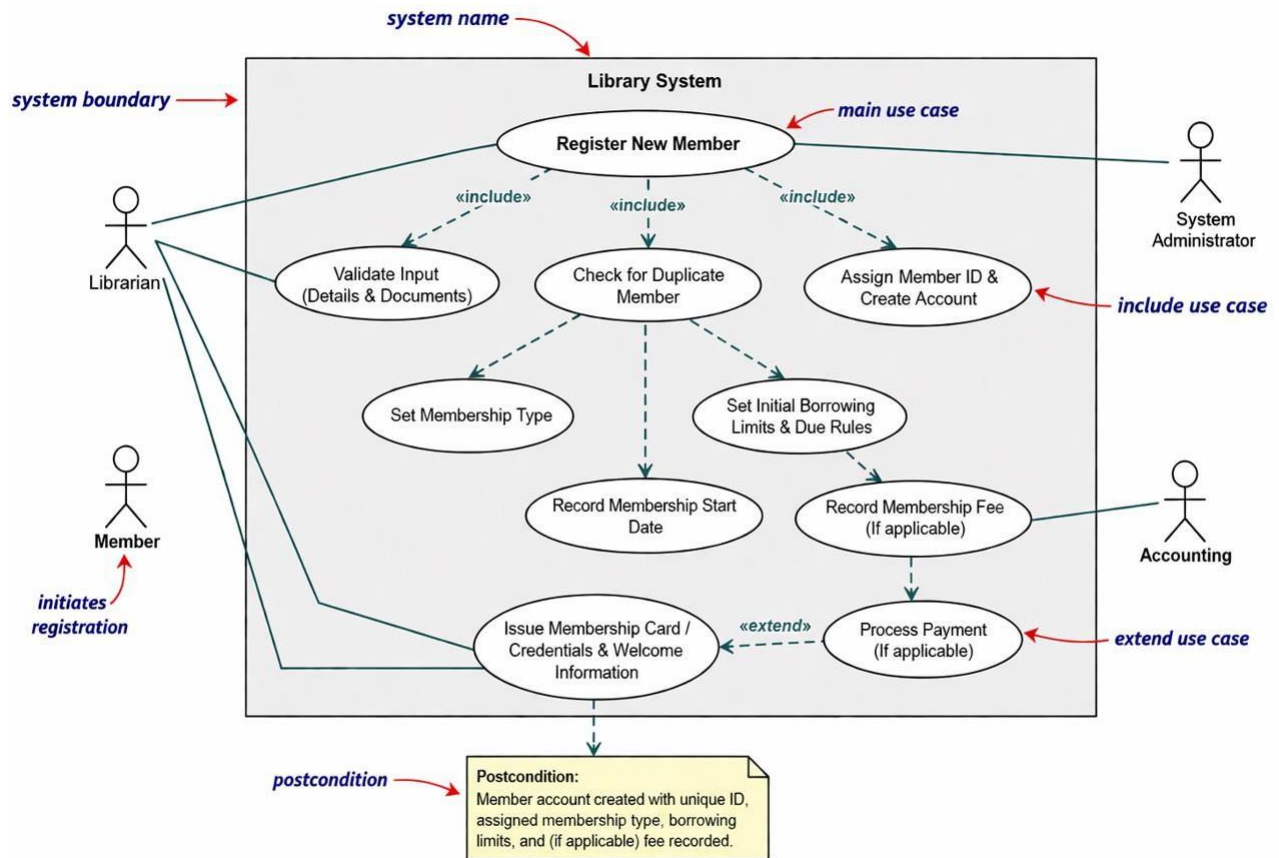


Figure 6: Register New Member

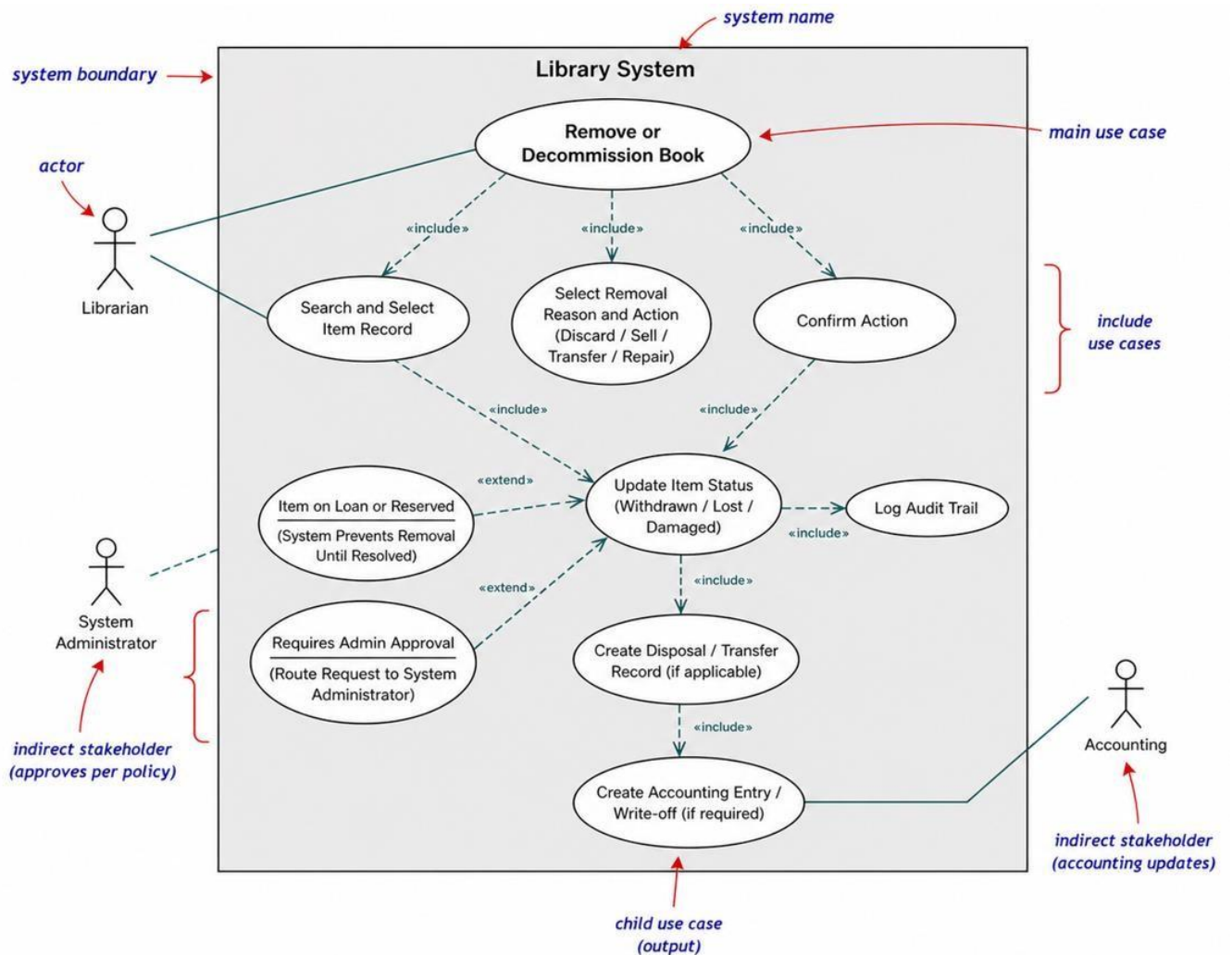


Figure 7: Remove Book

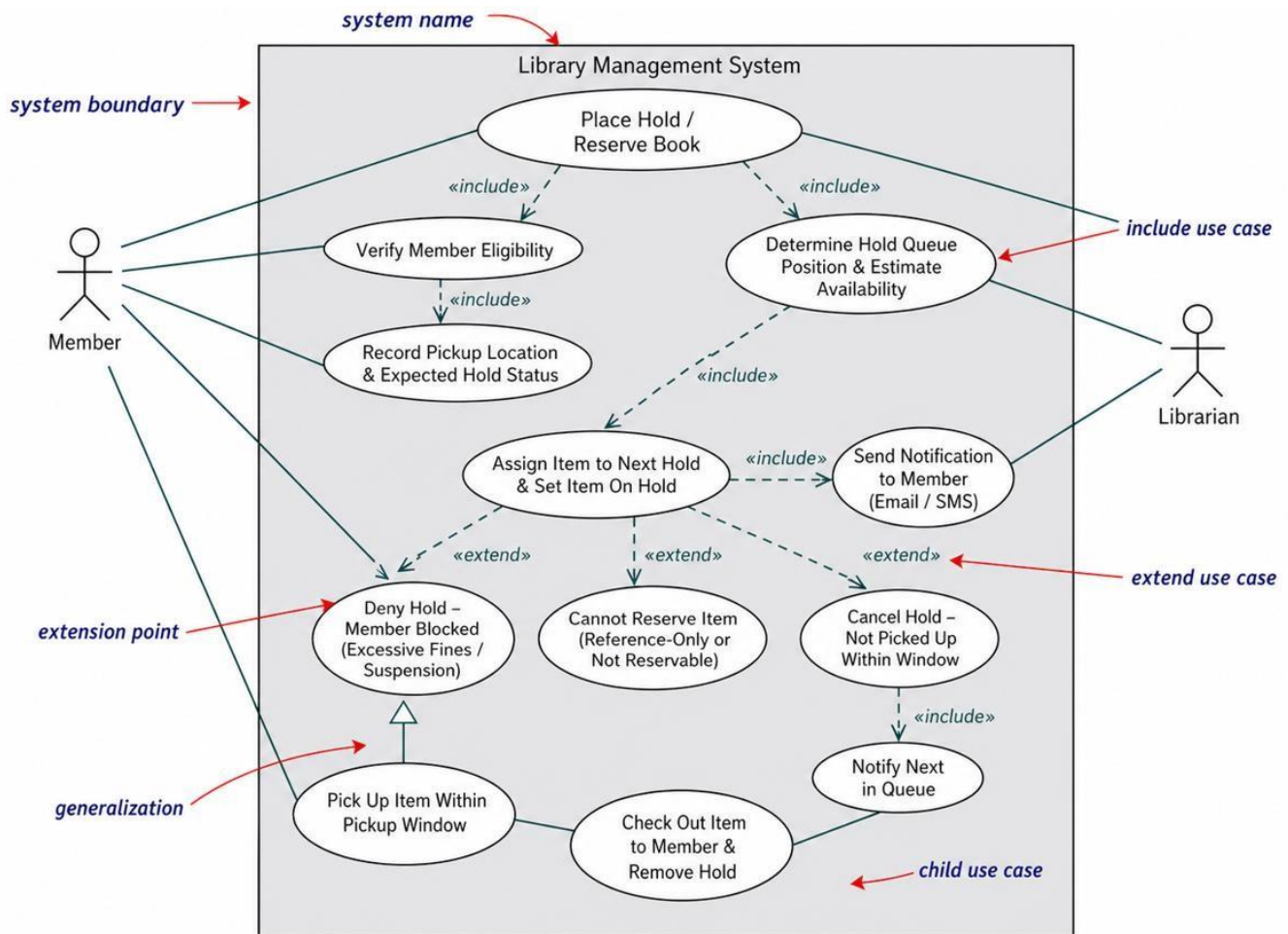


Figure 8: Reserve Book

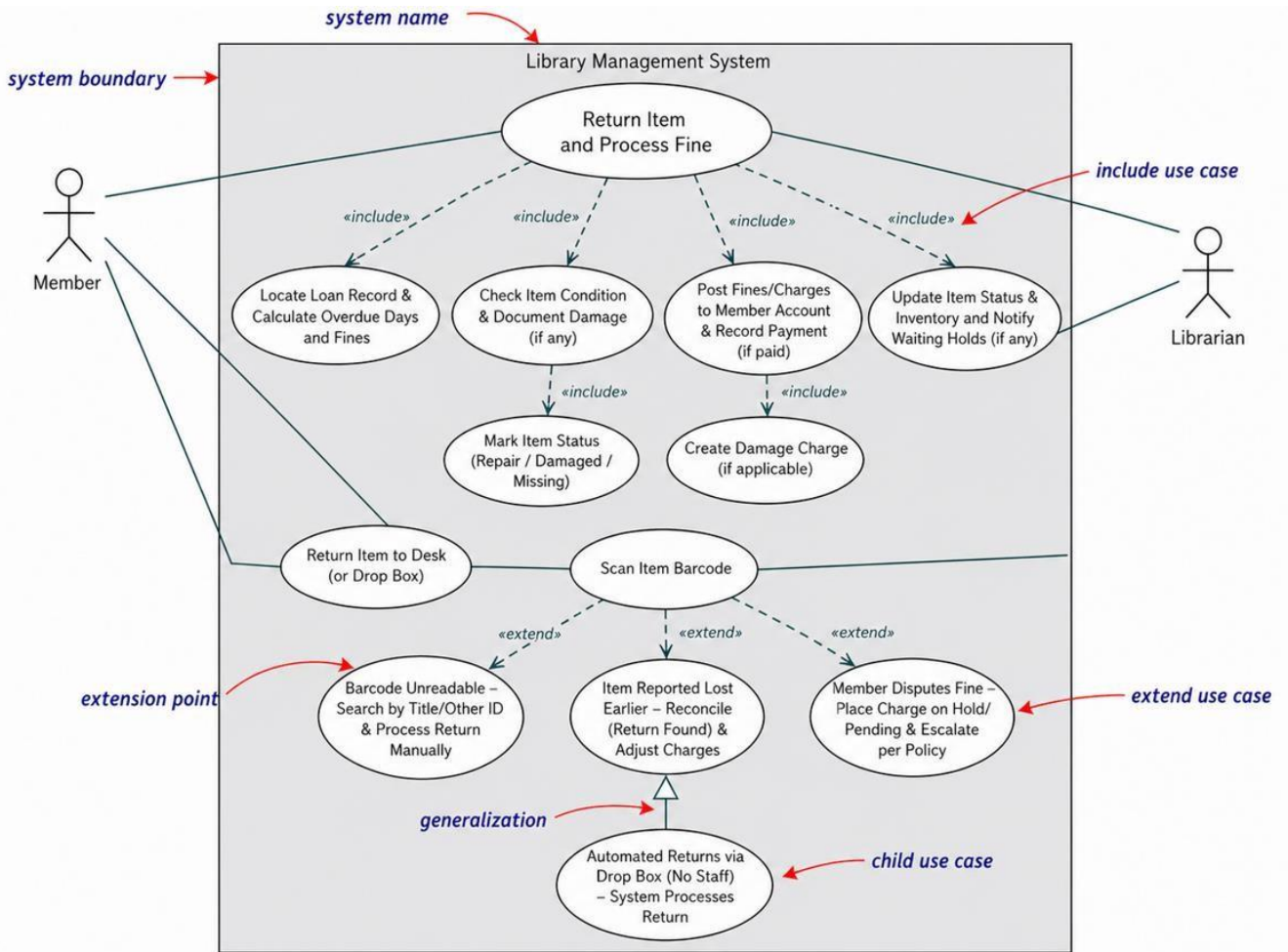


Figure 9: Return Item

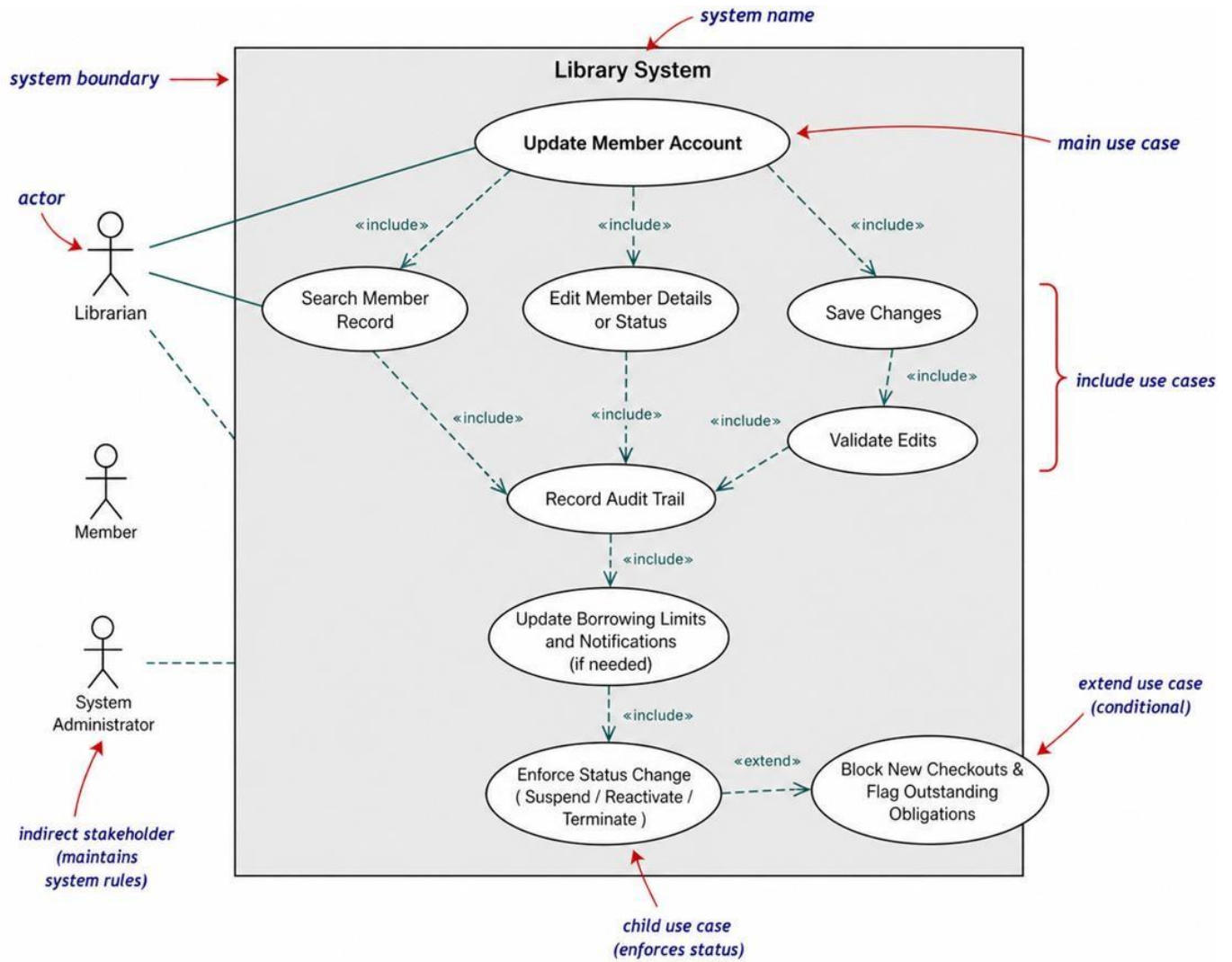
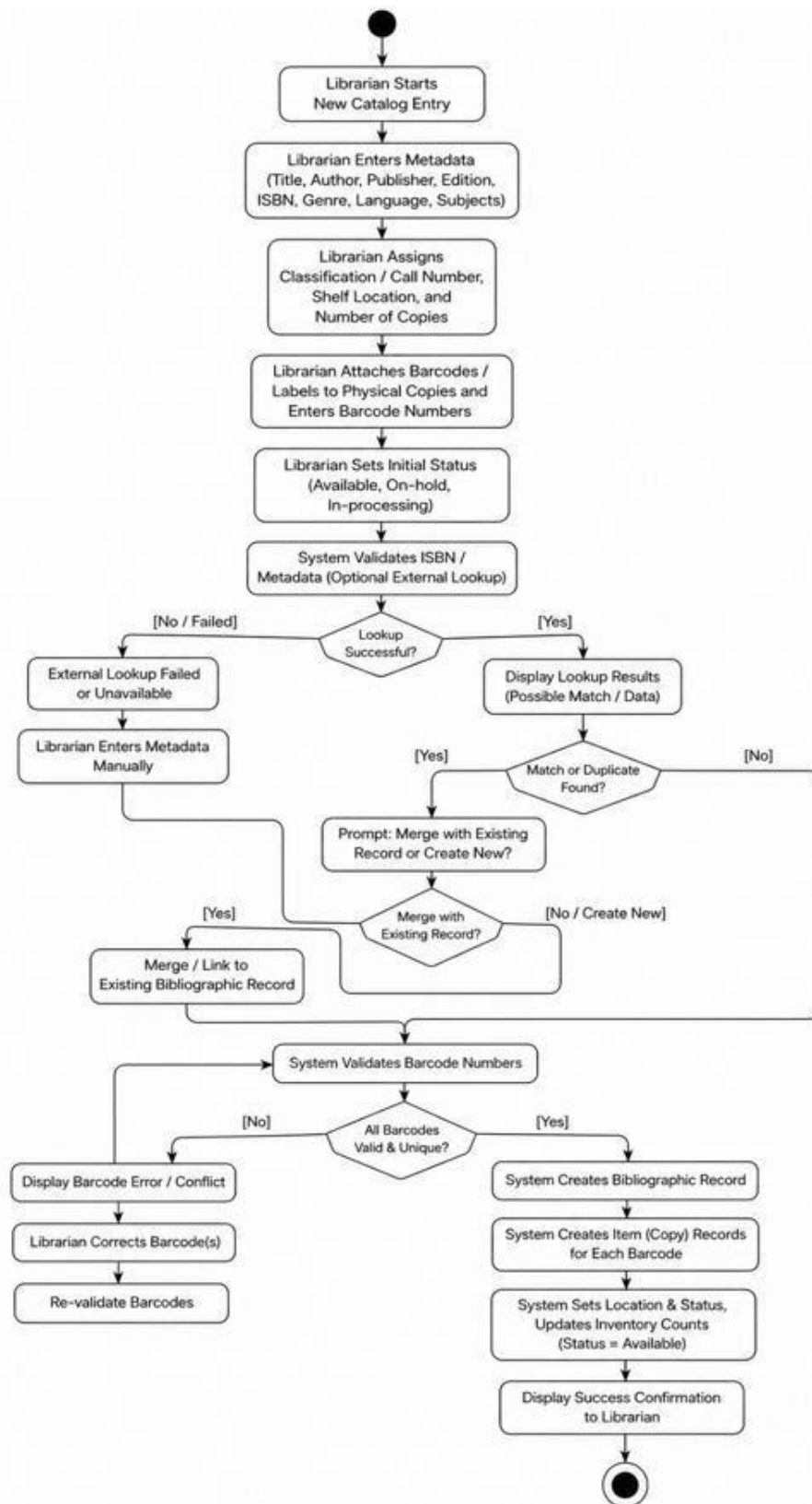
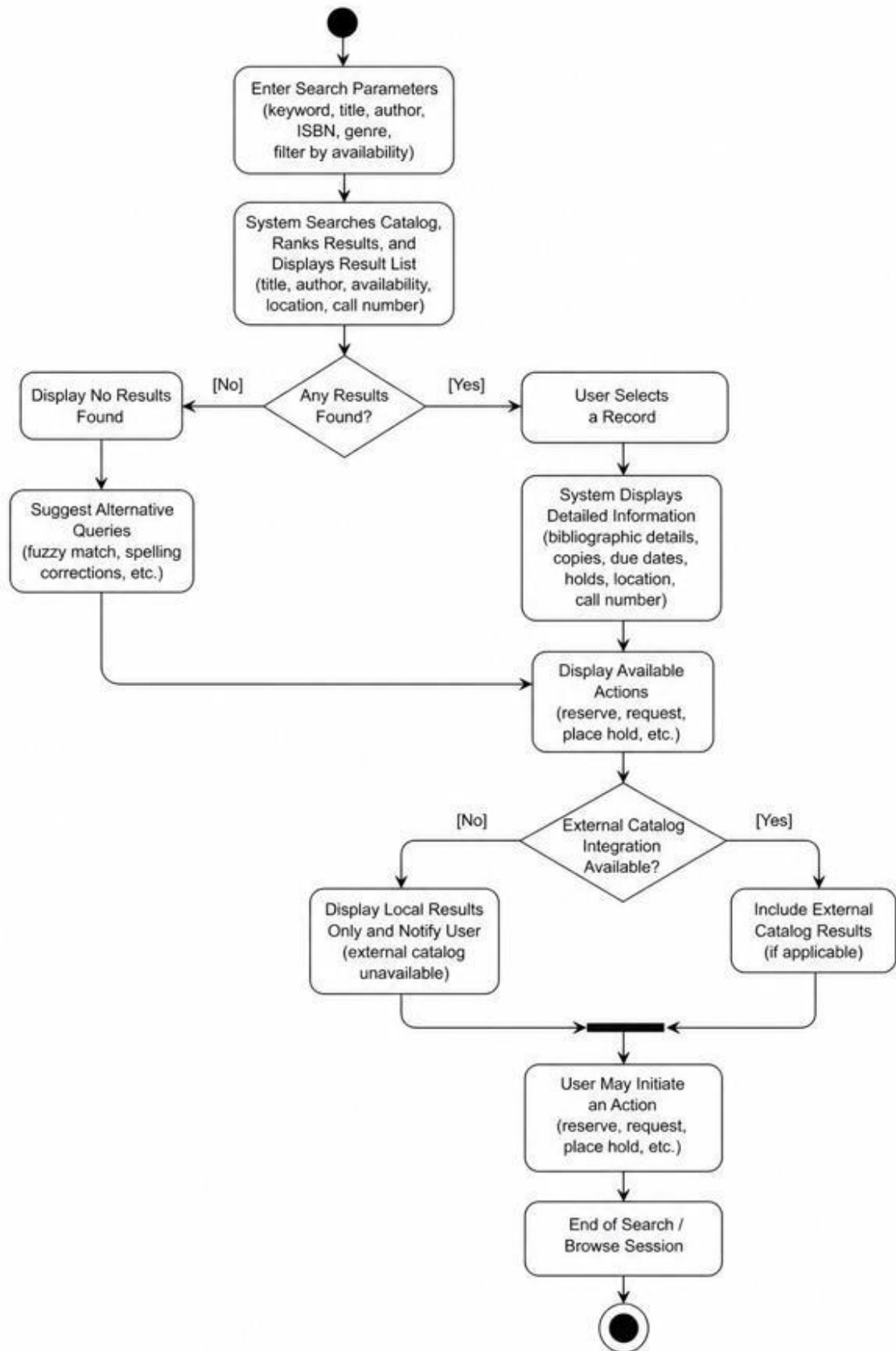
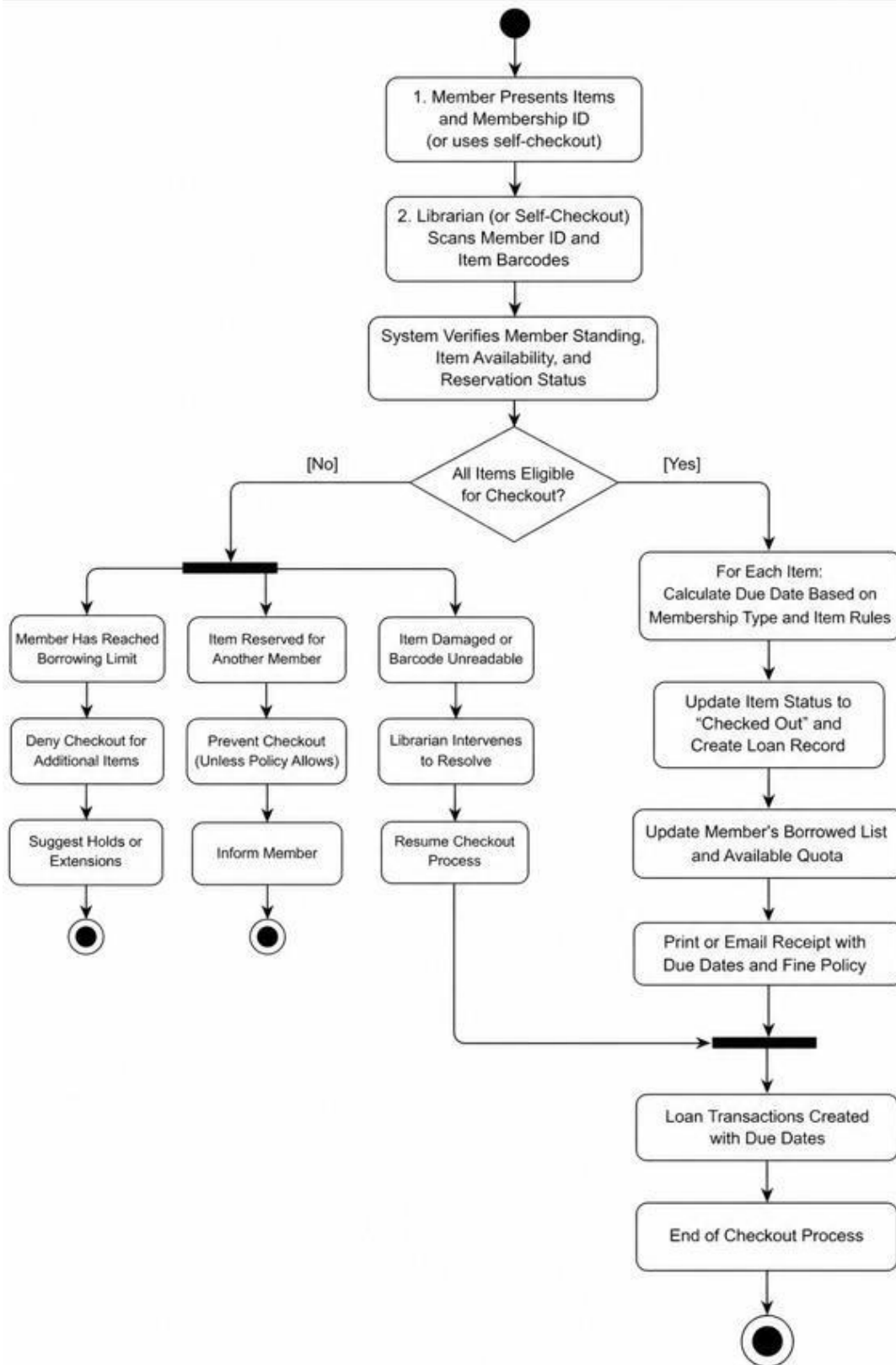


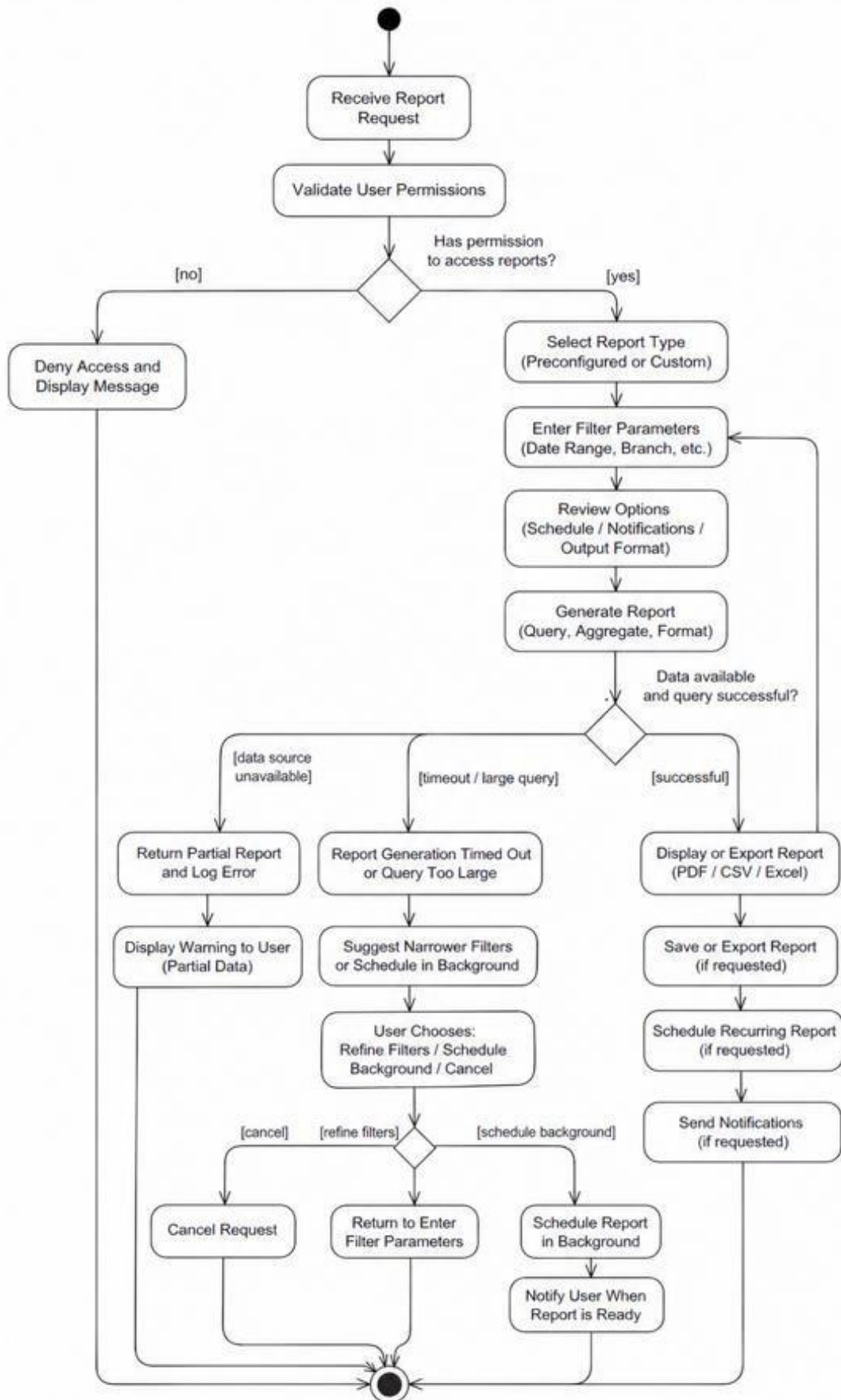
Figure 10: Update Member Account

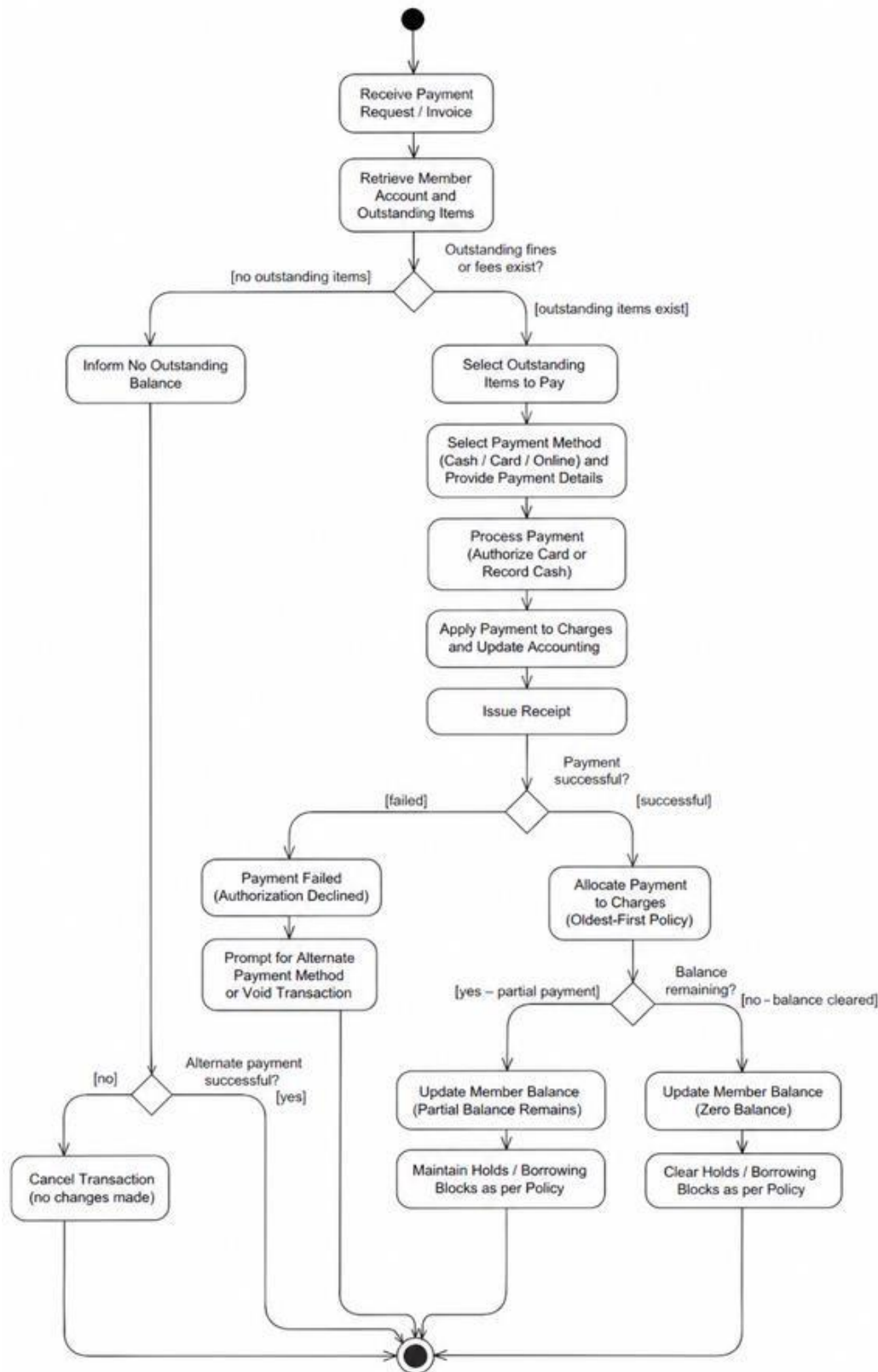
7.2 Activity Diagrams

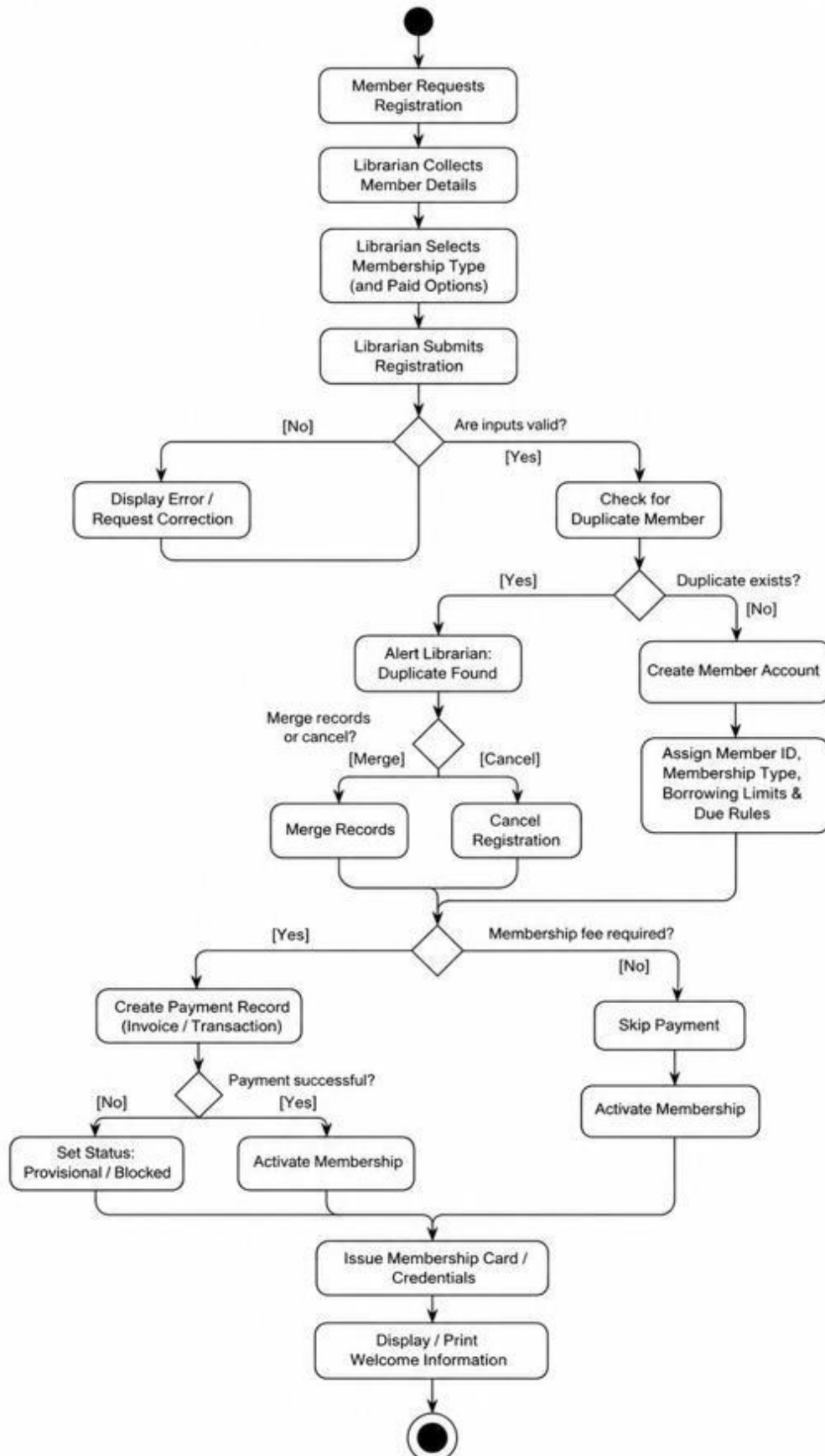


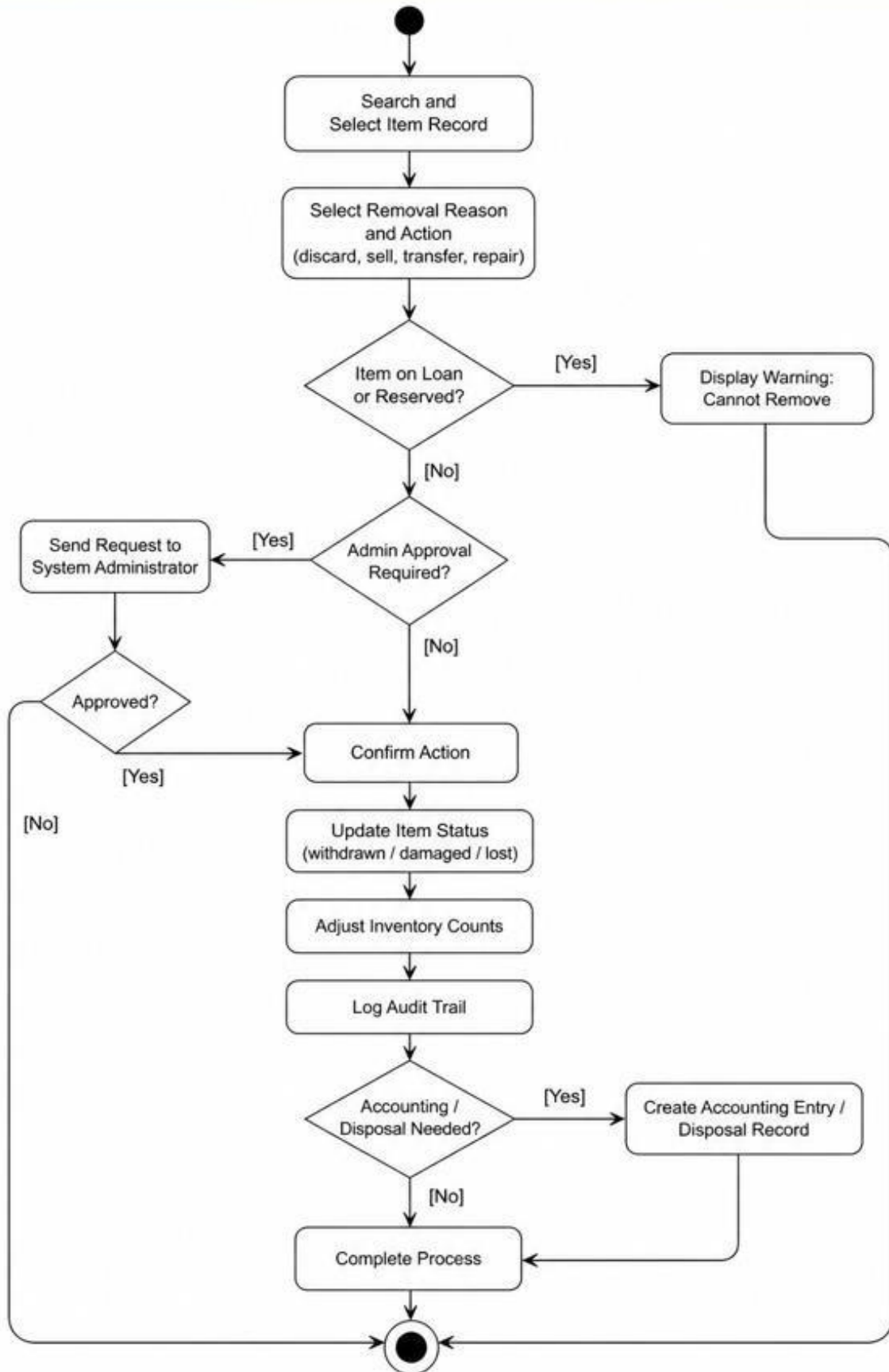


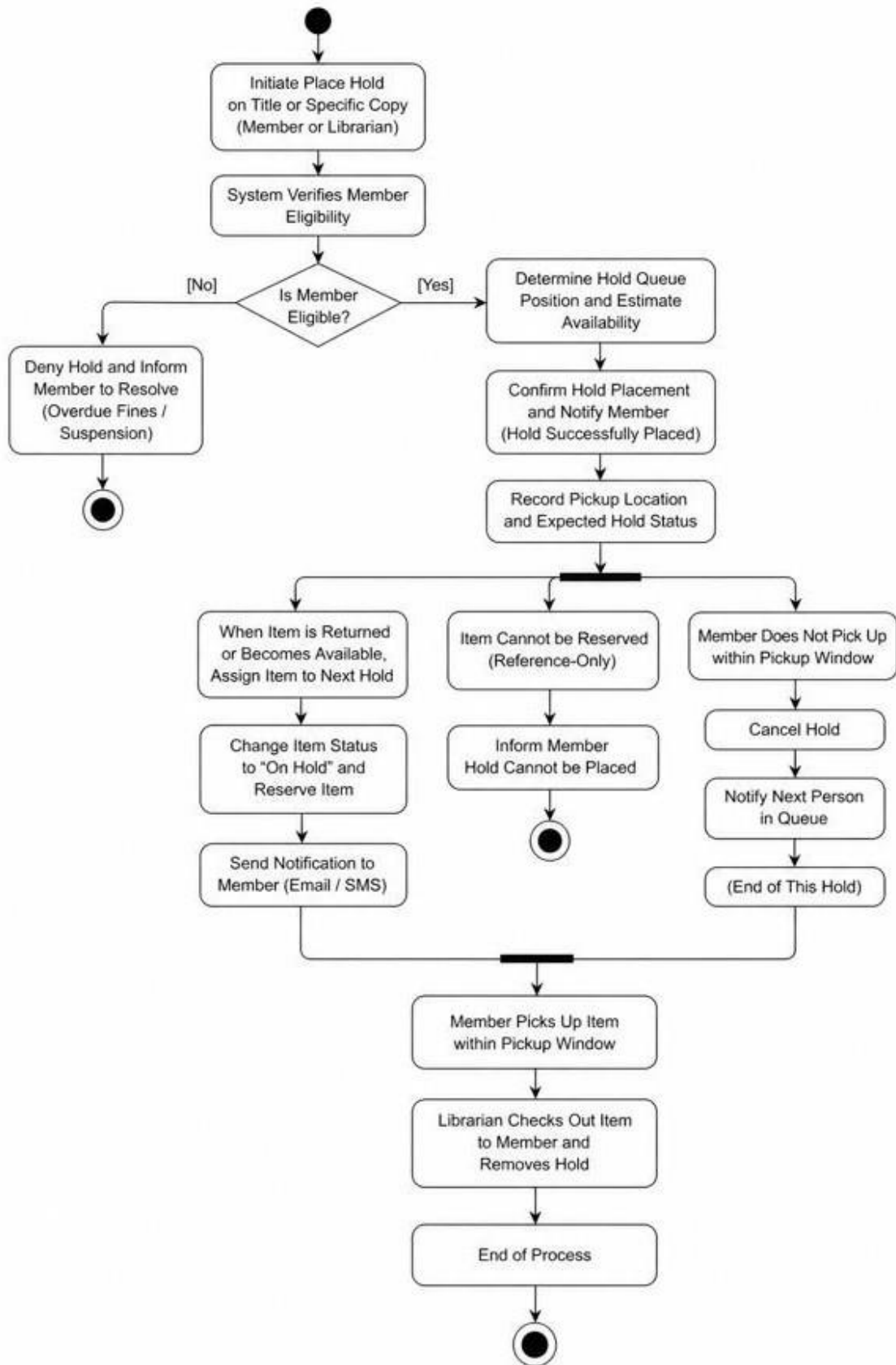


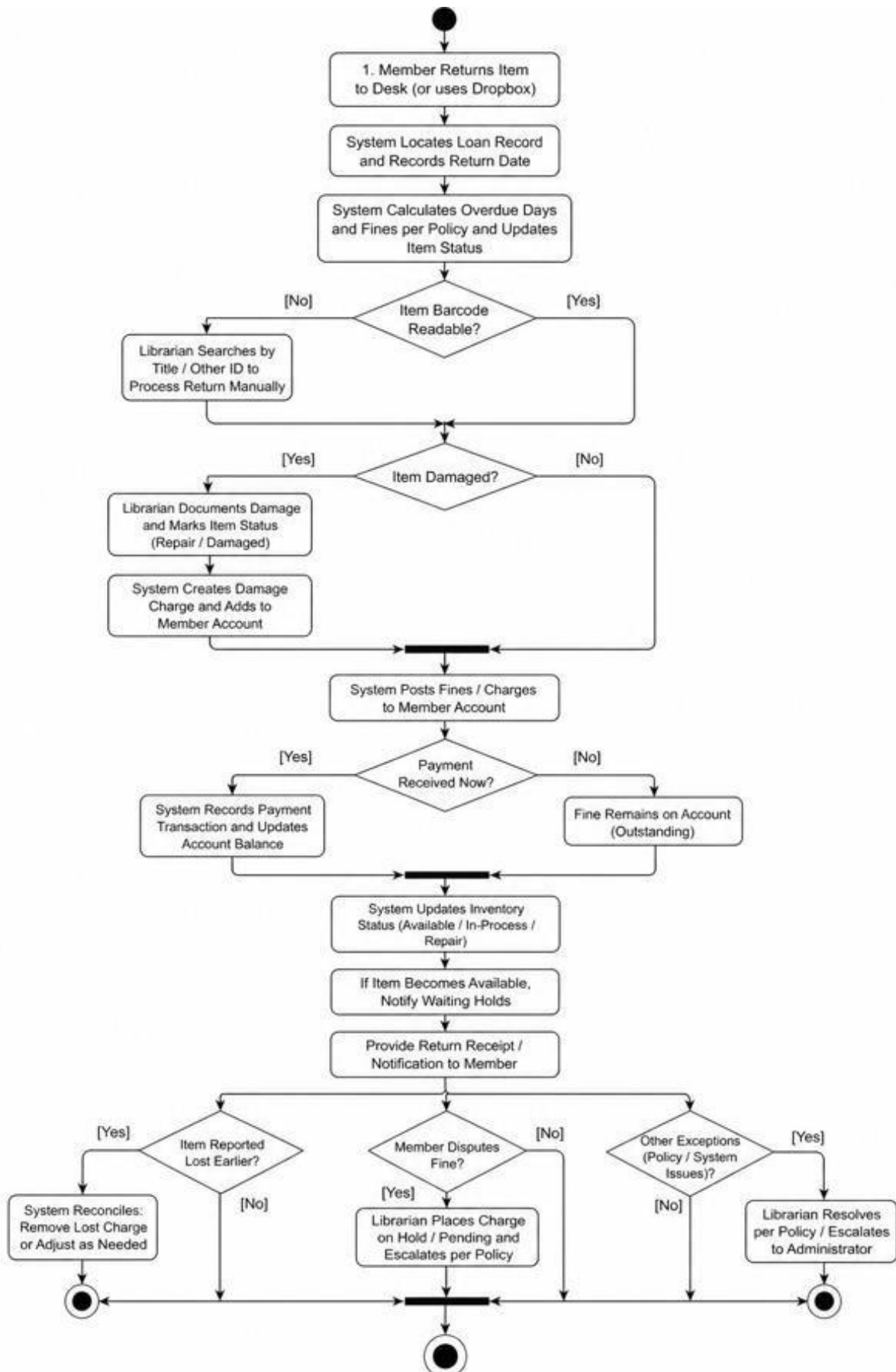


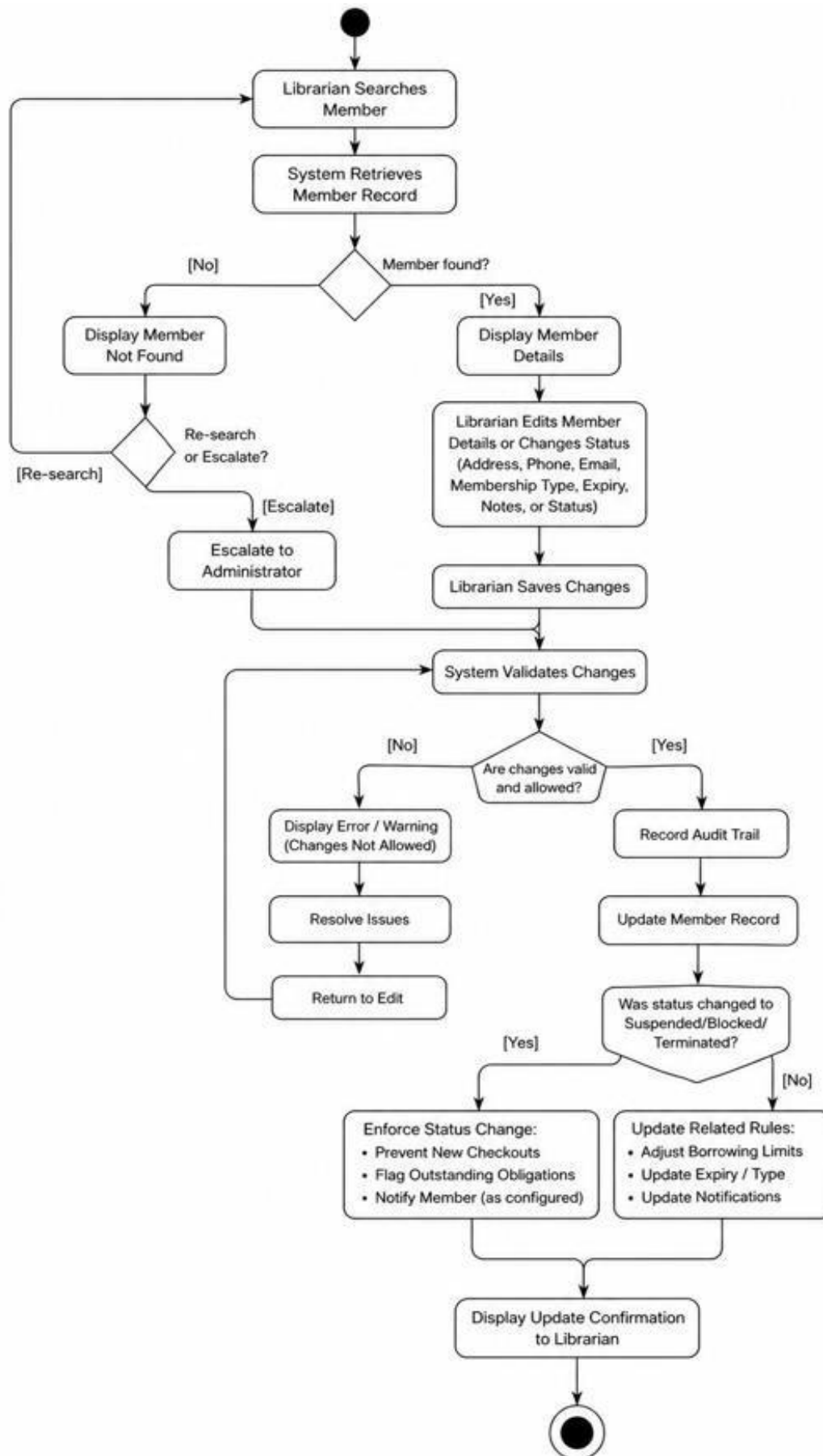




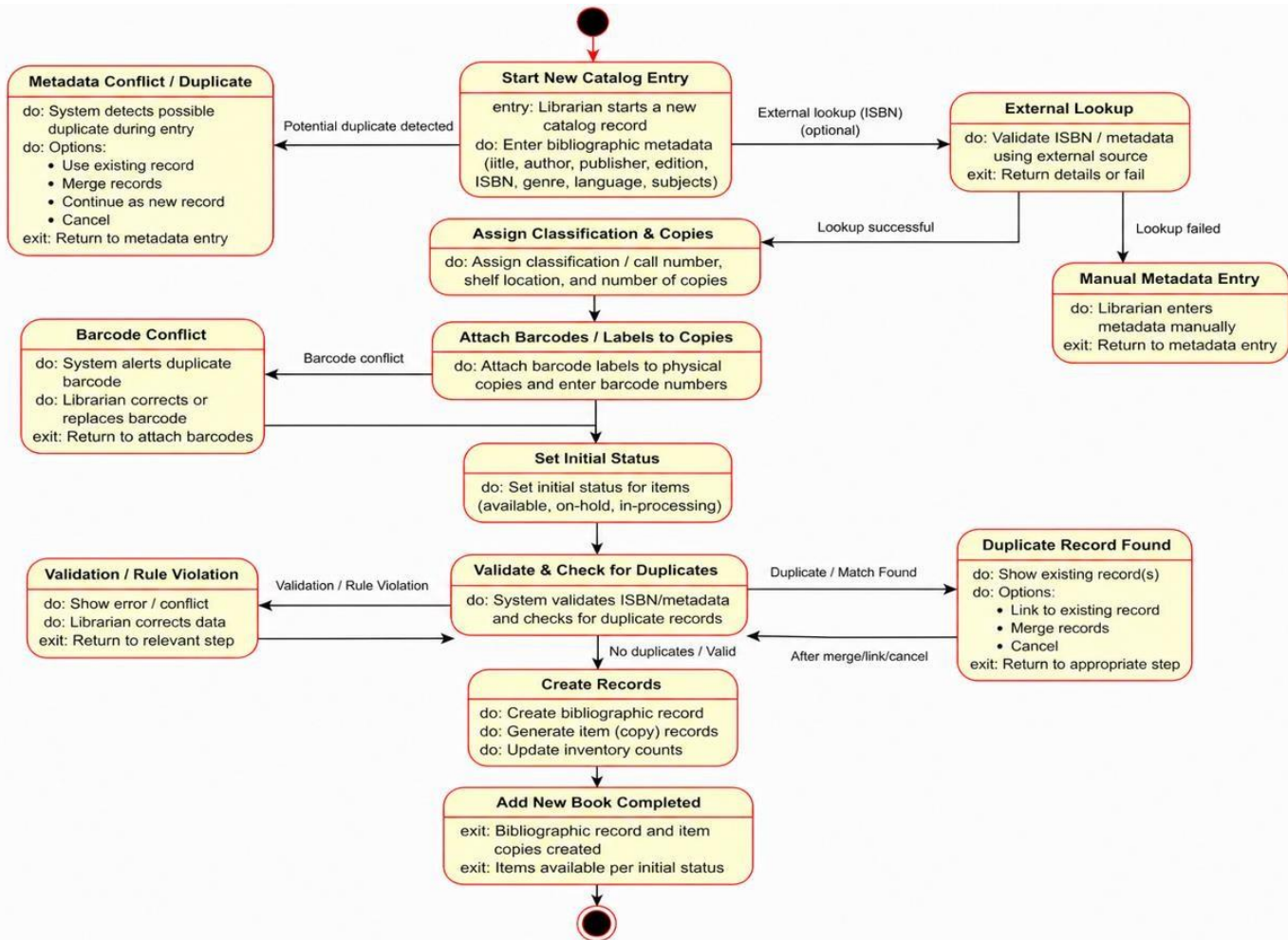


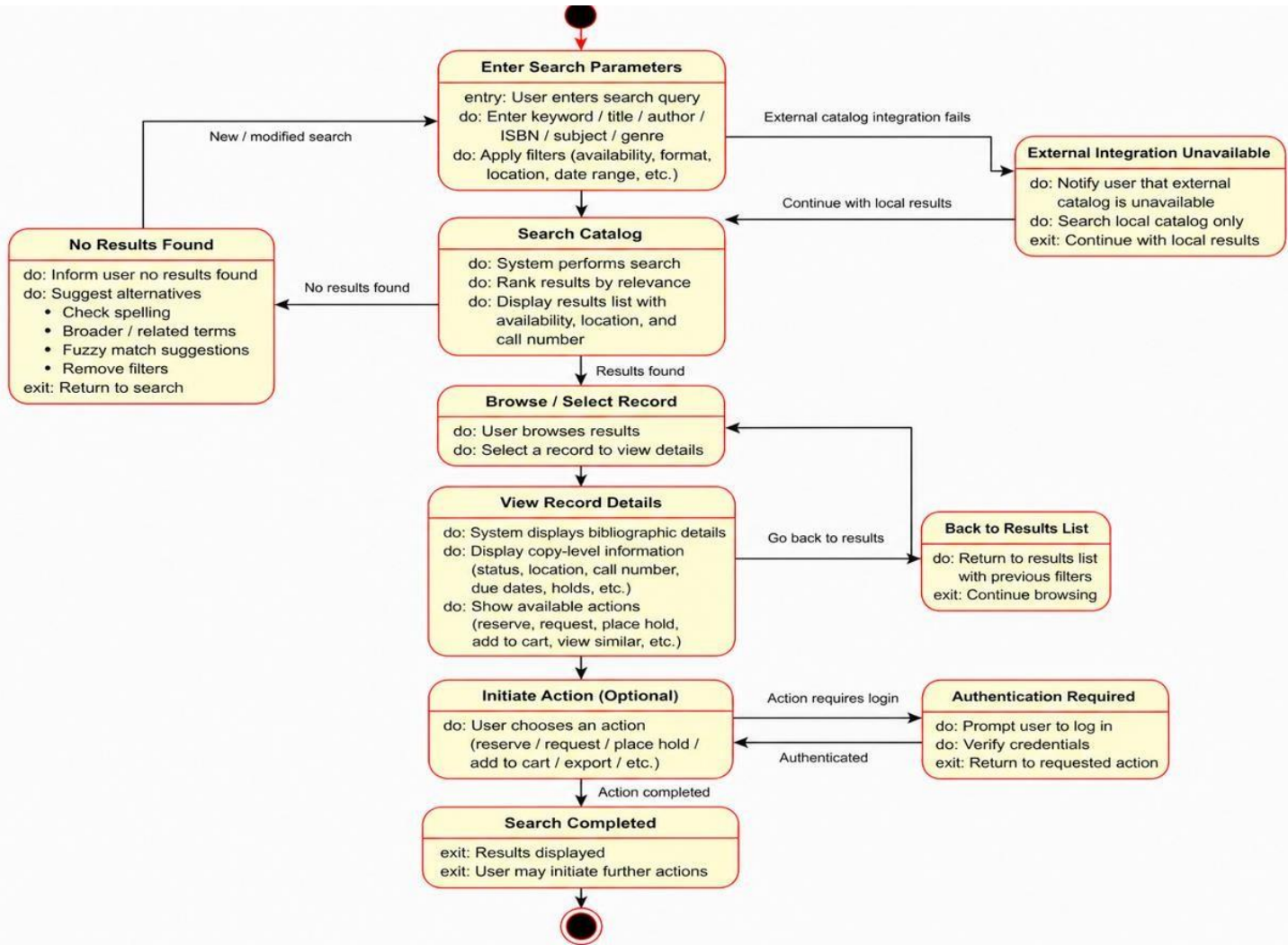


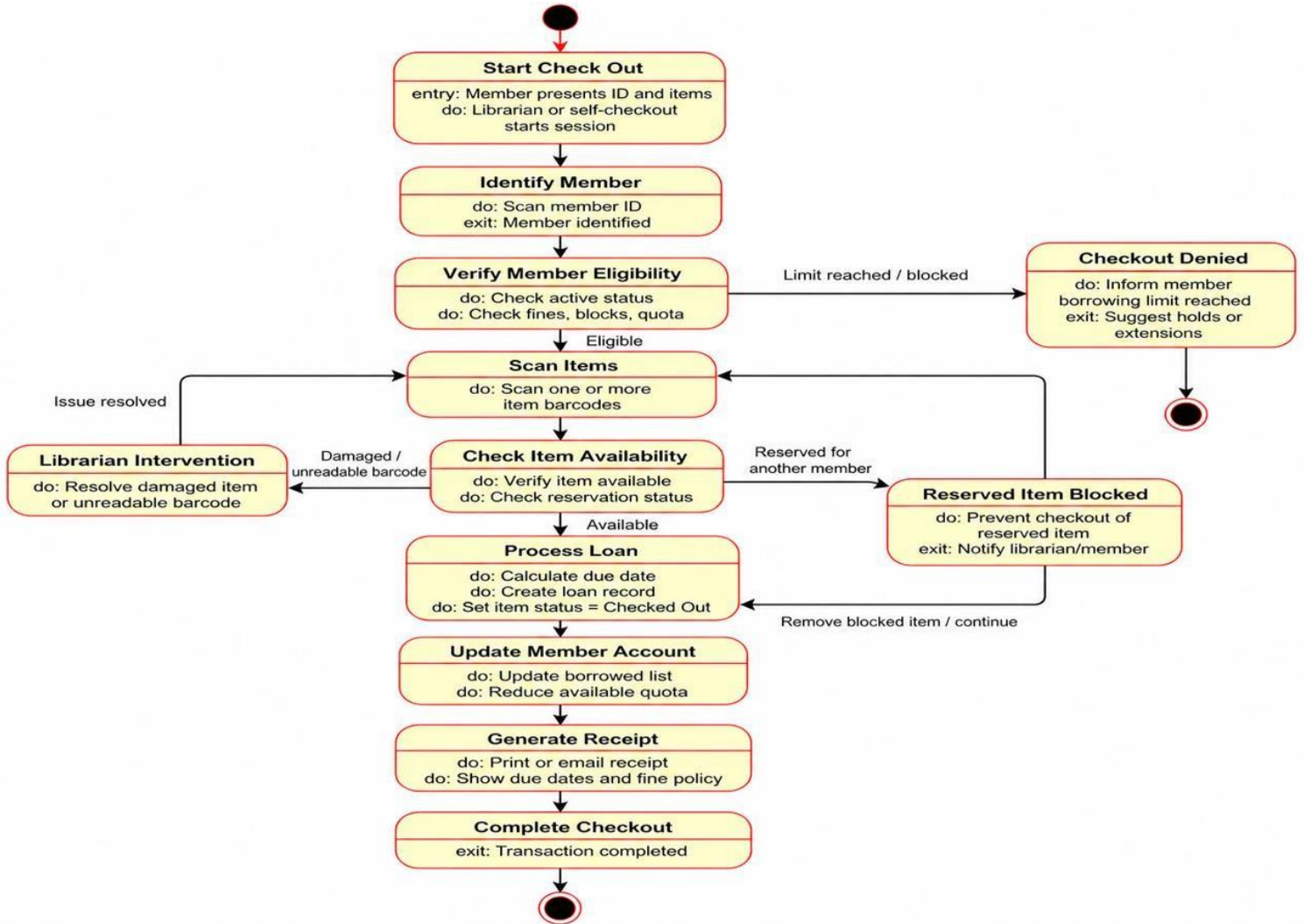


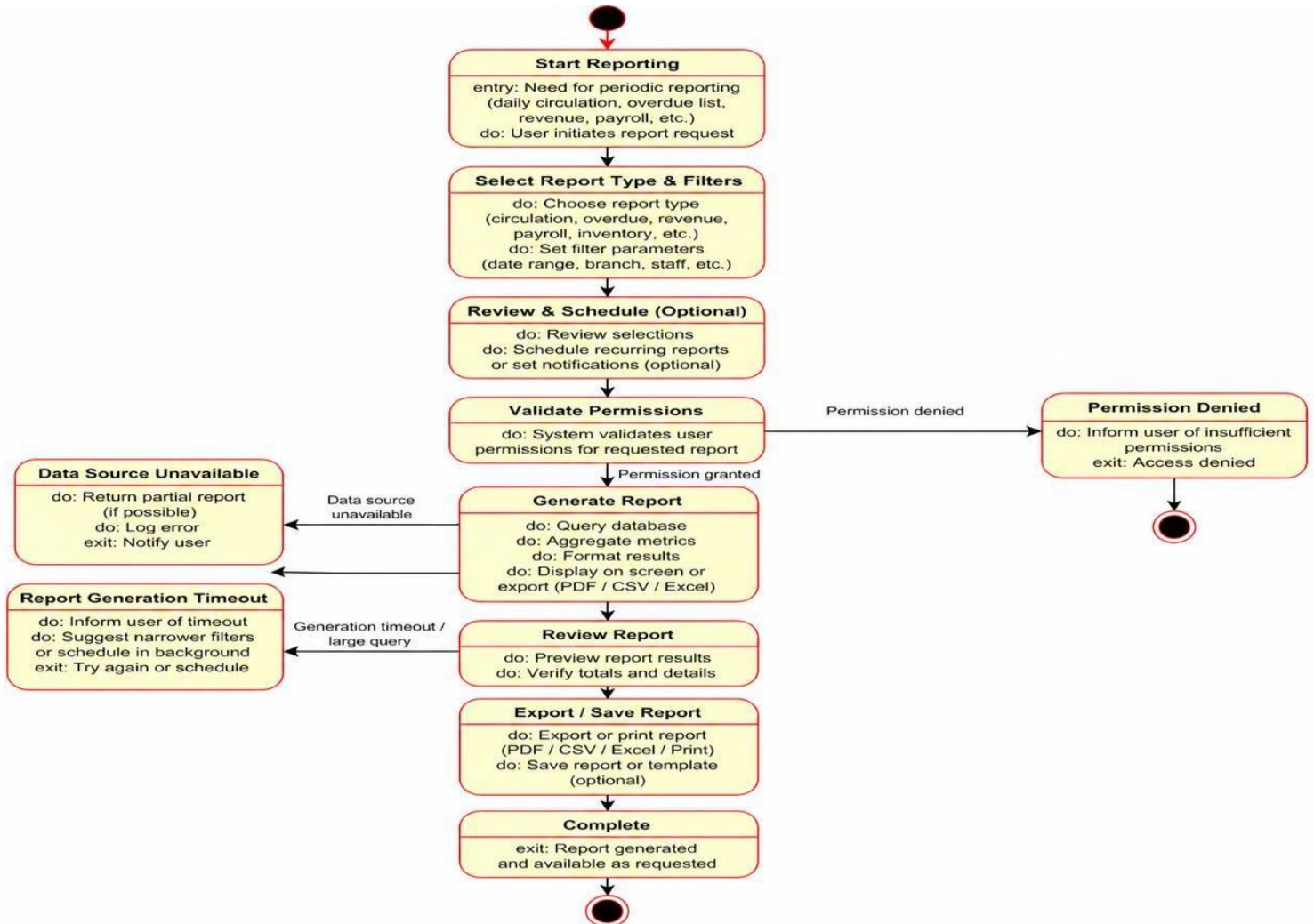


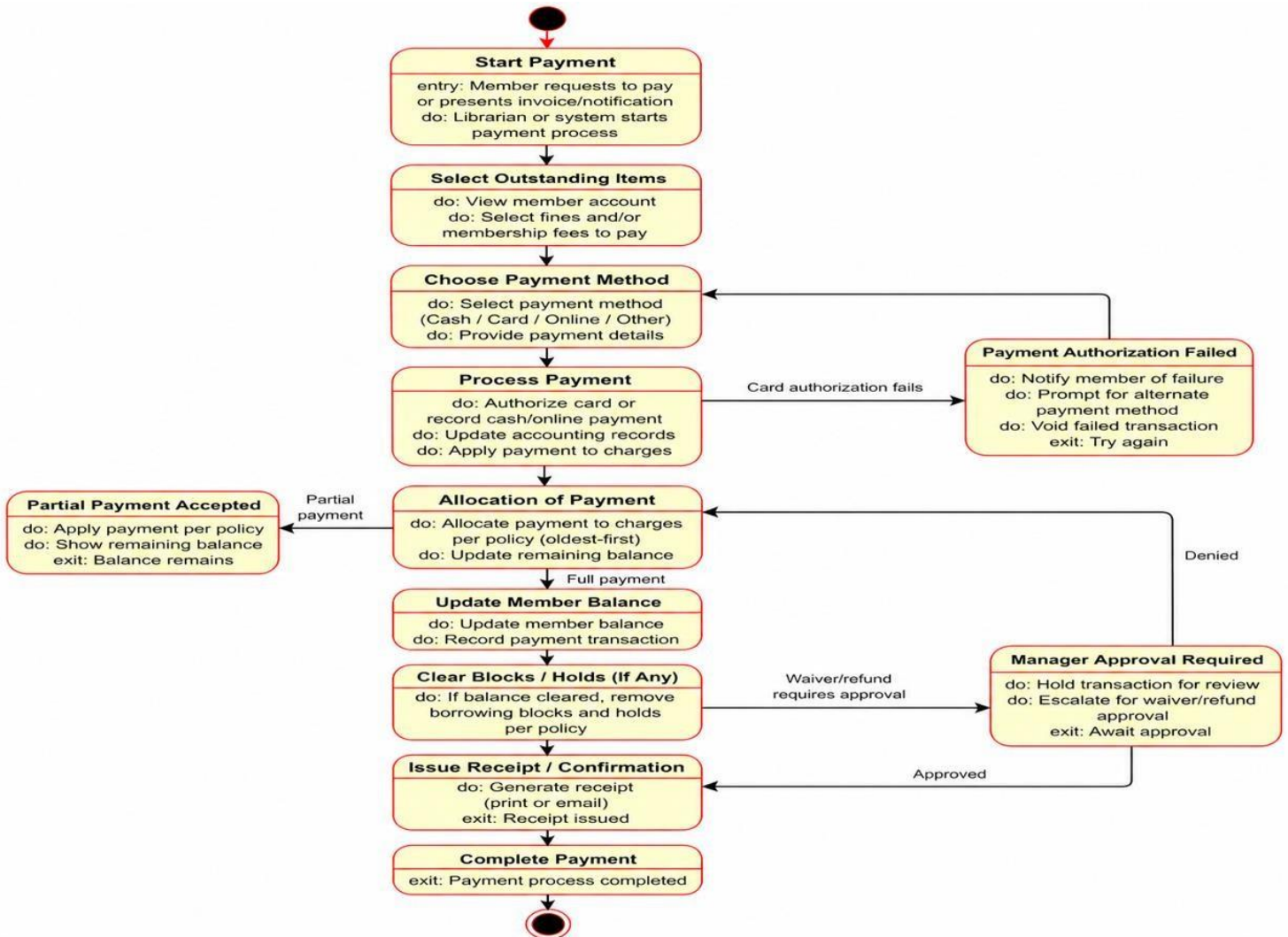
7.3 State Machine Diagrams

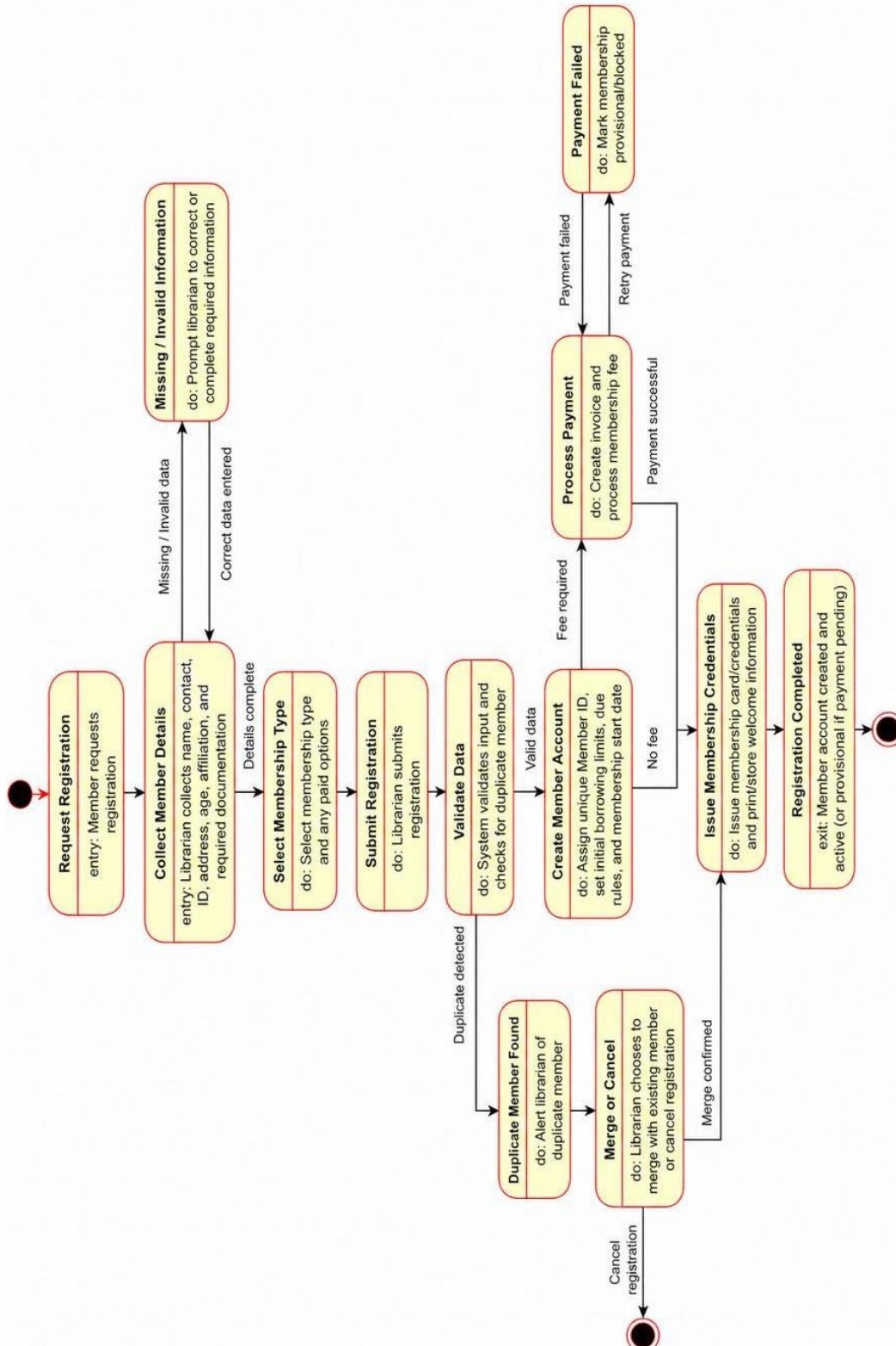


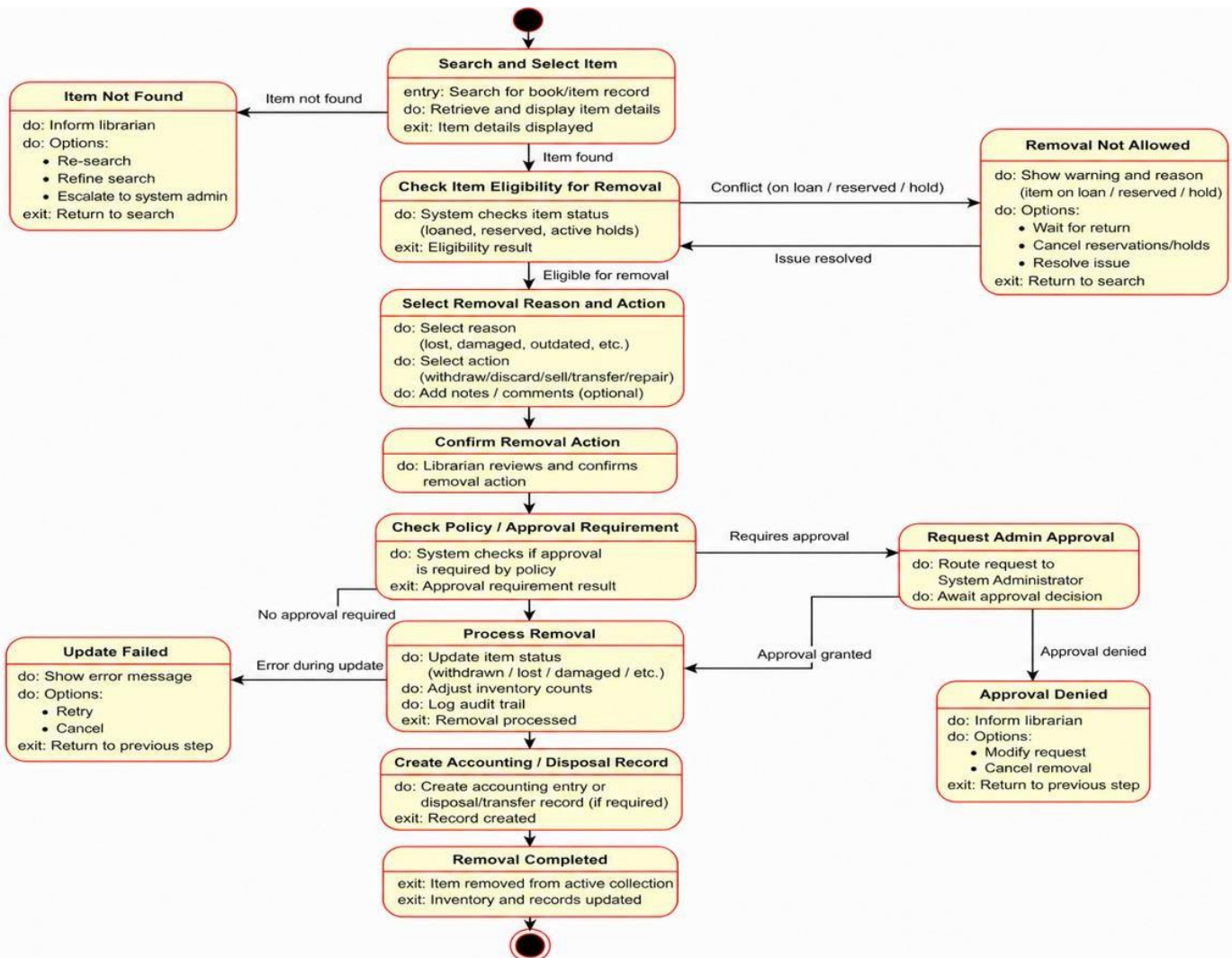


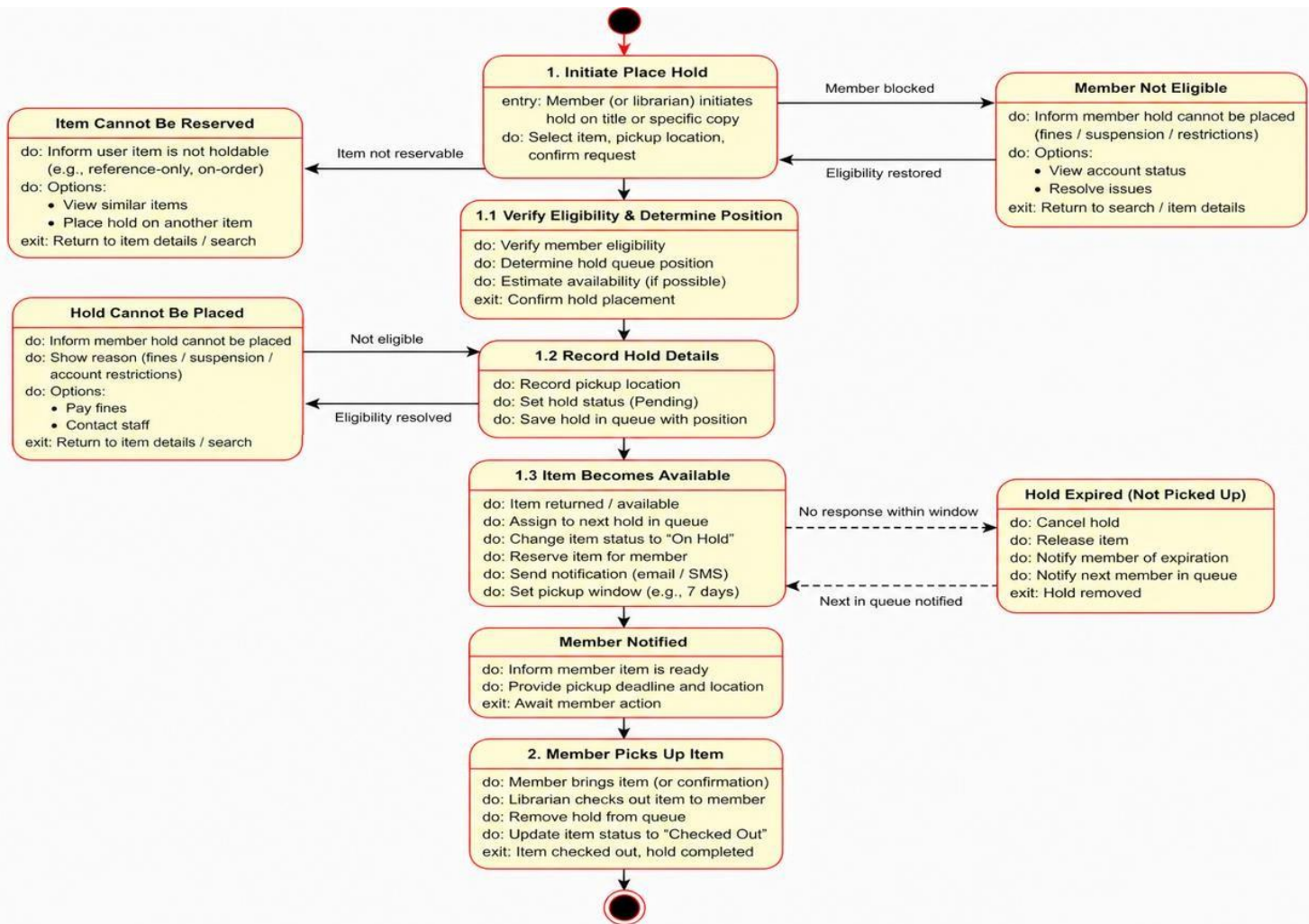


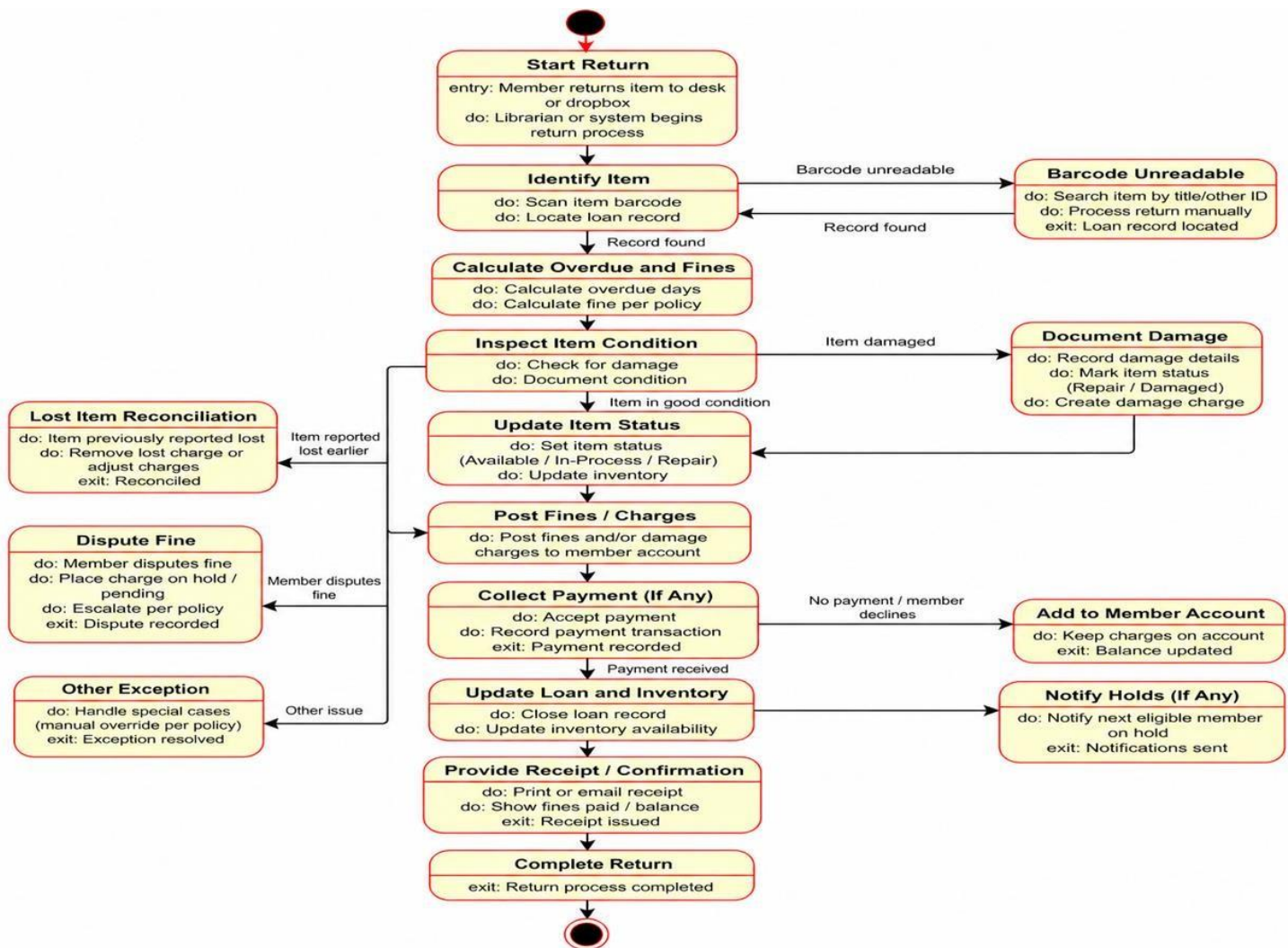


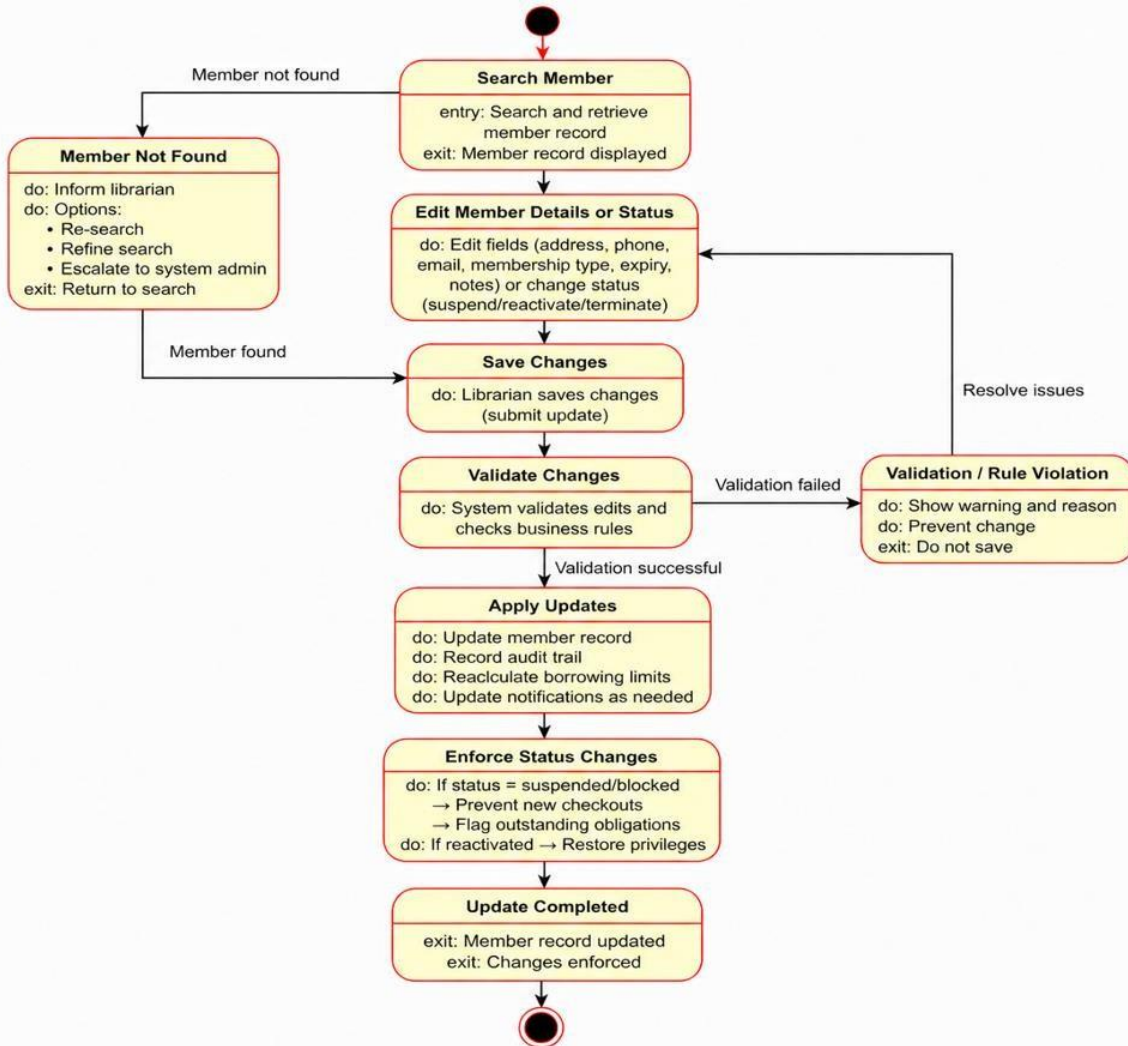






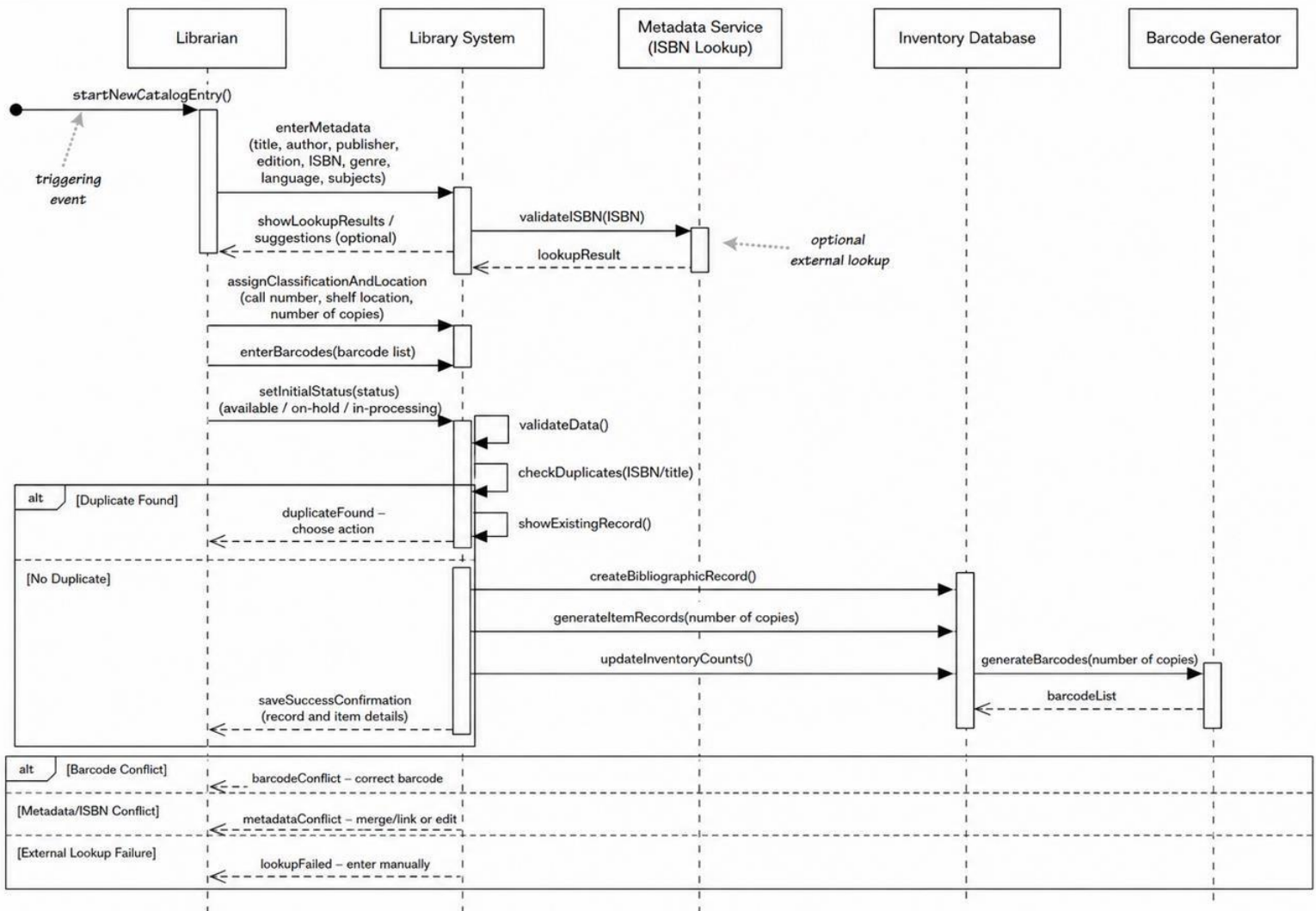




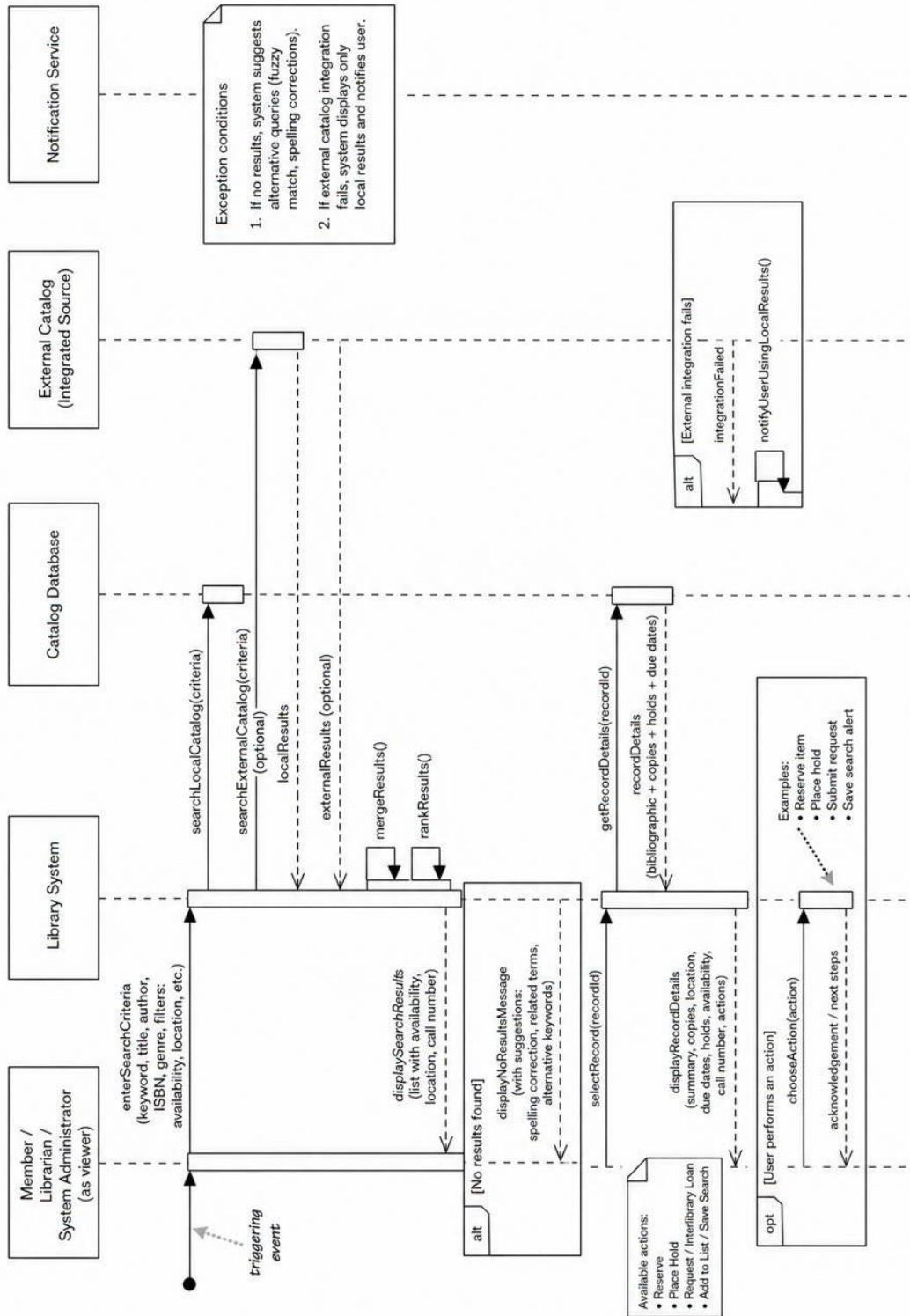


8 Interaction diagrams

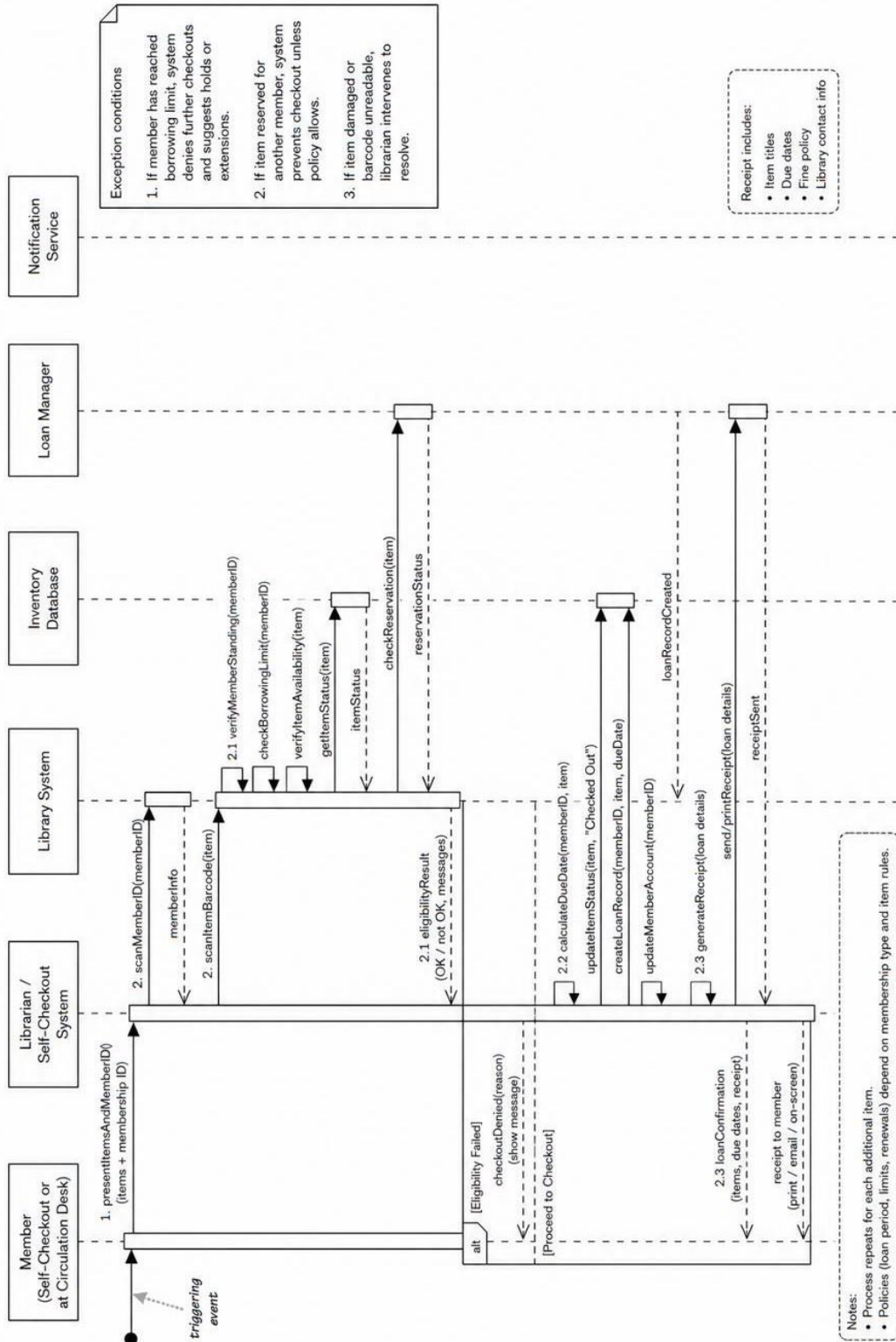
8.1 Sequence Diagrams



Add new book

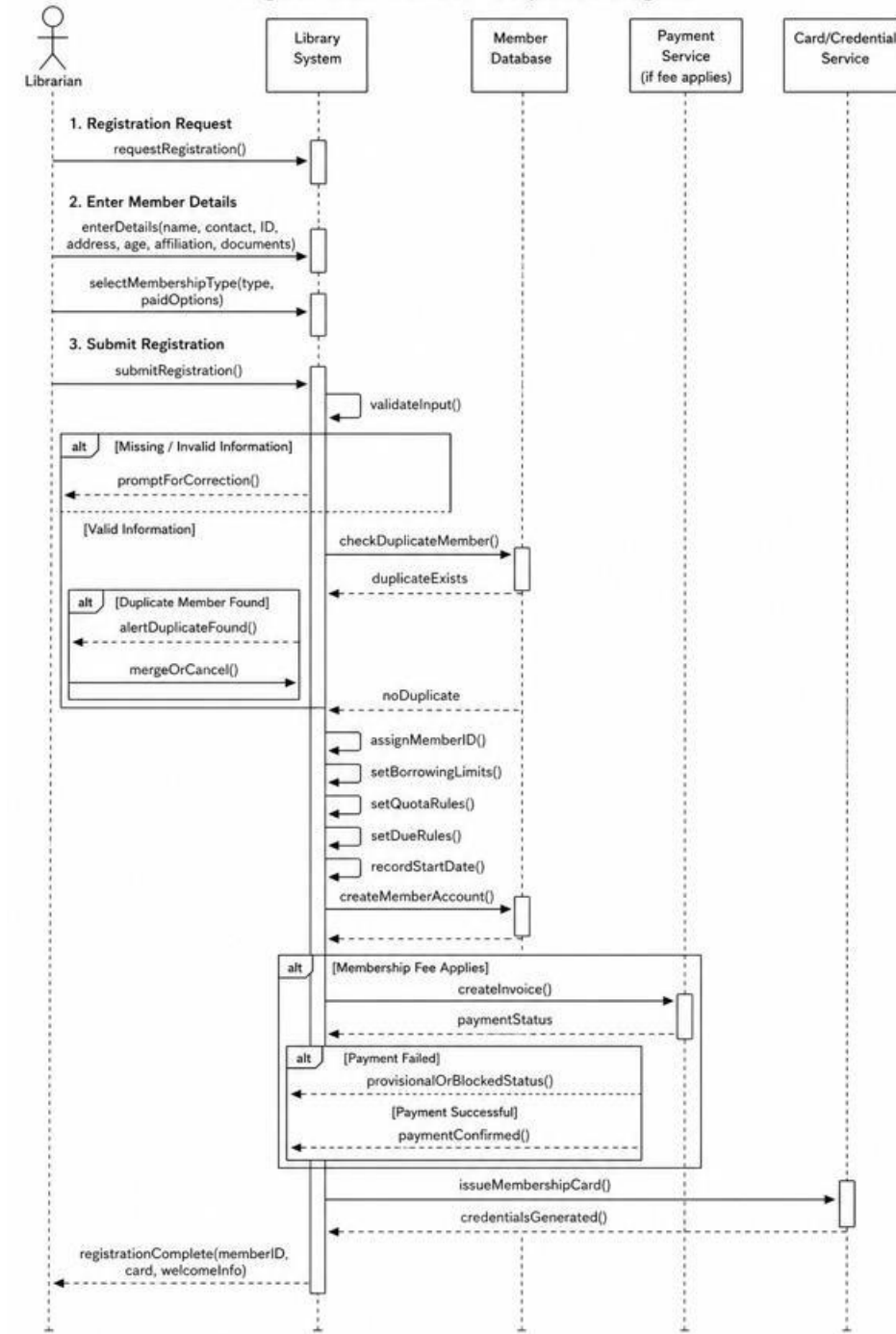


Browse Catalog

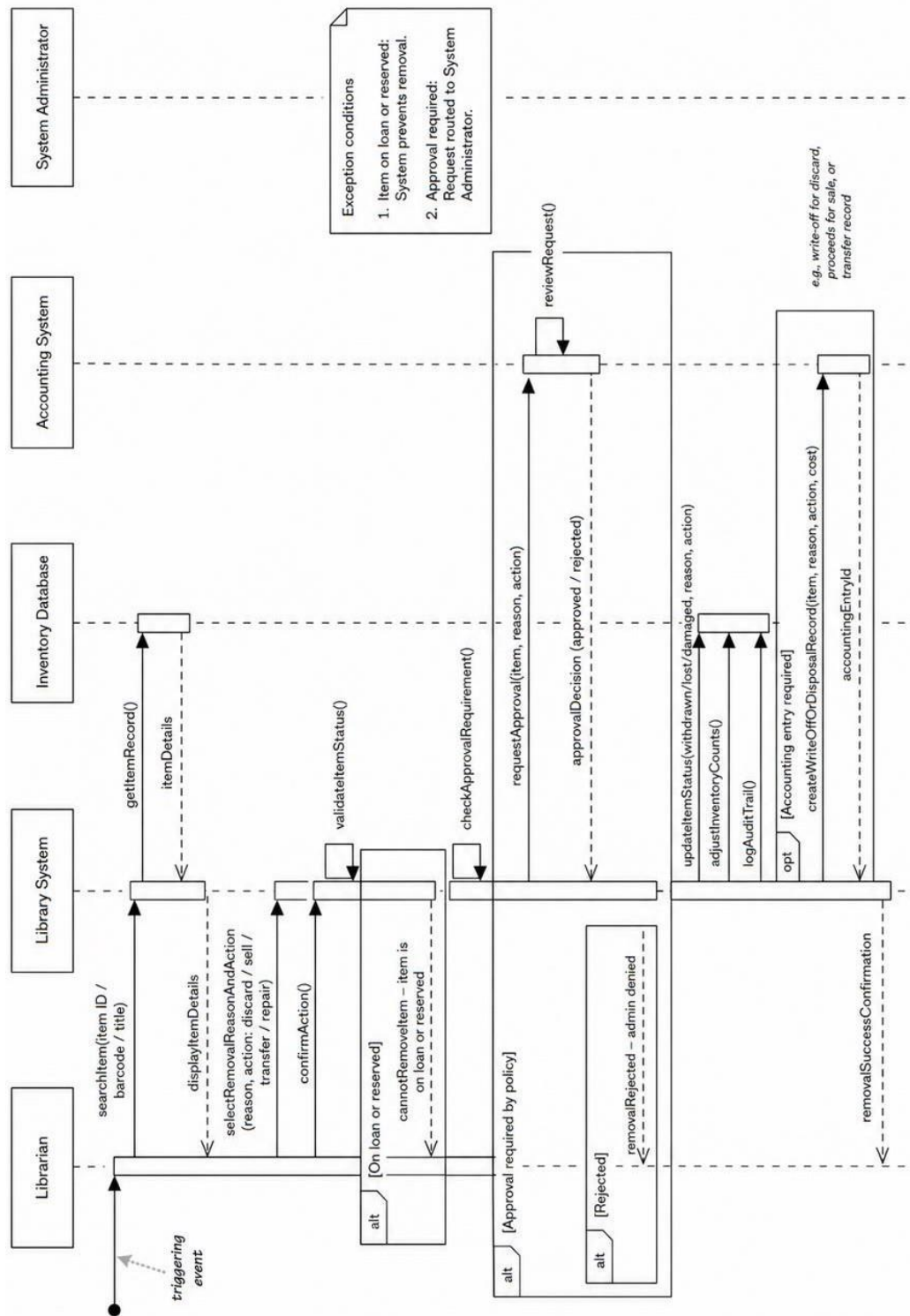


Check out (Borrow)

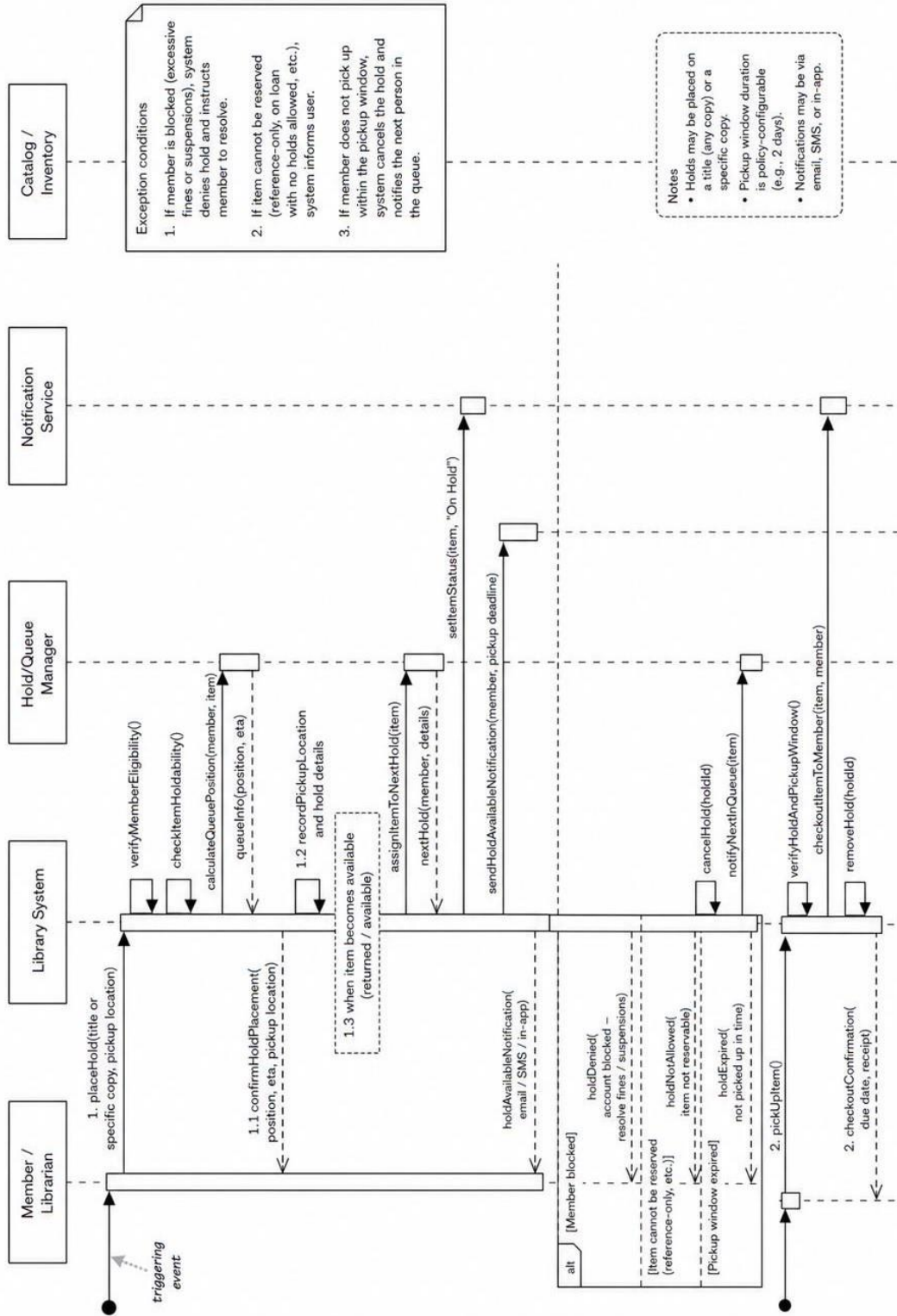
Register New Member - Sequence Diagram

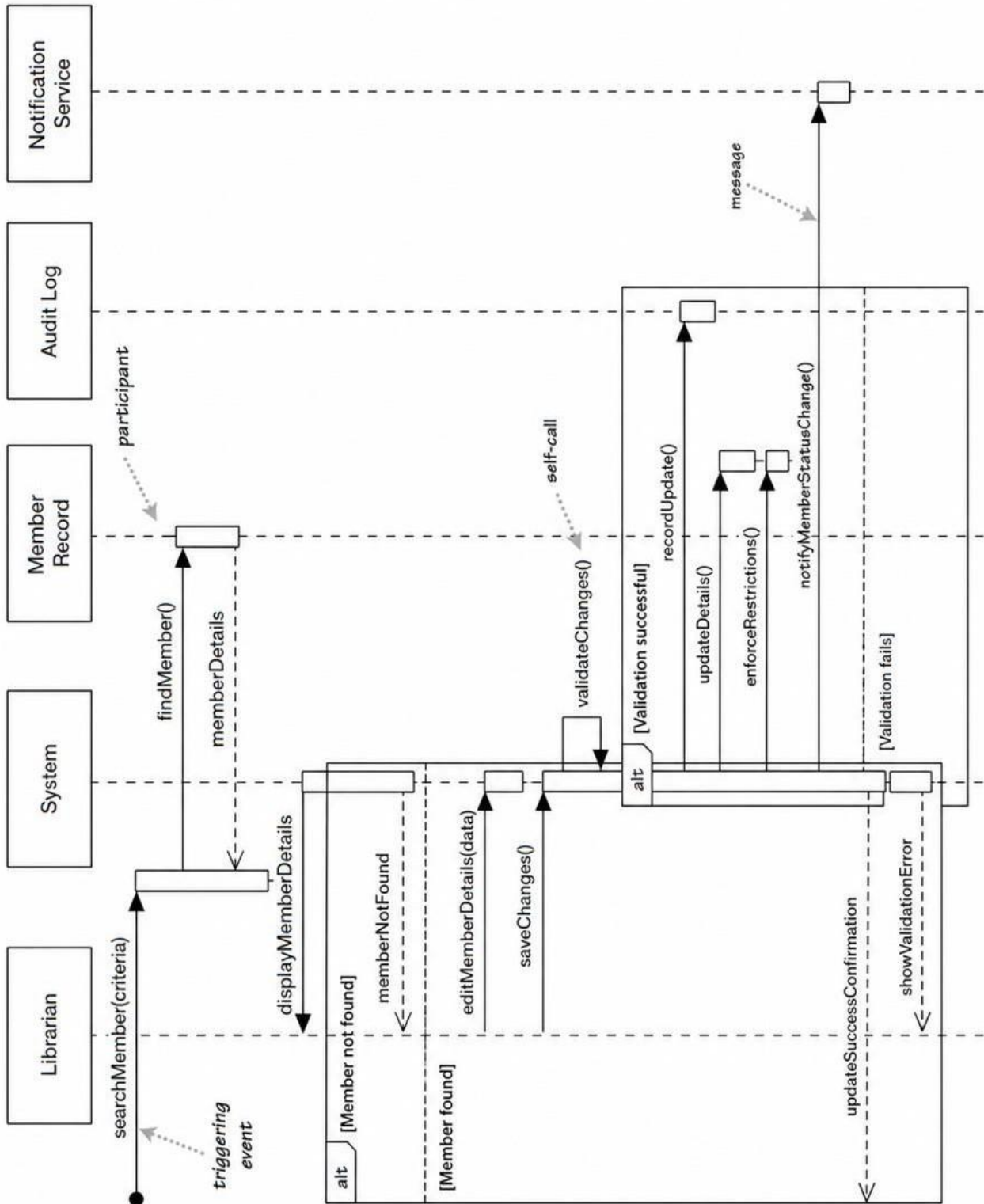


Register New Member



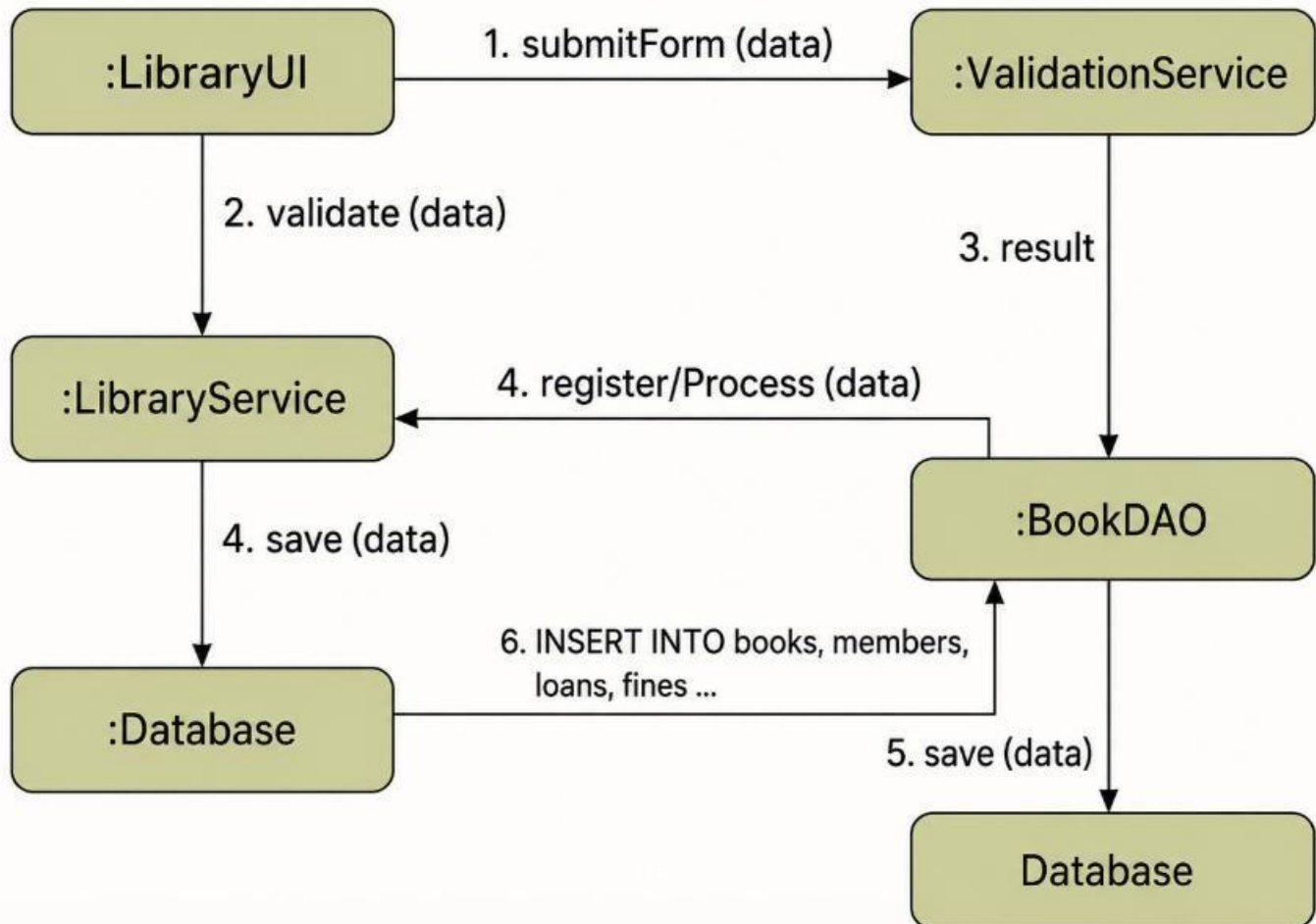
Remove Book



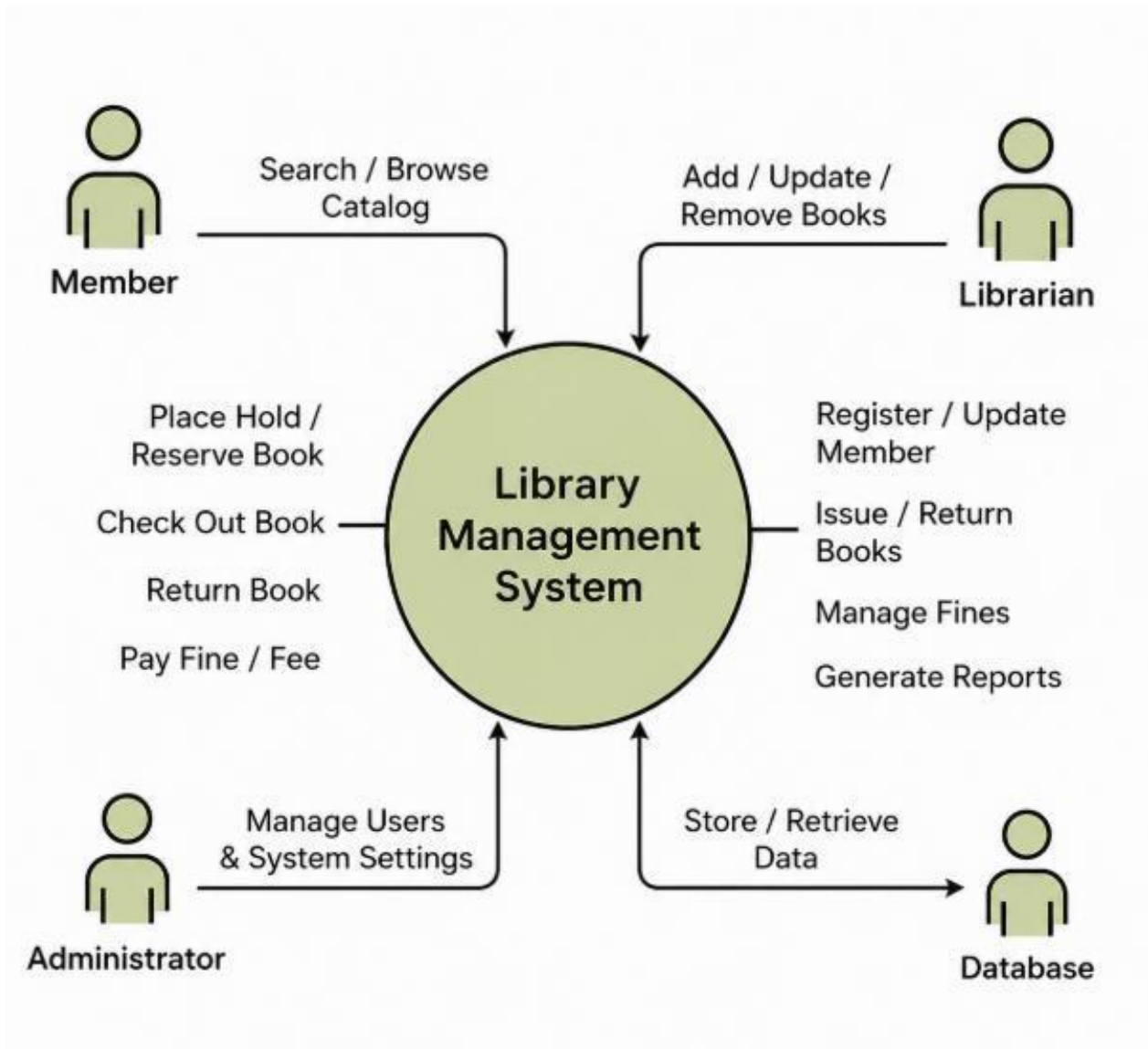


Update Member Account

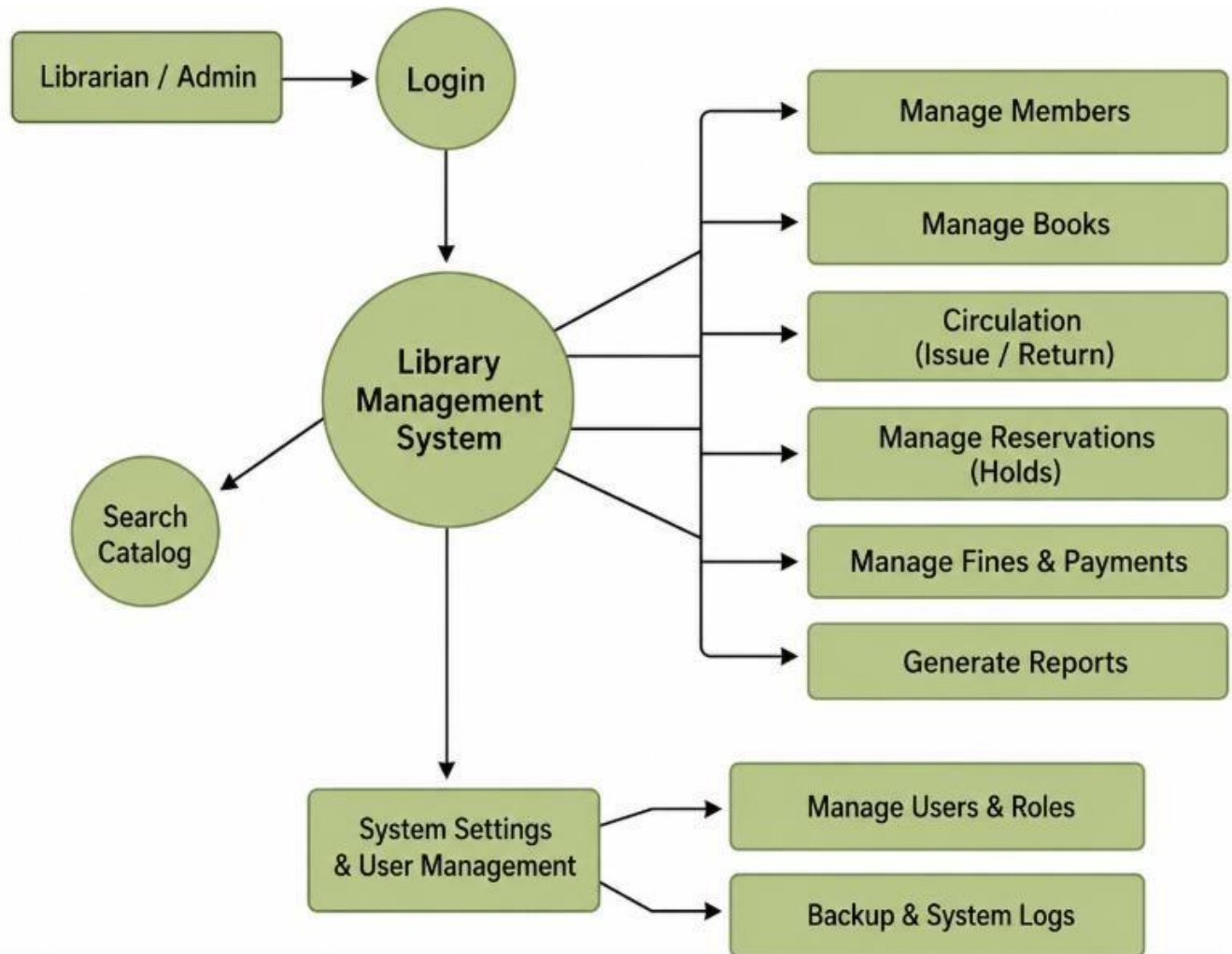
8.2 Communication Diagrams



9 Data Flow Diagram (DFD)

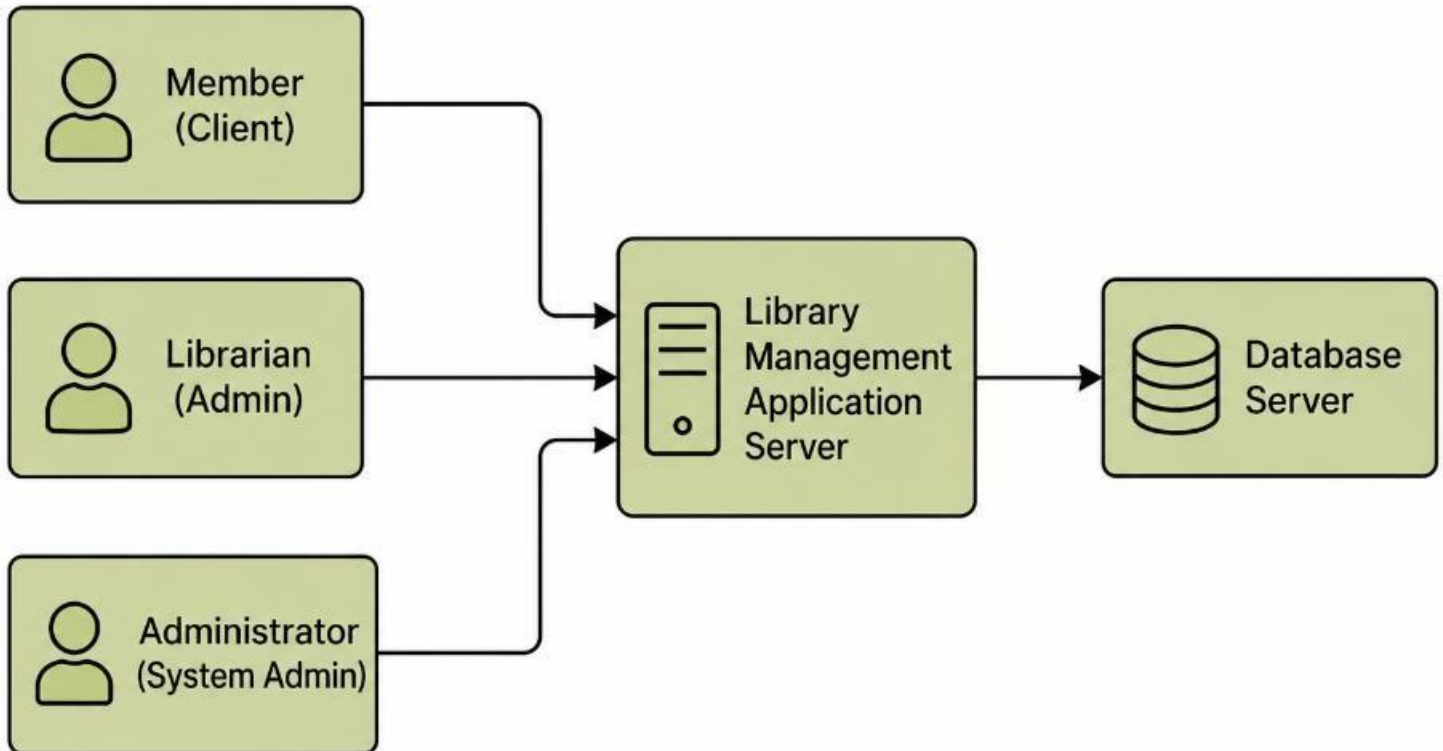


DFD Diagram 0

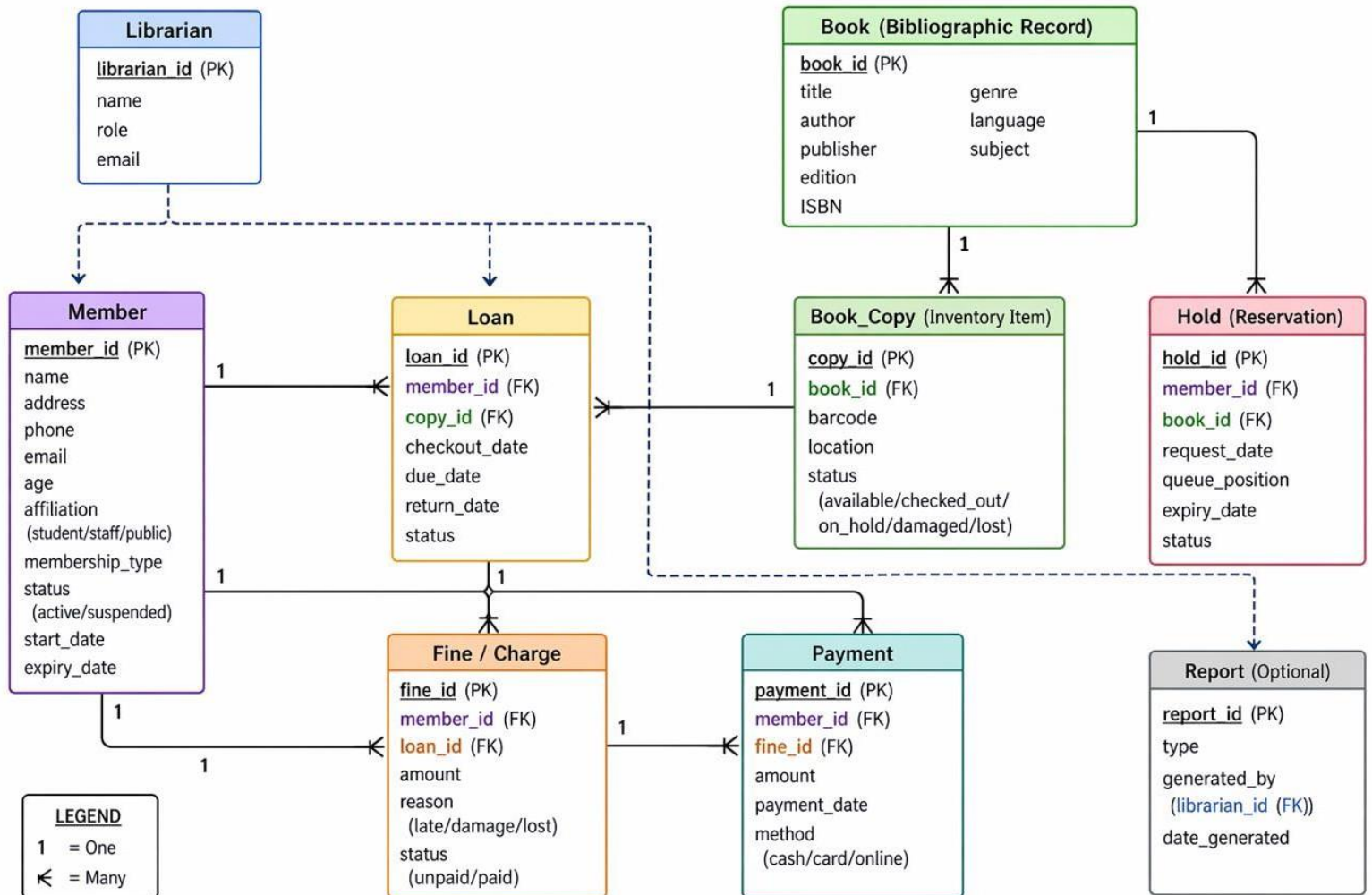


DFD Diagram 1

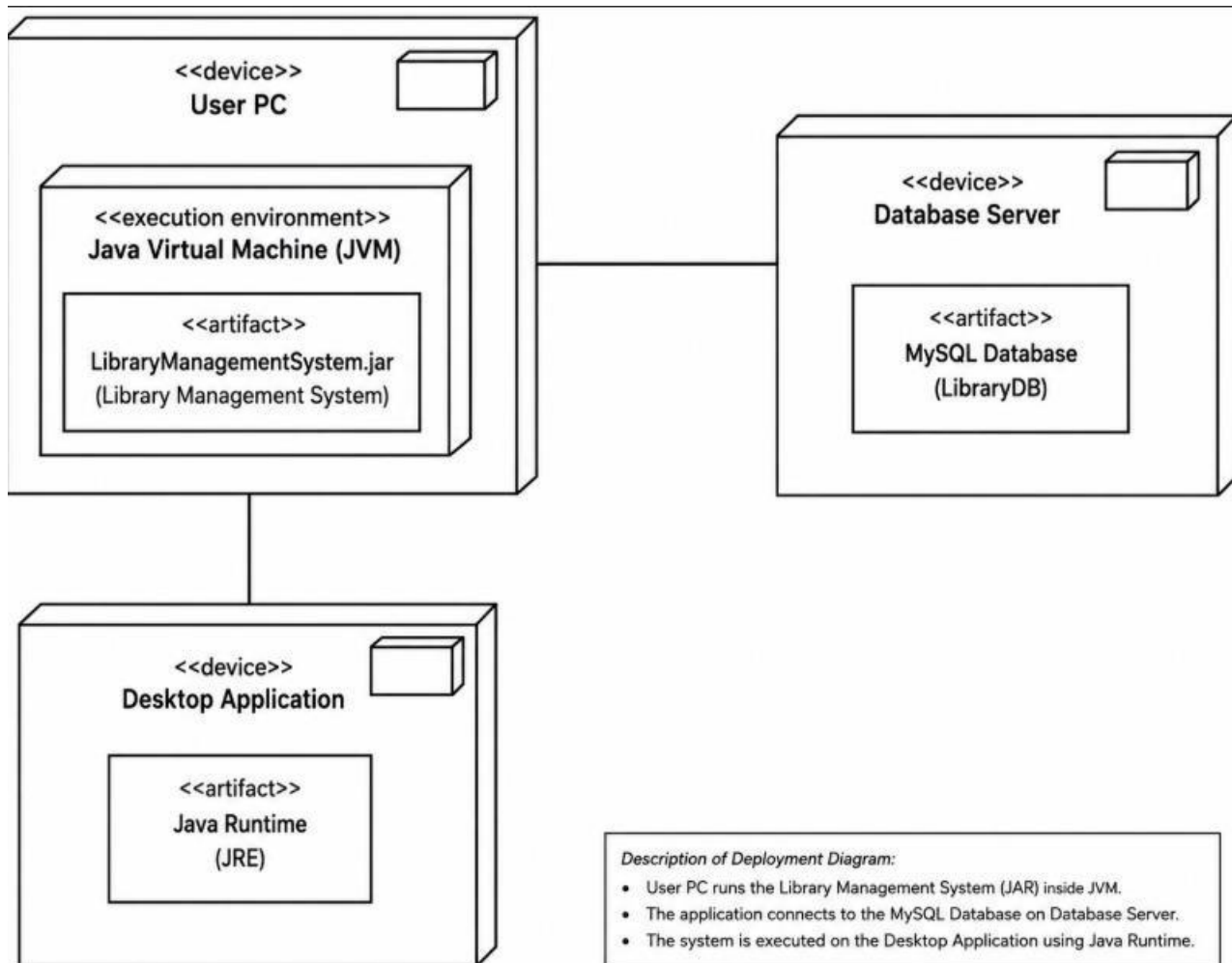
10 Architecture Diagram



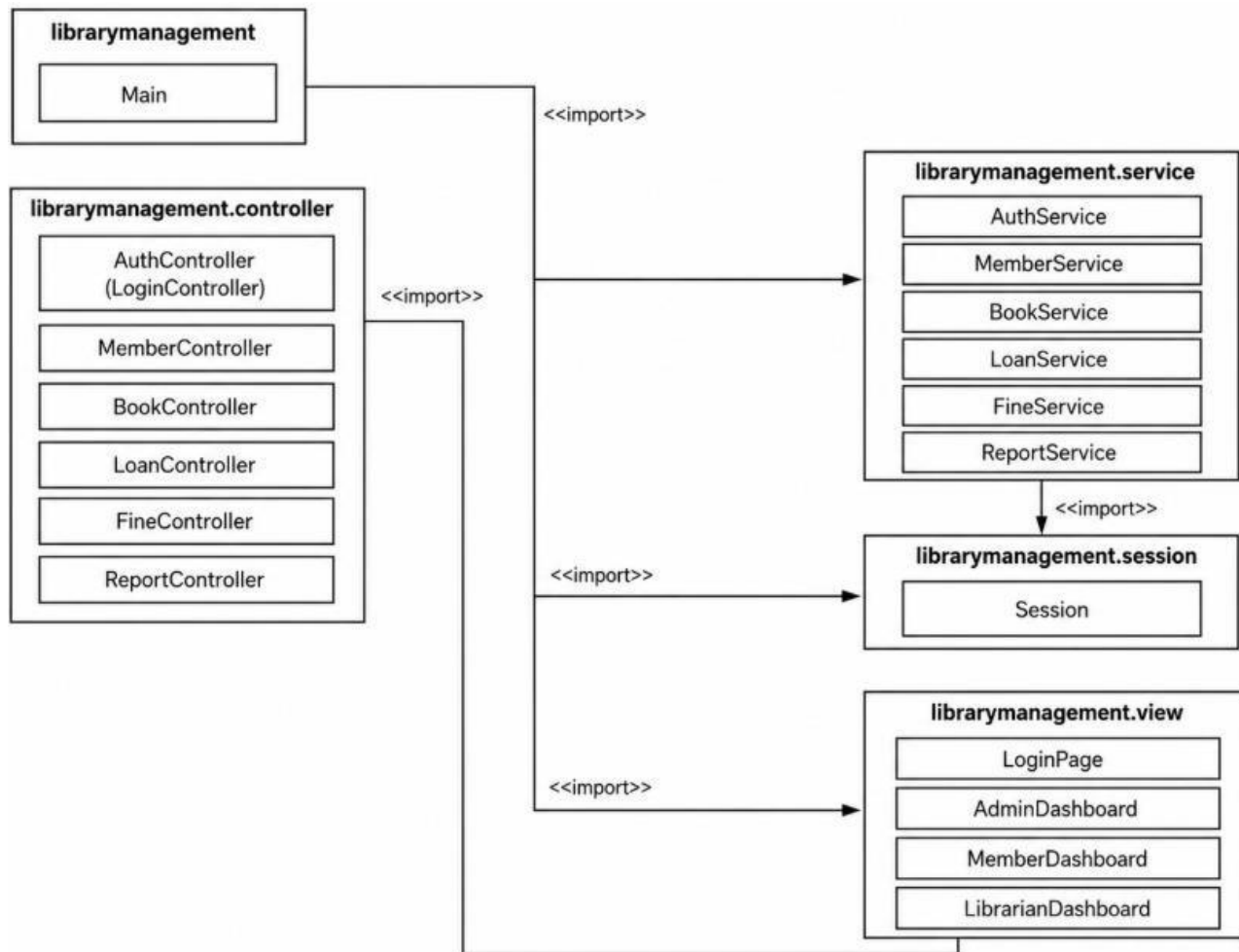
11 Component Diagram



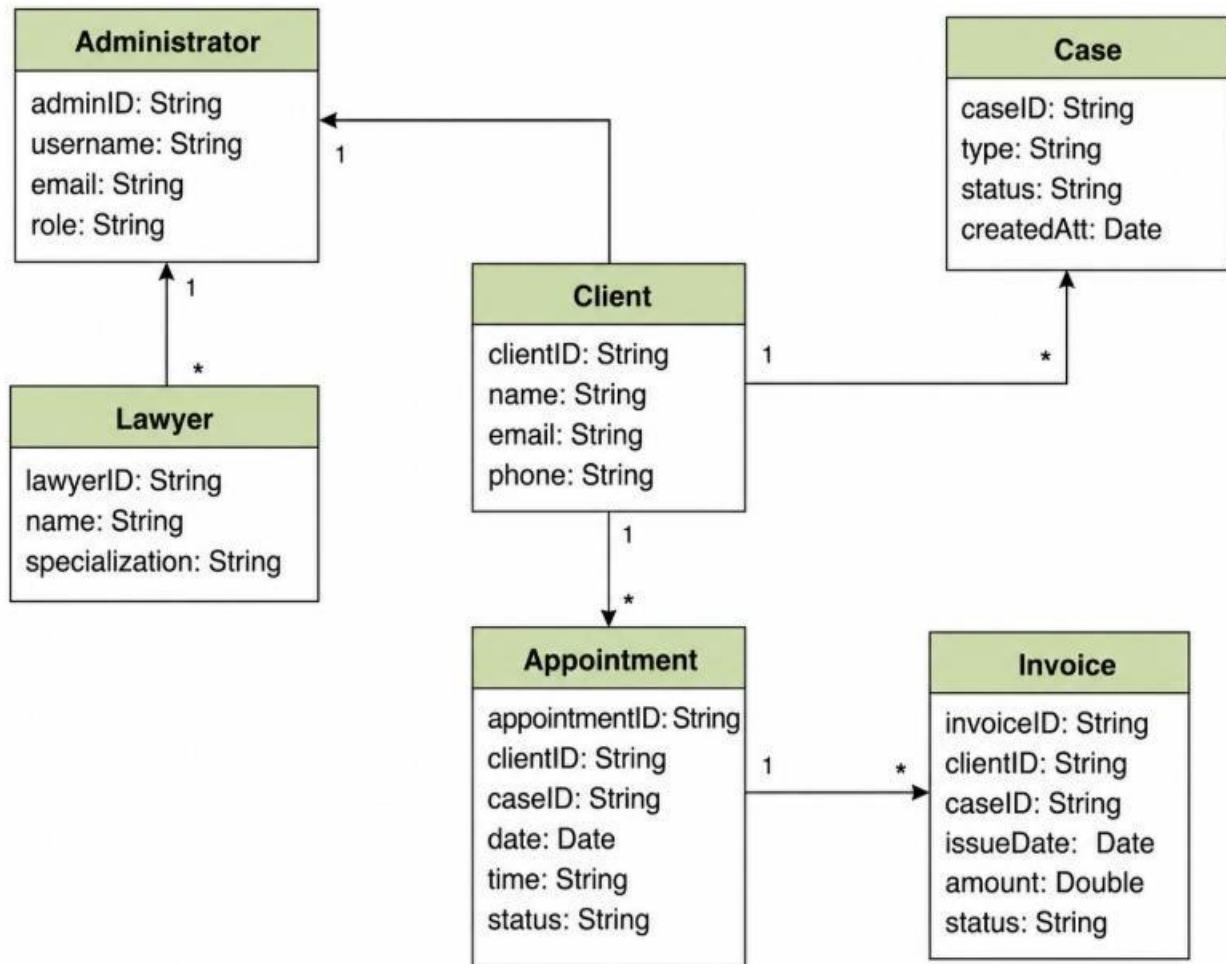
12 Deployment Diagram



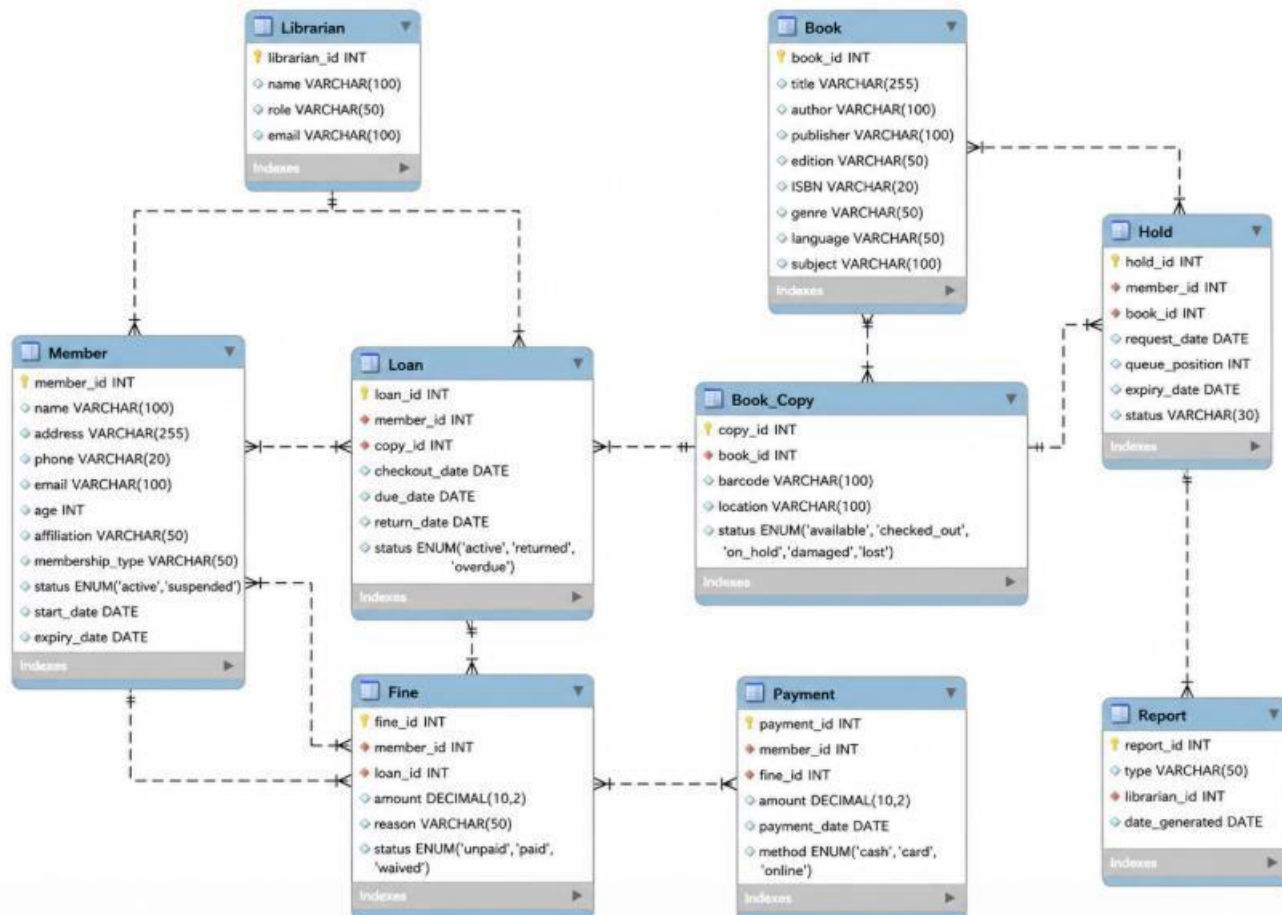
13 Package diagram



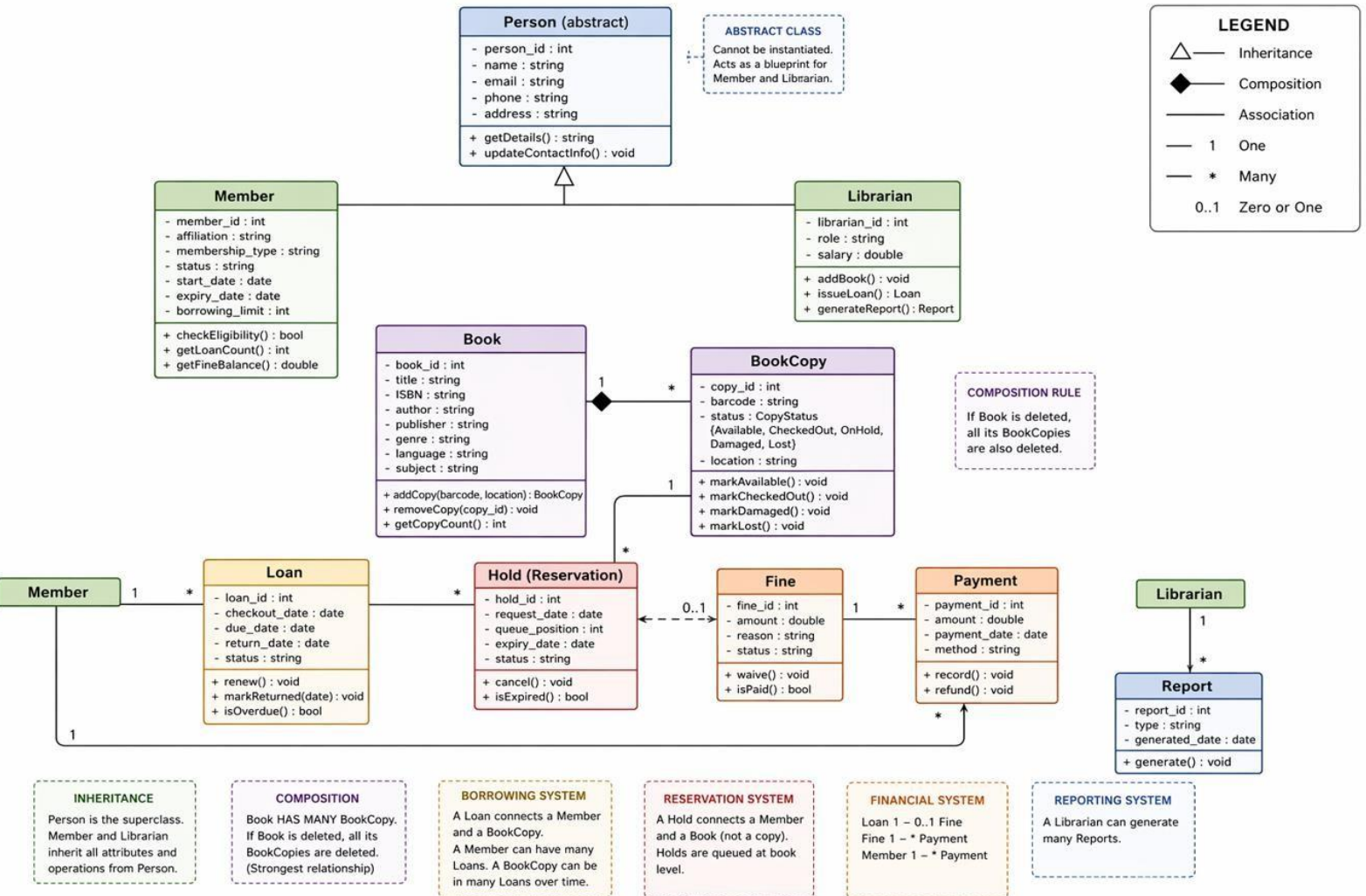
14 Object diagram



15 Entity-Relationship Diagram (ERD)



16 Class Diagram

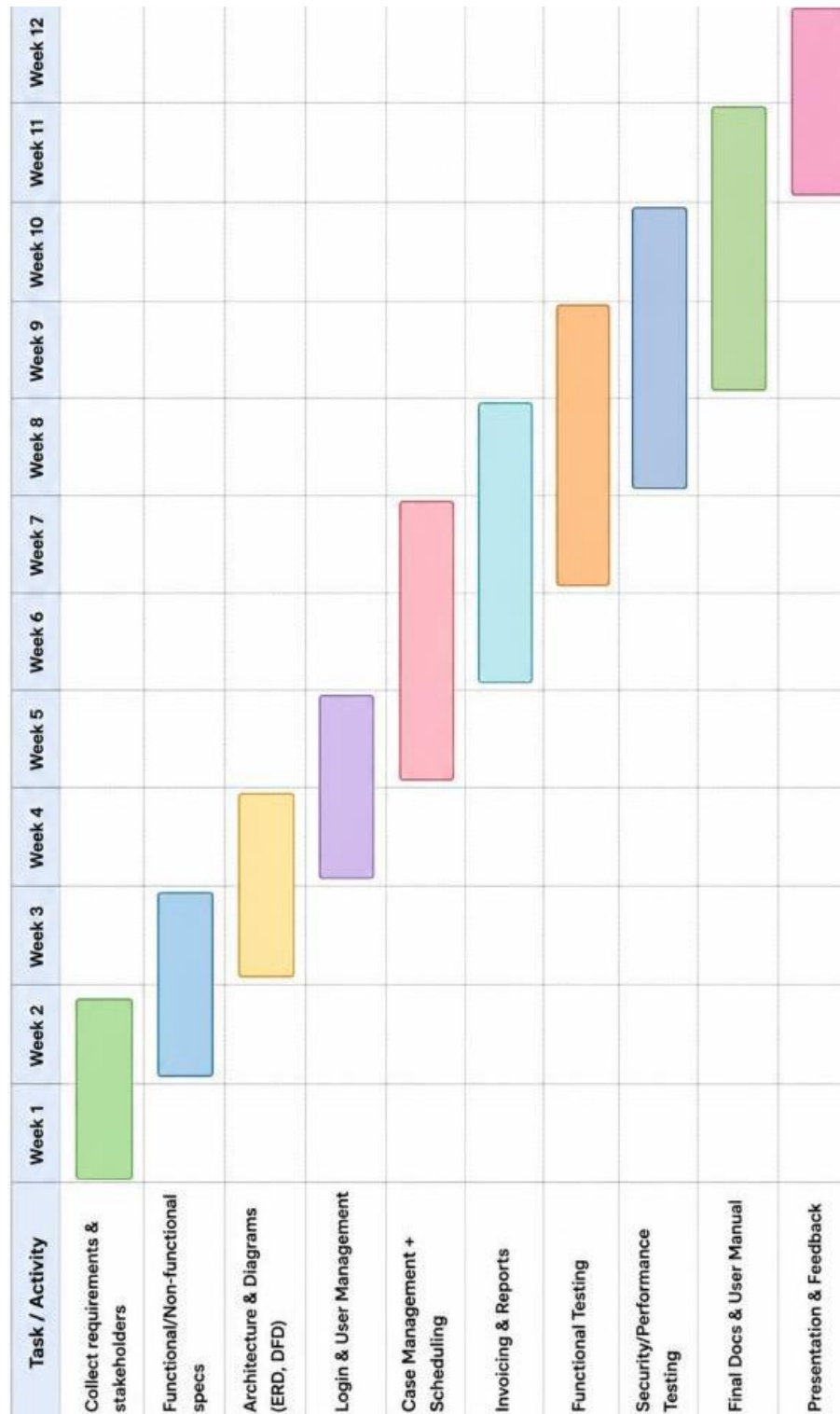


17 Project Planing

17.1 Risk Matrix

Risk	Likelihood	Impact	Mitigation Strategy
System Integration Failure	Medium	High	Conduct thorough testing of all modules such as Member, Loan, Payment, and Hold before deployment. Use proper integration planning and debugging techniques.
Data Loss	Medium	High	Implement regular database backups and recovery procedures. Store backups securely and test restoration processes periodically.
Security Breach	High	High	Apply strong authentication, role-based access control, encryption, and regular security audits to protect sensitive member and payment information.
Unauthorized Access	Medium	High	Restrict access using librarian and administrator privileges. Use secure passwords and session management techniques.
Database Corruption	Low	High	Use transaction management, database constraints, and integrity checks to maintain consistency among entities such as Loan, Fine, and Payment.
Server Downtime	Medium	Medium	Maintain reliable hosting infrastructure and use backup servers or cloud recovery solutions to minimize downtime.
Payment Processing Errors	Medium	Medium	Validate payment transactions carefully and maintain transaction logs for auditing and troubleshooting.
Incorrect Book Availability Status	Medium	Medium	Ensure automatic updates of Book_Copy status during loan, return, and hold operations to prevent inconsistencies.
Loss of Reservation Data	Low	Medium	Regularly back up Hold (Reservation) records and implement validation checks for reservation processing.
Human Error	High	Medium	Provide proper training for librarians and staff on how to use the system correctly and efficiently.

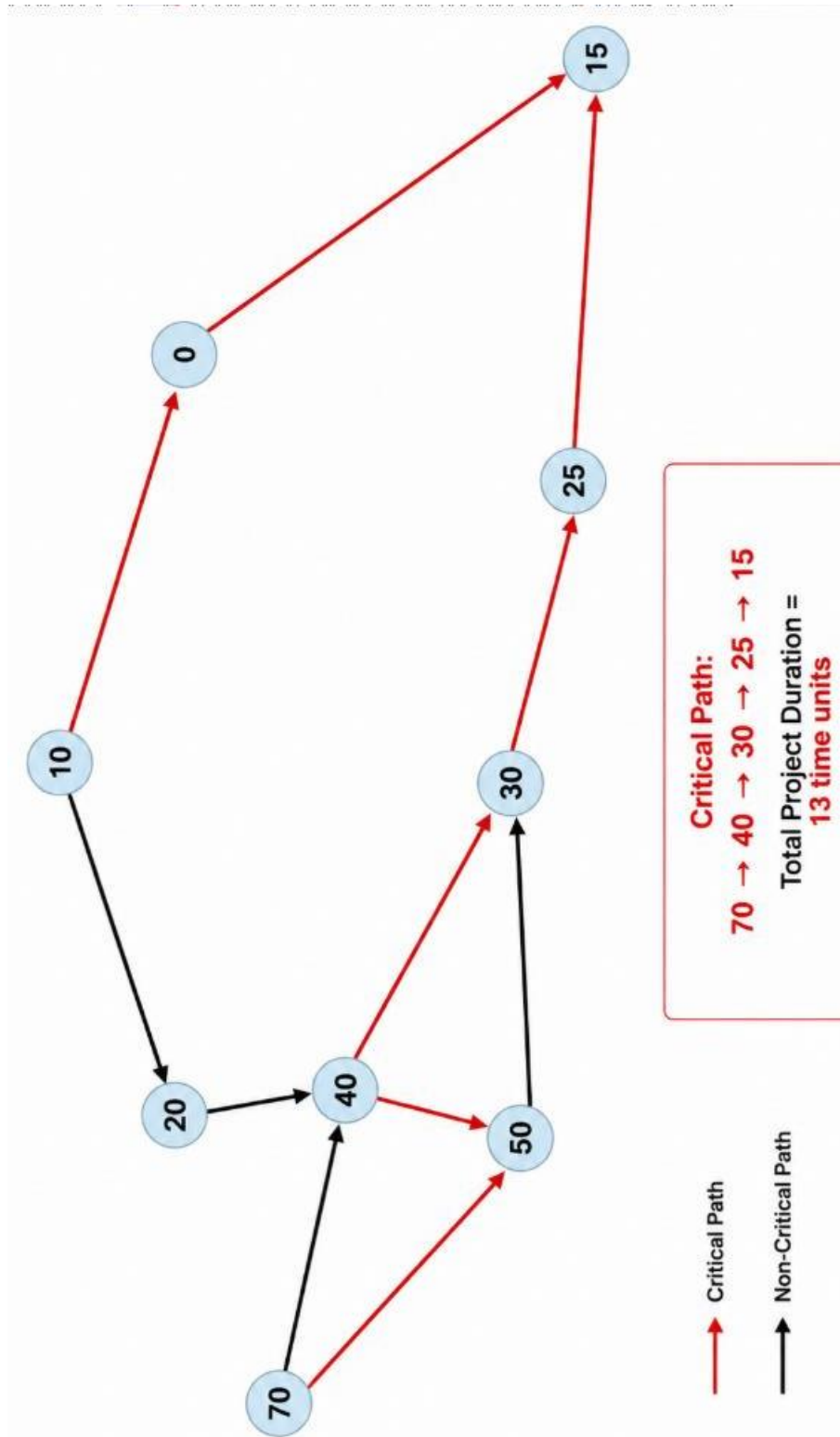
17.2 Gantt Chart



17.3 Project Timeline

Phase	Description	Duration	Weeks	Key Deliverables
Requirements	Gathering system requirements from librarians, staff, and members. Analyze current manual process.	2 weeks	Week 1-2	Requirements Document, Use Cases
Specifications	Define functional and non-functional requirements in detail.	1 week	Week 2-3	SRS Document
Database Design	Design ERD, DFD, database schema for all modules (Member, Book, Loan, Fine, Payment, etc.).	2 weeks	Week 3-5	ERD, DFD, Database Schema
System Architecture & UI/UX Design	Plan system architecture and design user interfaces for all modules.	2 weeks	Week 4-5	Architecture Diagram, UI Mockups
Development	Develop modules: Member, Book, Book Copy, Loan, Hold, Fine/Charge, Payment, Reports, etc.	4 weeks	Week 4-8	Source Code, Working Modules
Testing	Perform functional, security, and performance testing to ensure system reliability.	2 weeks	Week 7-9	Test Cases, Test Reports, Bug Fixes
Documentation & Reports	Generate reports (inventory, overdue, fine, etc.) and prepare technical documentation.	2 weeks	Week 9-10	Reports, Technical Documentation
User Manual & Deployment	Prepare user manual and deploy the system in the production environment.	2 weeks	Week 10-11	User Manual, Deployed System
Presentation & Feedback	Present the system, collect feedback, and finalize the project.	2 weeks	Week 11-12	Final Presentation, Feedback Report

17.4 Critical Path



18 Design Patterns

Software Design Patterns in the Library Management System

In modern software engineering, the use of well-established design patterns is essential for building scalable, maintainable, and reliable applications. These patterns provide standardized solutions to common software design problems and improve the organization, flexibility, and efficiency of the system. The Library Management System, developed using JavaFX and MySQL, incorporates several important software design patterns that define the overall architecture and behavior of the application. These patterns help separate responsibilities, simplify maintenance, and enhance communication between different system components.

1. Model–View–Controller (MVC) Pattern

The Library Management System is primarily structured using the Model–View–Controller (MVC) architectural pattern. This pattern divides the application into three major components: the Model, the View, and the Controller. The **Model** represents the business logic and database-related operations involving entities such as Member, Book, Loan, Fine, and Payment. The **View** defines the graphical user interface using JavaFX and FXML files. The **Controller** handles user actions and connects the interface with the application logic.

The MVC pattern ensures a clear separation of concerns, making the system easier to manage, maintain, and expand. For example, modifications to the user interface can be implemented without affecting the underlying business logic or database structure. This approach also improves code reusability and testing efficiency.

2. Singleton Pattern

The Singleton pattern is used within the database connection management component of the Library Management System. A dedicated database utility class, commonly called `DBUtil`, ensures that only one database connection instance is created and shared across the application.

3. Data Access Object (DAO) Pattern

The Data Access Object (DAO) pattern is implemented to separate database operations from business logic. Dedicated DAO classes manage CRUD (Create, Read, Update, Delete) operations for system entities such as: MemberDAO, BookDAO , LoanDAO. These classes handle all SQL queries and database interactions independently from the controllers and user interface components.

4. Service Layer Pattern

The Library Management System uses a Service Layer to organize and centralize business logic. Service classes act as intermediaries between the DAO layer and the controller layer.

5. Observer Pattern

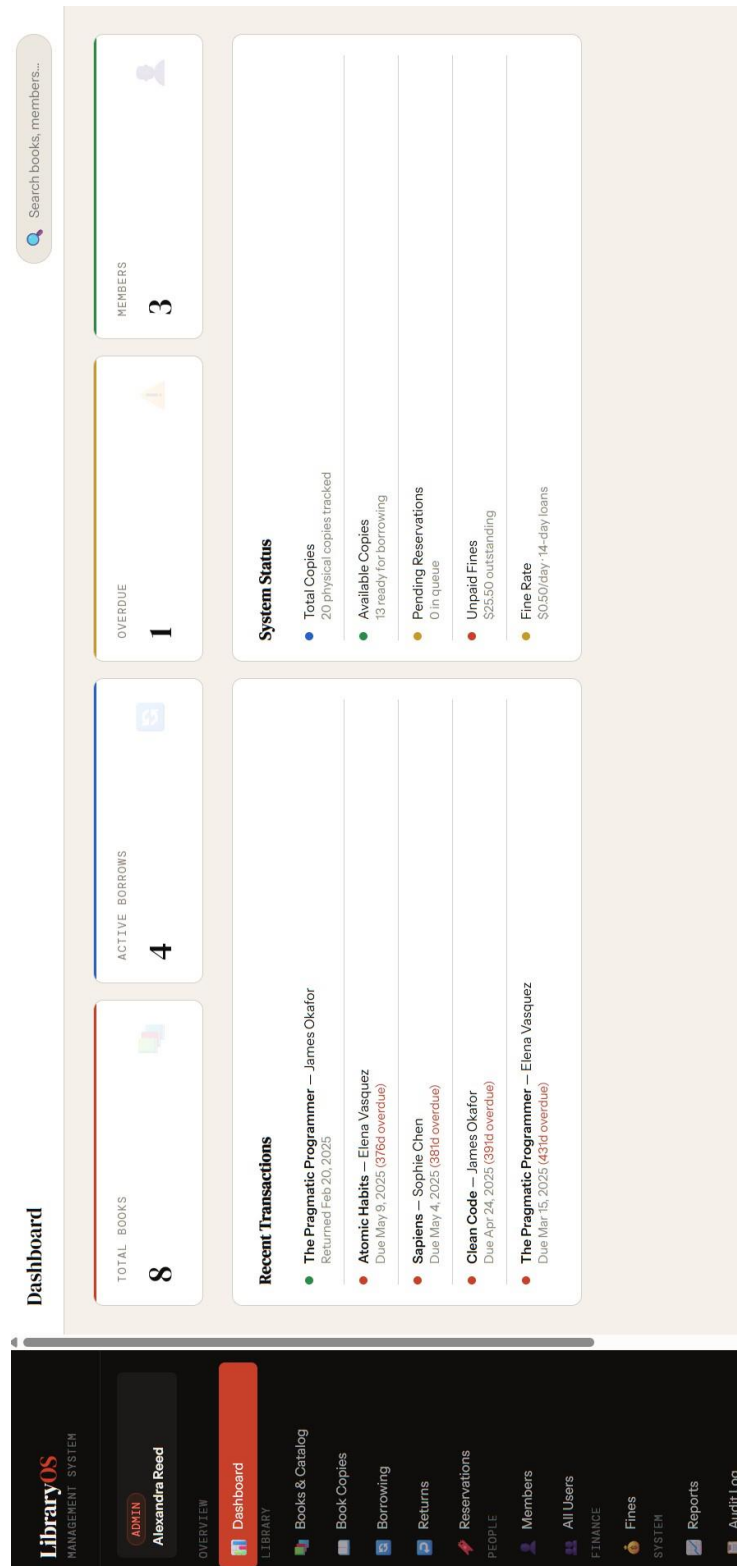
The Observer pattern is also incorporated through JavaFX's built-in property binding and observable features. JavaFX supports observable classes such as `SimpleStringProperty`, `SimpleIntegerProperty`, and `ObservableList`, which automatically synchronize changes between the data model and user interface components. For example, when a book's status changes from "available" to "checked out," the corresponding user interface components update automatically without requiring manual refresh operations. This reactive behavior enhances the responsiveness of the application and improves the user experience while reducing repetitive event-handling code.

Conclusion

Overall, the Library Management System demonstrates the effective use of several important software design patterns to create a well-structured and professional application. The integration of MVC, Singleton, DAO, Service Layer, and Observer patterns, together with the planned use of the Factory pattern, establishes a strong architectural foundation for the system. These design decisions improve code organization, reusability, scalability, and maintainability while aligning the project with modern software engineering standards suitable for both academic and real-world environments.

19 Appendix

Here are some screenshots from our software



LibraryOS
MANAGEMENT SYSTEM

ADMIN
Alexandra Reed

OVERVIEW

Dashboard

LIBRARY

Books & Catalog

Book Copies

Borrowing

Returns

Reservations

PEOPLE

Members

All Users

FINANCE

Fines

SYSTEM

Reports

Audit Log

Books & Catalog

Search books, members...

Books & Catalog (8)

+ Add Book

Search by title, author, ISBN, genre...

All Genres

BOOK	AUTHOR	ISBN	GENRE	YEAR	COPIES	AVAILABLE	ACTIONS
The Pragmatic Programmer	David Thomas	978-0135957059	Technology	2019	3	2/3	Edit Del
Clean Code	Robert C. Martin	978-0132350884	Technology	2008	2	1/2	Edit Del
Dune	Frank Herbert	978-0441013593	Science Fiction	1965	4	3/4	Edit Del
Sapiens	Yuval Noah Harari	978-0062316110	History	2015	2	0/2	Edit Del
The Great Gatsby	F. Scott Fitzgerald	978-0743273565	Classic	1925	3	2/3	Edit Del
Design Patterns	GoF	978-0201633610	Technology	1994	1	1/1	Edit Del
Atomic Habits	James Clear	978-0735211292	Self-Help	2018	2	1/2	Edit Del
1984	George Orwell	978-0451524935	Classic	1949	3	3/3	Edit Del

Add New Book



TITLE *

Book title

AUTHOR *

Author name

ISBN *

978-XXXXXXX

GENRE

e.g. Technology

PUBLISHER

Publisher

YEAR

2024

NUMBER OF COPIES TO ADD

1

Cancel

Add Book

LibraryOS
MANAGEMENT SYSTEM

ADMIN

Alexandra Reed

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Search books, members...

+ Add Copy

Search by book, status, location...

All Status

COPY ID	BOOK TITLE	STATUS	LOCATION	CURRENT BORROWER	ACTIONS
#1	The Pragmatic Programmer	AVAILABLE	A-01	—	Status ▼ Del
#2	The Pragmatic Programmer	BORROWED	A-01	Elena Vasquez	Status ▼ Del
#3	The Pragmatic Programmer	AVAILABLE	A-01	—	Status ▼ Del
#4	Clean Code	BORROWED	A-02	James Okafor	Status ▼ Del
#5	Clean Code	AVAILABLE	A-02	—	Status ▼ Del
#6	Dune	AVAILABLE	B-01	—	Status ▼ Del
#7	Dune	RESERVED	B-01	—	Status ▼ Del
#8	Dune	AVAILABLE	B-01	—	Status ▼ Del
#9	Dune	AVAILABLE	B-01	—	Status ▼ Del
#10	Sapiens	BORROWED	C-01	Sophie Chen	Status ▼ Del

LibraryOS

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ADMIN

Alexandra Reed

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LIBRARY

Books & Catalog

Book Copies

Borrowing

Returns

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PEOPLE

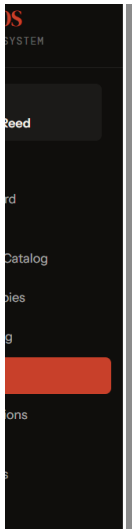
Issue Books

Search books, members...

+ Issue Book

Active Borrows (4)

TRANS ID	MEMBER	BOOK	COPY	ISSUE DATE	DUE DATE	STATUS	OVERDUE
#1	Elena Vasquez	The Pragmatic Programmer	#2	Mar 1, 2025	Mar 15, 2025	OVERDUE	431 days
#2	James Okafor	Clean Code	#4	Apr 10, 2025	Apr 24, 2025	OVERDUE	391 days
#3	Sophie Chen	Sapiens	#10	Apr 20, 2025	May 4, 2025	OVERDUE	381 days
#4	Elena Vasquez	Atomic Habits	#17	Apr 25, 2025	May 9, 2025	OVERDUE	376 days



returns

Search books, members...

Book Returns

Process Return

Active (4) Returned (Recent)

TRANS	MEMBER	BOOK	DUE DATE	OVERDUE	EST. FINE	ACTION
#1	Elena Vasquez	The Pragmatic Programmer	Mar 15, 2025	431d	\$215.50	Return
#2	James Okafor	Clean Code	Apr 24, 2025	391d	\$195.50	Return
#3	Sophie Chen	Sapiens	May 4, 2025	381d	\$190.50	Return
#4	Elena Vasquez	Atomic Habits	May 9, 2025	376d	\$188.00	Return

Reservations

Search books, members...

Reservations (2)

ID	MEMBER	BOOK	REQUESTED	STATUS	ACTIONS
#1	Sophie Chen	Dune	Apr 28, 2025	CANCELLED	
#2	Elena Vasquez	Design Patterns	Apr 15, 2025	CANCELLED	

Members

Search books, members...

Members (3)

+ Register Member

Search members...

NAME	EMAIL	PHONE	JOINED	ACTIVE BORROWS	FINES	ACTIONS
Elena Vasquez	member@library.edu	555-0102	Feb 10, 2024	2	\$25.50	Edit Del
James Okafor	jokafor@library.edu	555-0103	Mar 15, 2024	1	None	Edit Del
Sophie Chen	schen@library.edu	555-0104	Apr 1, 2024	1	None	Edit Del

Register New Member



FULL NAME *

Full name

EMAIL *

email@example.com

PASSWORD *

Min 6 chars

PHONE

555-XXXX

Cancel

Register

Fines

 Search books, members...

Fines

TOTAL FINES

2



UNPAID

\$25.50



COLLECTED

\$2.50



FINE ID	MEMBER	BOOK	DAYS OVERDUE	AMOUNT	STATUS	ACTION
#1	Elena Vasquez	The Pragmatic Programmer	51	\$25.50	UNPAID	Mark Paid Waive
#2	James Okafor	The Pragmatic Programmer	5	\$2.50	PAID	—

Reports & Analytics

 Search books, members...

Reports & Analytics



Books Report

Inventory, genres, copy status



Members Report

Activity, borrows, fines



Transactions

All borrowing history



Fines Report

Overdue, collected, outstanding



Overdue Report

Currently overdue books



Inventory Report

Copy availability by status

Reports & Analytics

 Search books, members...



Books Report

Inventory, genres, copy status



Members Report

Activity, borrows, fines



Transactions

All borrowing history



Fines Report

Overdue, collected, outstanding



Overdue Report

Currently overdue books



Inventory Report

Copy availability by status

Inventory Report

Generated: 5/20/2026, 7:40:47 AM

STATUS	COUNT	%
AVAILABLE	13	65.0%
BORROWED	4	20.0%
RESERVED	1	5.0%
MAINTENANCE	1	5.0%
LOST	1	5.0%

System Settings

 Search books, members...

System Configuration

Library Name Display name for the institution	University Library
Fine Rate Per Day USD charged per overdue day	0.5
Borrow Duration (days) Default loan period	14
Borrow Limit Per Member Max books a member can hold	5

Save Settings

Security Info

Password Hashing bcrypt with salting (industry standard)	ENABLED
Role-Based Access Control Admin / Librarian / Member permissions enforced	ACTIVE
Audit Logging All transactions logged with timestamps	ACTIVE

 Search books, members...

My Profile



Alexandra Reed
admin@libraryedu · ADMIN

PHONE
555-0100

MEMBER SINCE
Jan 1, 2024

USER ID
#1

Edit Profile

LibraryOS

MANAGEMENT SYSTEM

LIBRARIAN

Marcus Thorn

OVERVIEW

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Search books, members...

TOTAL BOOKS

8

ACTIVE BORROWS

4

OVERDUE

1

MEMBERS

3

Recent Transactions

The Pragmatic Programmer — James Okafor

Returned Feb 20, 2025

Atomic Habits — Elena Vasquez

Due May 9, 2025 (376d overdue)

Sapiens — Sophie Chen

Due May 4, 2025 (381d overdue)

Clean Code — James Okafor

Due Apr 24, 2025 (391d overdue)

The Pragmatic Programmer — Elena Vasquez

Due Mar 15, 2025 (431d overdue)

System Status

Total Copies

20 physical copies tracked

Available Copies

13 ready for borrowing

Pending Reservations

1 in queue

Unpaid Fines

\$25.50 outstanding

Fine Rate

\$0.50/day - 14-day loans

LibraryOS

MANAGEMENT SYSTEM

MEMBER

Elena Vasquez

OVERVIEW

My Dashboard

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Browse Catalog

My Borrowers

My Reservations

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Search books, members...

ACTIVE BORROWS

2

OUTSTANDING FINES

\$25.50

NOTIFICATIONS

2

My Current Borrowers

The Pragmatic Programmer

Due: Mar 15, 2025 (431 days overdue)

Atomic Habits

Due: May 9, 2025 (376 days overdue)

Recent Notifications

Your reservation for "Design Patterns" is now ready for pickup!

May 1, 2025

Reminder: "The Pragmatic Programmer" is overdue. Fine accumulating.

Mar 16, 2025

You have outstanding fines totalling \$25.50. Please visit the library to settle.